

TPS System Site Planning

SW02-550

System Site Planning - 1

TPS System Site Planning

SW02-550

**Release 500
CE Compliant
10/2011**

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About This Publication

This publication assists the user in planning an overall Honeywell TPS site. The publication specifies the requirements and recommendations for system environment, ac power, and grounding. If these requirements and recommendations are followed, the customer will achieve the high level of performance and satisfaction expected from Honeywell products.

This publication is intended to be used in conjunction with the planning manual for each of the following:

- Local Control Network (LCN) Planning
- Process Manager/Advanced Process Manager (PM/APM) Planning
- High-Performance Process Manager (HPM) Planning
- Logic Manager (LM) Planning
- Universal Control Network (UCN) Planning
- Data Hiway (DH) Planning

These publications are referenced in Section 1 and are found in two *System Site Planning* binders.

This publication supports **TotalPlant** Solution (TPS) system network Release 520 and CE Compliant hardware. TPS is the evolution of TDC 3000^X.

Any equipment designated as "CE Compliant" complies with the European Union EMC and Health and Safety Directives. All equipment shipping into European Union countries after January 1, 1996 require this type of compliance—denoted by the "CE Mark."

Section 4 applies to CE Compliant systems (with a mix of pre-existing Data Hiway systems).

Section 5 applies to systems that are not required to be CE Compliant.

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Standard Symbols

Scope

The following defines standard symbols used in this publication.

ATTENTION

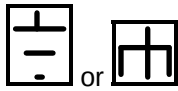
Notes inform the reader about information that is required, but not immediately evident.

CAUTION

Cautions tell the user that damage may occur to equipment if proper care is not exercised.

WARNING

Warnings tell the reader that potential personal harm or serious economic loss may happen if instructions are not followed.



Ground connection to building safety ground.



Ground stake for building safety ground.

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Acronyms

US¥Universal Station
LCN¥Local Control Network
I/O¥Input/Output
MRG¥Master Reference Ground
UPS¥Uninterruptible Power System
ESD¥Electrostatic Discharge
APM¥Advanced Process Manager
PM¥Process Manager
LM¥Logic Manager
UCN¥Universal Control Network
MG¥Motor Generator
UAC¥Uninterrupted Automatic Control
EMI¥Electromagnetic Interference
NEC¥National Electric Code
NFPA¥National Fire Protection Association
HLAI¥High Level Analog Interface
LLAI¥Low Level Analog Interface
LLMUX¥Low Level Multiplexer
DI¥Digital Input
DO¥Digital Output
AO¥Analog Output
PI¥Pulse Input
STI¥Serial Transmitter Interface
SI¥Serial Interface
FTA¥Field Termination Assembly
IOP¥Input Output Processor
KVA¥Kilo Volt Amperes
RMS¥Root Mean Square
THD¥Total Harmonic Distortion
LCNE¥Local Control Network Extender
NEMA¥National Electronics Manufacturers' Association;
ANSI¥American Nation Standards Institute
OSHA¥Occupational Safety and Health Authority
NIM¥Network Interface Module
CS/R¥Clock Source Repeater
ANSI¥American Nation Standards Institute
OSHA¥Occupational Safety and Health Authority
NIM¥Network Interface Module
CS/R¥Clock Source Repeater
RH¥Relative Humidity
HVAC¥Heating, Ventilation and Air Conditioning
BTU¥British Thermal Unit
MOV¥Metal Oxide Varistor
RTD¥Resistive-Temperature Detector
PPB¥Parts per Billion
MHz¥Mega Hertz
Hz¥Hertz
FM¥Factory Mutual
PCE¥Process Control Equipment
PLTC¥Power Limited Tray Cable
CEC¥Canadian Electrical Code

References

Publication Title	Publication Number	Binder Title	Binder Number
<i>Local Control Network Planning</i>	SW02-501	System Site Planning - 1	TPS 3020-1
<i>Universal Control Network Planning</i>	UN02-501	System Site Planning - 1	TPS 3020-1
<i>Process Manager/Advanced Process Manager Planning</i>	PM02-501	System Site Planning - 1	TPS 3020-1
<i>High-Performance Process Manager Planning</i>	HP02-500	System Site Planning - 2	TPS 3020-2
<i>Logic Manager Planning</i>	LM02-501	System Site Planning - 2	TPS 3020-2
<i>Data Hiway Planning</i>	DH02-501	System Site Planning - 2	TPS 3020-2
<i>Process Manager I/O Installation</i>	PM20-520	Implementation/PM/APM Implementation/HPM - 3	TPS 3043 TPS 3066-3
Site Planning Manual for Basic Products	Various	BASIC Product	—

Section 1 – Introduction

1.1 Overview

Section contents

These are the topics covered in this section:

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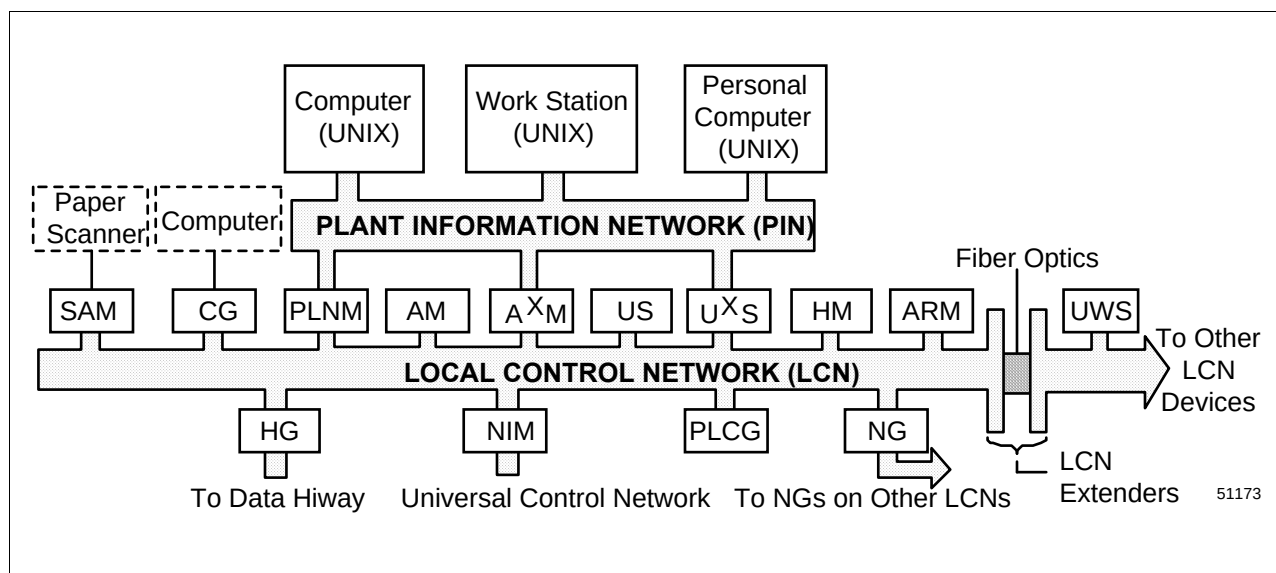
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System planning

This publication provides information needed to prepare facilities for the installation of your Honeywell **TotalPlant** Solution (TPS) System equipment before its delivery. It will guide you through the planning and preparation phases necessary for proper operating system equipment installation regardless of whether your engineering personnel or outside consultants formalize the site preparation plans.

The system typically includes Universal Stations and equipment cabinets that support the Local Control Network (LCN) nodes and gateways. See Figure 1-1.

Figure 1-1 TPS System Overview



1.2 Scope

Scope

This publication defines the planning of the overall site control room and input/output area. The following subjects discussed are:

- placement of control room
- layout of control rooms and I/O areas
- grounding
 - safety ground
 - Master Reference Ground (MRG)
 - lightning grounding
- power requirements and uninterruptible power system (UPS) sizing
- environmental controls and requirements
- planning for a hazardous environment
- Electrostatic Discharge

The planning of each individual control system is discussed in the publications listed below.

Control System	Publication	Publication Number
Local Control Network (LCN)	<i>Local Control Network Planning</i>	SW02-501
Process Manager (PM)/ Advanced Process Manager (APM)	<i>Process Manager/Advanced Process Manager Planning</i>	PM02-501
High-Performance Process Manager (HPM)	<i>High-Performance Process Manager Planning</i>	HP02-500
Logic Manager (LM)	<i>Logic Manager Planning</i>	LM02-501
Universal Control Network (UCN)	<i>Universal Control Network Planning</i>	UN02-501
Data Hiway	<i>Data Hiway Planning</i>	DH02-501

These publications are located in two *System Site Planning* binders.

1.3 TPS Site Selection

Guidelines

The selection of the site for your Honeywell TPS System is probably the most important single factor in planning and preparing for your system's installation.

The choice of a location for the system site should be based on the applicable requirements for aggregate space, clearance, and floor load for the free-standing equipment, air conditioning, electrical power, safety, and fire prevention equipment, support functions and equipment, noise levels, adequate delivery capabilities, storage space, and expansion of the system. These considerations apply to both new and old buildings.

Outlined below are some major site considerations that should be addressed before selecting your system site.

- Is adequate space provided for efficient operation of the system-related work stations and allowance made for any anticipated expansion?
- Is space available to house the heating and air-conditioning equipment (the compressor, air handlers, evaporators, condensers, and cooling-tower equipment), if needed?
- Is the ceiling a minimum of 2.44 m (8 ft.) from the top of the raised floor to accommodate the Honeywell systems cabinets?
- Is the floor of the control-room area structurally adequate to sustain the weight of the system components and any other loads that may be imposed upon it?
- Is the site relatively free from process dust and explosive or corrosive fumes?
- Is the air-filtration system (air cleaner) adequate?
- Is electrical power available and adequate? Are room electrical outlets sufficient as to type and location for any free-standing peripherals? Do electrical power and communications equipment provide for future expansion?
- Is there a separate room for the motor generator (MG)? These are not required when the customer is furnishing an uninterruptible power supply (UPS) and batteries, or battery backup units are installed in the system.
- Is the fire and smoke-detection system adequate?
- Is the fire extinguishing system adequate?
- Are support-function areas (office space, media storage, and equipment maintenance area) satisfactory?
- Is the lighting adequate? Proper natural and artificial lighting is essential for efficient and productive operation and to minimize operator fatigue.
- Is there telephone access?

Continued on next page

1.3 TPS Site Selection, Continued

Guidelines, continued

- Do the control room surrounding walls and ceilings have any type of existing acoustical treatment to help reduce noise?
 - Are the size and location of entrances and exits adequate for equipment delivery?
 - Is there room to unload the delivery van?
 - Is there year-around access where heavy flooding or snow are possibilities?
 - If the building (or facility) is more than one story high, is the elevator accessible and its capacity adequate?
 - Are there any building modifications necessary before moving in the TPS equipment?
-

Additional factors

The following are some additional factors you might consider:

- Insurance costs can be affected by the type of building construction used and by the location of the site relative to fire hazards and fire fighting facilities.
 - Zoning regulations may affect site location, construction, use, and expansion.
 - Before finalizing your site plans, investigate the building codes for your area. These often require that your drawings be approved by a locally licensed architect and/or city engineer.
 - Existing customer facilities and/or operations may be temporarily disturbed by the installation of the TPS System equipment and communications equipment.
 - Electrostatic or electromagnetic interference (EMI) sources in the proximity of the system equipment may have an adverse effect on the system's operation.
 - Your computer and operator areas will be a focal point of interest for employees and visitors — keep them uncluttered and attractive.
 - Security from unauthorized personnel.
-

1.4 Code Requirements

Reference

It is important that both local and National Electrical Code (NEC) requirements be adhered to in the construction of your site. The national codes are contained in the National Electrical Code, NFPA 70, and the Standards for Protection of Data Processing Equipment, NFPA 75. Any power and signal cables installed by Honeywell shall be installed in accordance with NEC and local regulations and requirements.

In the European Community, CENELEC requirements must be adhered to.

1.5 Preinstallation Schedule

Schedule

After you have selected the equipment complement that suits your system needs, establish a schedule incorporating all phases of site preparation and system installation work.

1.6 Optional Services

Overview	In addition to the standard services provided, optional Project Management and Project Engineering functions can be provided to meet customer requirements on a broad range of project services. These could include planning and coordination of an overall project or any of the specific services listed.
Documentation	Special or customized system-specific documentation and drawings can be provided in addition to the standard system documentation.
Kickoff meeting	Special project kickoff meetings can be conducted at a Honeywell location or customer site to review project design, establish communications paths, and finalize project commercial and technical details.
Special wiring	To accommodate special options and wiring requirements, engineering, wire, wiring labor, testing, and documentation can be provided.
Application engineering	Studies can be provided for feasibility and economic analysis. System-design work involves the development of a functional process-control strategy to meet customer plant and process requirements.
Data configuration and programming services	A broad range of optional data services can be provided to create, modify, and maintain the operating system database or specific programs.
Special testing	A wide variety of special factory and field testing can be provided in addition to the standard system-level testing.
Outside purchase	Honeywell can provide procurement services for other vendor's equipment that may be part of the total project.
Power-on support	Assistance can be provided during site preparation, unloading, installation, and power-up of the TPS Process Control System.
Field services	A variety of field services can be provided, including construction engineering, installation, loop check, calibration, commissioning, startup, and hot cutover.

Continued on next page

1.6 **Optional Services,** Continued

Technical training

A full range of quality technical courses for Operations, Engineering, and Maintenance is available to support customer requirements.

Maintenance support services

Honeywell, through its Industrial Services Center (ISC), provides customer's Maintenance Support Services for the TPS System. These services range from parts support and hourly maintenance to comprehensive maintenance contracts to reduce the overall life-cycle cost of both the hardware and software.

Section 2 – Site Layout

2.1 Overview

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2.2 Placement

TPS System

Before your **TotalPlant** Solution (TPS) system equipment arrives, you must prepare to house the system equipment and operators. At this point, the system equipment for each area and the area environment must be defined. Figures 2-2 and 2-3 show process control system-layout examples (unscaled) with three major areas:

- The computer room or computer area
- The control room or operator area
- The equipment room or I/O area

Every site requires a unique review of the layout. For example, the computer room may not be on the same floor, or it may be part of the control room (i.e., the Operating Center), or it may not be needed at all. Likewise, in many sites, the I/O equipment will be installed in remote or satellite equipment rooms to take full advantage of the TPS distributed architecture.

Continued on next page

2.2 Placement, Continued

TPS general environmental classes

Environmental specifications for all of the TPS equipment are grouped in four classes that reflect the allowable environmental ranges for the equipment.

- Class A (General Industrial)—For severe applications where dust, corrosive gases, wide temperature and humidity ranges, and general industrial conditions exist.
 - Class A1 (Mild Industrial)—for mild applications in an industrial atmosphere without corrosive gases with dust and wide temperature and humidity ranges.
 - Class A2 —same as A1 except for reduced operating temperature ranges.
 - Class C (Office)—For applications generally within the comfort zone.
-

Application

Table 2-1 lists the environmental specifications for each class and subclass (e.g., A1, A2). In some systems there will be equipment from different classes, in which case, the more restrictive environment shall apply to the whole system unless the more sensitive components are protected by a special enclosure or room that meets the higher class. For example, if a system is made up of equipment from Class A and Class C that reside in the same room, the more restrictive environment of Class C shall apply to the whole system.

Figure 2-1 graphically shows the relationship between the relative humidity and operating temperature external to the cabinet for each of the classes. Each of the environmental classes and subclasses is represented on the graph. The data for plotting humidity versus the external cabinet temperature comes from Table 2-1. Figure 2-1 also contains the environmental ranges for paper media. (See Note 3 on Table 2-1.)

Environmental specification

CAUTION This equipment is for use in a controlled environment. Although TPS will operate at 0-50°C (32-122°F), Honeywell recommends a normal environment of a nominal 25°C (77°F) with a relative humidity of 40-50% to realize maximum life and enhanced reliability.

Continued on next page

2.2 Placement, Continued

Environmental specification, continued

Table 2-1 Environmental Specifications for LCN Equipment

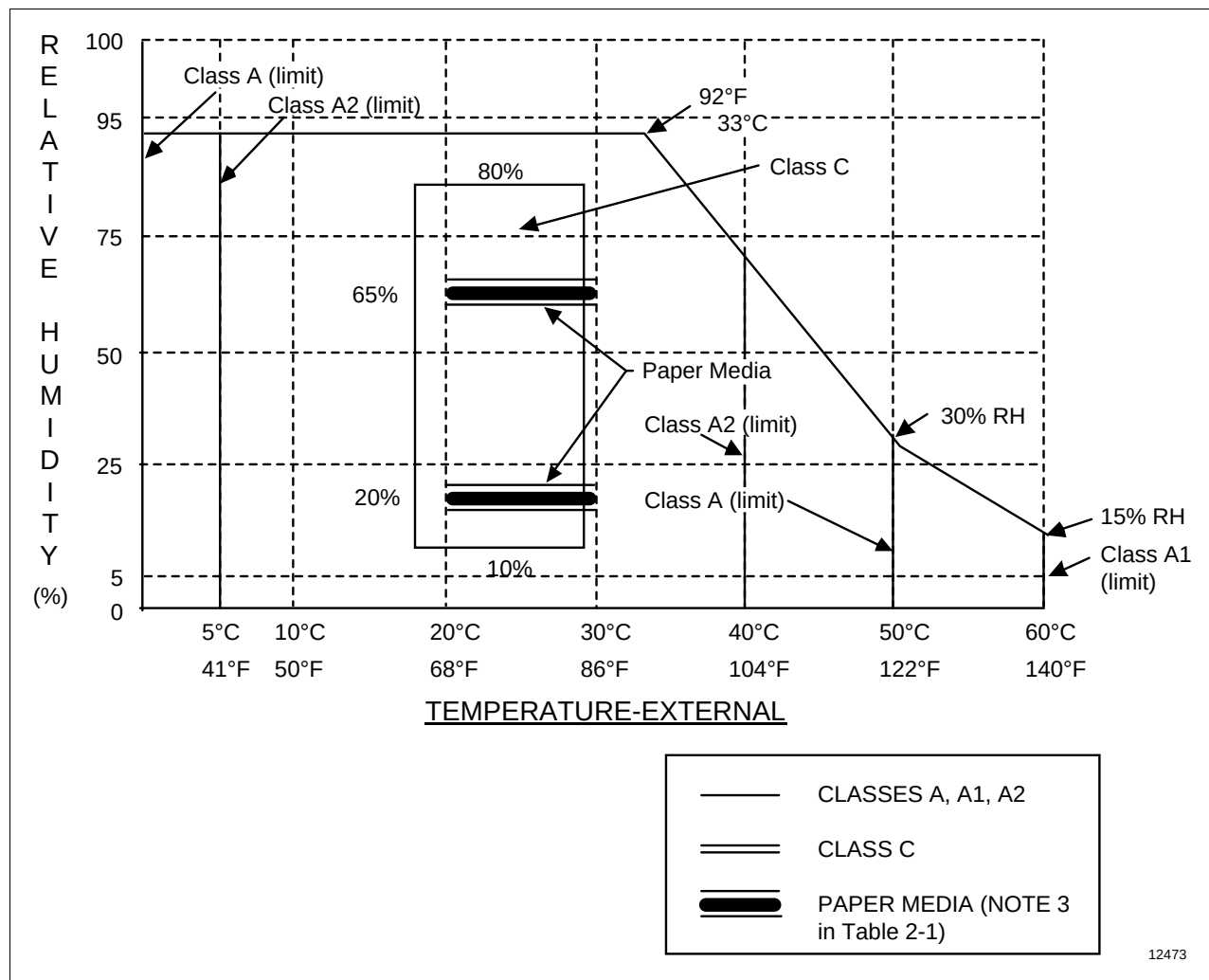
Environmental Classes	General Industrial			Office
	A	A ₁	A ₂	C
Temperature (See Note 1) Operating Cabinet - External Cabinet - Internal Operational Transient (0.25°C/Min. for 1 Hr. Max.) Storage/Shipment	0° to 50°C 0° to 70°C ±15°C -35° to 70°C	0° to 60°C 0° to 70°C ±15°C -35° to 70°C	5° to 40°C 5° to 60°C ±15°C -35° to 70°C	18° to 29°C 18° to 44°C +11° -15°C -35° to 70°C
Humidity (R.H.) (See Notes 2 & 3) MAX. WET BULB	5 to 95% @ 32°C	5 to 95% @ 32°C	5 to 95% @ 32°C	10 to 80% See Figure 2-1
Vibration (See Notes 4 & 5) Operating (14 to 150 Hz) Shipping/Storage (14 to 150 Hz)	1.0 g 1.0 g	1.0 g 1.0 g	1.0 g 1.0 g	1.0 g 1.0 g
Shock Shipping/Storage (See Note 4)	3.0 g	3.0 g	3.0 g	3.0 g
Altitude Operating - Maximum Shipping/Storage - Maximum	7500 Ft. 35000 Ft.	7500 Ft. 35000 Ft.	7500 Ft. 35000 Ft.	7500 Ft. 35000 Ft.
Dust Filters are effective down to:	10 Microns	10 Microns	10 Microns	10 Microns
Notes: 1. The specifications listed are "Room Ambient" conditions except for "Cabinet Internal Temperature," which is the maximum temperature of cooling air that electronic components inside the cabinet will be exposed to. 2. The temperature and relative humidity should not be cycled such that moisture and condensation occurs on the equipment. The rate-of-change of relative humidity should be less than 6% per hour. 3. For Class C paper media, R.H. = 20 to 65%, temperature = 20 to 30°C. 4. The storage and shipping requirements are for one year duration only, provided the equipment is properly packaged and contains an adequate amount of desiccant (moisture removing agent).				

Continued on next page

2.2 Placement, Continued

Sample layout

Figure 2-1 Relative Humidity vs. Temperature



2.3 Electronics Room Design

Guidelines—physical requirements

Follow these guidelines when designing a TPS control room:

- A Universal Station (US) in standard furniture is wedged shaped and can be complexed in a circular console form with stations that contain 10 US nodes. See the *Local Control Network Planning* manual.
- A Universal Station in ergonomic furniture is rectangular and can be complexed in a console to form:

Continued on next page

2.3 Electronics Room Design, Continued

-
- a circular console (using wedge shaped spacers) that has up to 10 Universal Station nodes
 - straight line console that has up to 10 Universal Station nodes.
- See the *Local Control Network Planning* manual.
- LCN modules are placed in a double entry cabinet that is 83.82 cm x 77.47 cm x 196.21 cm (33 in. x 30.5 in. x 77.25) and can weigh up to 476 kg. (1050 lb). See the *Local Control Network Planning* manual.
 - Advanced Process Manager, Process Manager, High-Performance Process Manager, and Logic Manager modules are placed in a double entry cabinet that is nominally 0.8 x 0.8 x 2 meters (32 x 32 x 82 inches). Lifting eye-bolts, which are removable, add about 5 cm (2 inches) to the height. The weight varies according to options ordered and can go as high as 363 kg (800 pounds). A factory assembled complex uses up to four such cabinets bolted together. See the *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Planning*, and *Logic Manager Planning* manuals.
 - Data Hiway equipment uses two different cabinet sizes:
 - the standard-sized Electronics Industries Association (EIA) cabinet, that is designed to mount 484 cm (19 in.) equipment, is 55.9 cm x 64.6 cm x 189.2 cm (22 in. x 25.5 in. x 74.5)
 - the larger cabinet, that is used for optional hardware for intrinsically safe systems, is 68.6 cm x 77.5 cm x 207.0 cm (27 in. x 30.5 in. x 81.5).
- See the *Data Hiway Planning* manual.
- Structural Requirements
 - Is space available to house air conditioning equipment such as compressors, air handling equipment, evaporators, condensers, and cooling tower equipment?
 - Can new equipment be added without causing radical changes to the current allocation of space?
 - Is the lighting and power adequate for expansions?
 - Is there space for an Uninterruptible Power Supply (UPS)?
 - Is the floor loading of existing equipment areas adequate?
 - Will it withstand future loads?
-

Continued on next page

2.3 Electronics Room Design, Continued

Guidelines—access

- Convenience Receptacles—When space permits, place one 15-ampere duplex receptacle in one cabinet in each row. This provides power for maintenance without having to stretch cables across aisles.
- Telephone Receptacles—Provide telephone service as required.
- Emergency Shutdown—If local electrical codes require an emergency shutdown capability, place the instrument ac distribution panel near a principal exit.
- Electrostatic (ESD) or Electromagnetic Interference (EMI) sources in the area.

Follow these guidelines when designing a TPS control room:

- Consider the number and size of cabinets, including allowances for door clearance, aisles, etc. Aisles must have a minimum width of 1370 mm (54 inches) and should allow access to all equipment, both front and rear, as shown in Figure 2-2.
- Doorways, stairwells, elevators, hallways, ramps, and aisles (including turns) must be of sufficient size to permit movement of equipment.
- Local Process Interface—An operator CRT or other interface equipment may be needed on a regular or intermittent basis.

Guidelines—cabling and wiring

Follow these guidelines when designing a TPS control room:

- Cabling—System cabinets are designed to allow cable entry from either the top or bottom. Conduit or cable trays can be used above or below the cabinets. If the trays are overhead, use conduits or tray extensions for the drop from the tray to the equipment cabinet.

Remember, 50-conductor FTA cables cannot be pulled through conduit because of the size of the connectors.

- Intrinsic Safety—I/O areas that handle signals to a hazardous area should be located so that wire runs are kept to a minimum.
- Terminal Panels—There may be a need for two terminal panels shown in Figures 2-2 and 2-3 as follows:
 - Use one terminal panel for all process signals that operate below 30 volts ac/dc. This includes thermocouple/millivolt, 1-5 V/4-20 mA, and 24 V limit switches.
 - Use another terminal panel for process signals that operate above 24 volts ac/dc, up to 240 volts. This keeps the electrical interference associated with high voltage away from the more sensitive, low voltage signals.

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2.3 Electronics Room Design, Continued

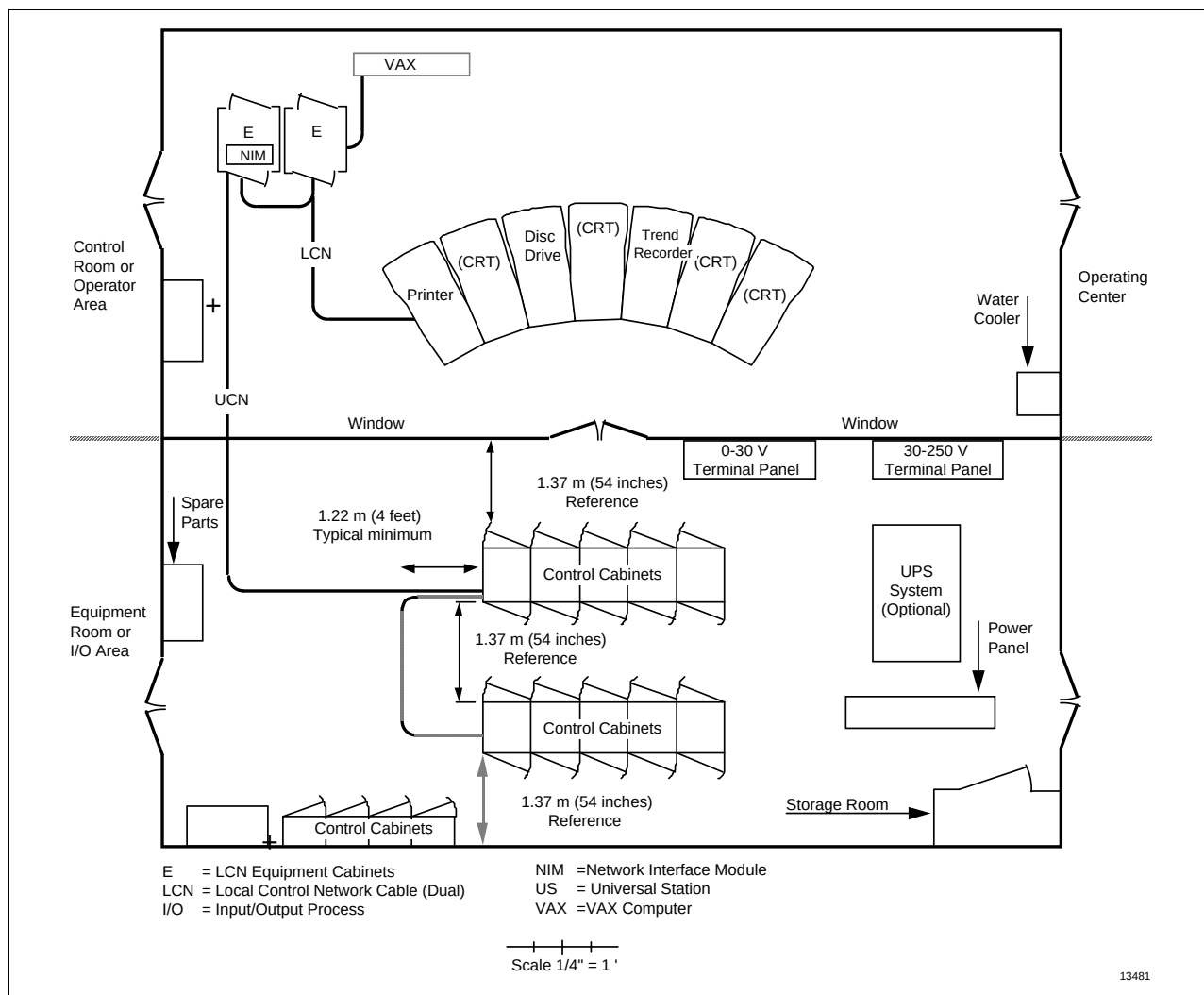
Guidelines—Environment

Follow these guidelines when designing a TPS control room:

- **Cooking Facilities**—Install any kitchen so that cooking fumes do not reach the electronics area. Is the possibility of a cooking fire worth the convenience of having a kitchen by the electronics?
- **Traffic Passages**—As shown in Figures 2-2 and 2-3, avoid routing foot traffic past electronic cabinets. The associated dirt, vibration, banging or pounding, and Electrostatic Discharge (ESD) unnecessarily stresses the equipment.
- **Emergency Lighting**—Wall-mounted rechargeable units work well and are economical.

Sample layout

Figure 2-2 Sample Control Room Layout Using New Console Furniture

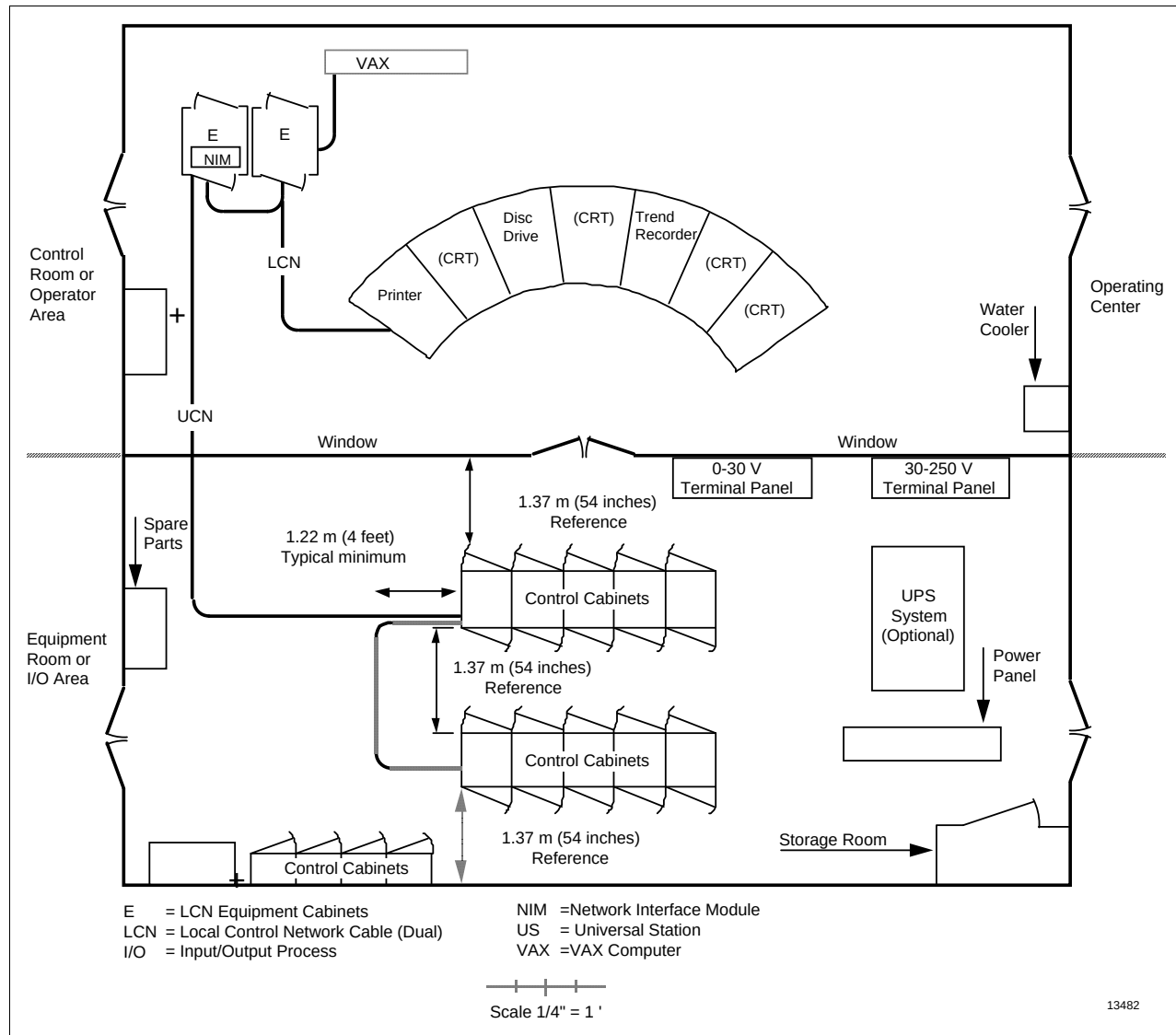


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2.3 Electronics Room Design, Continued

Sample layout

Figure 2-3 Sample Control Room Layout Using Classic Console Furniture



2.4 Floor Design

Overview

The floor of the TPS control room and process control input/output areas must consider the following:

- Ease of wiring—allows direct access to ac power wiring
- Ease of cable installation and maintenance—allows direct access to cabling
- Need to provide for future expansion—allows easy installation of new cables, grounds, and power wiring
- Electrostatic Discharge (ESD) protection—floor panels can be installed that are ESD protective
- Practicality
- Durability
- Floor Loading—must support cabinets and consoles—see specification for weights
- Cable ducts and troughs—cable ducts and troughs can provide electrical shielding for cables and wiring
- Drainage—water must not be allowed to accumulate or stand

Loading

TPS equipment cabinets place different loads on flooring. Table 2-2 lists the weight and the distribution pattern of the weight of each equipment type. The *Local Control Network Planning*, *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Installation*, *Logic Manager Planning*, and *Data Hiway Planning* manuals have more detailed specifications.

Table 2-2 Floor Loading

Type	Weight	Distribution Pattern
Universal Station	204 kg (450 lb) 592 kg/m ²	Leveling pads
LCN Equipment	476 kg (1050 lb) 592 kg/m ²	Rectangle edge of cabinet
PM, APM, HPM, LM	363 kg (800 lb) 567 kg/m ²	1 1/4–4 in. border around the bottom of the cabinet
Data Hiway	450 kg (1000 lb) 979 kg/m ²	2–4 in. border around the bottom of the cabinet

See the *Local Control Network Planning*, *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Installation*, *Logic Manager Planning*, and *Data Hiway Planning* manuals for scaled footprint templates.

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2.4 Floor Design, Continued

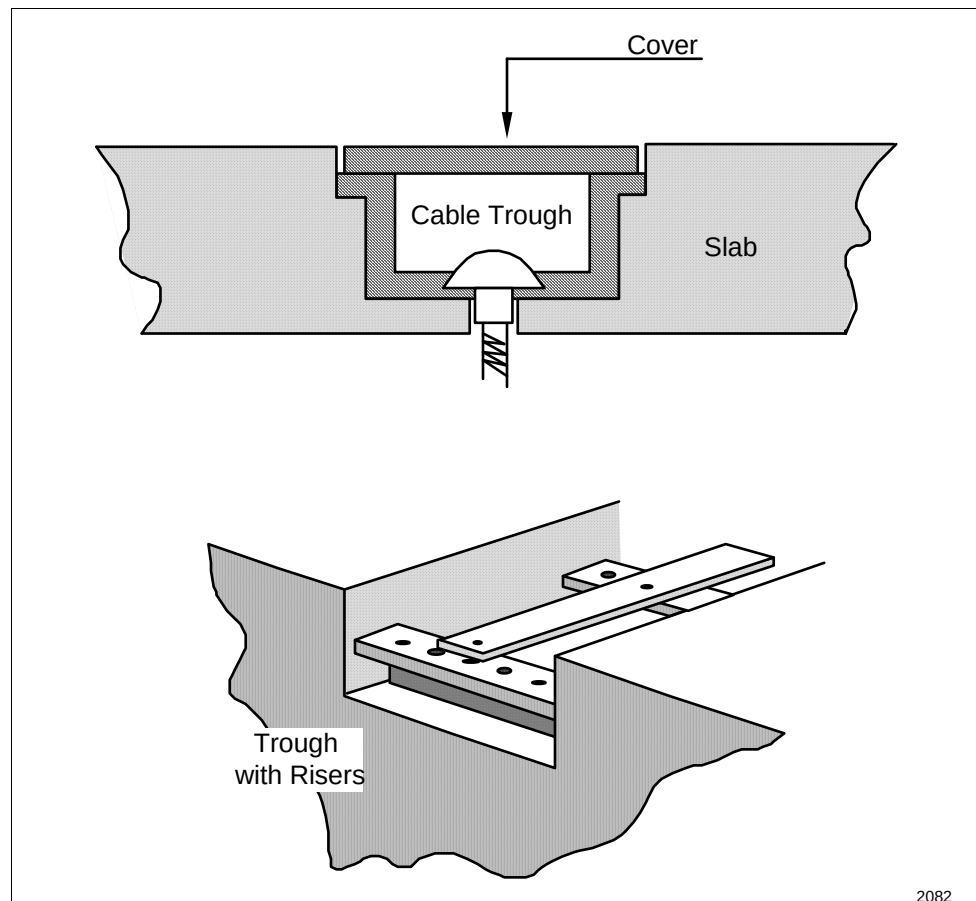
Types of floor construction

There are two types of floor construction in general use.

- Concrete floor with depressed cable troughs
- Raised floor (steel panels covered with ESD tile) on pedestal/raceway-type supports (stringers or screw jacks).

Examples

Figure 2-4 Concrete Floor with Cable Troughs



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2.4 Floor Design, Continued

Examples, continued

Figure 2-5 Raised Floor with Stringers

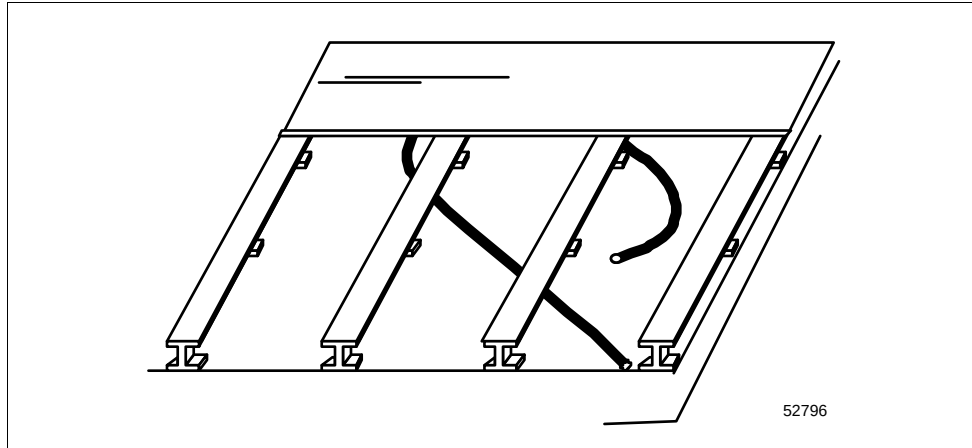
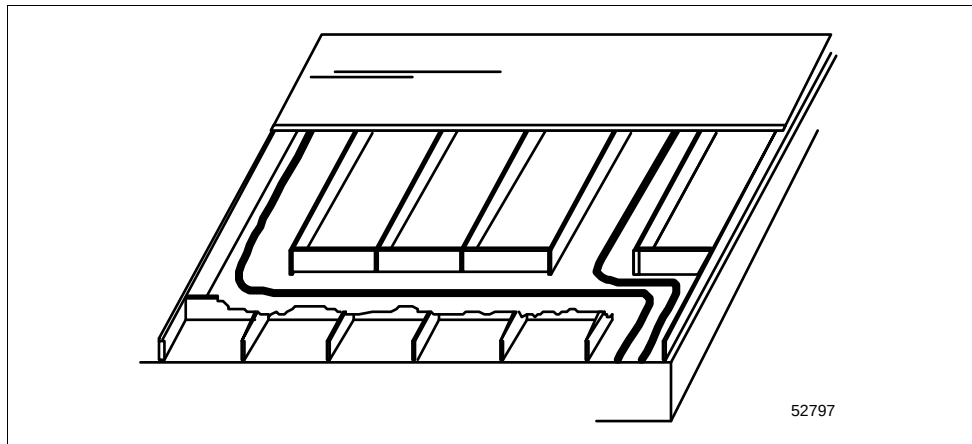


Figure 2-6 Raised Floor with Cable Raceways



Advantages of raised flooring

You may wish to consider a raised floor for the following reasons:

- A raised floor can compensate for a marginal floor by equally distributing the load, while adding only slightly to the total load.
- Installation of equipment wiring may be easier than using overhead trays.
- The free area below the raised flooring can house air ducts, water pipes, and cables or cable trays.

Raised floors commonly provide 40 cm (16 inches) of working space underneath. Construct them to a height of no less than 9 cm (4 inches). Provide drainage to guard against trapping water underneath. If flooding is possible, under-floor power receptacles should be waterproof.

Continued on next page

2.4 Floor Design, Continued

Floor covering

A variety of floor coverings are acceptable with the following precautions:

- All floor coverings must meet local fire codes.
 - Carpets must have antistatic construction that uses electrically conductive threads. This will reduce ESD damage to equipment. Periodic treatment of standard carpeting provides only temporary improvement.
 - Tiled floor should never be waxed. The wax creates and retains static electricity. Any new floor should be treated with a wax stripper and then only washed.
 - Concrete floors should be tiled or coated (sealed) to reduce dust generation and make cleaning easier.
-

Section 3 – AC Power

3.1 Overview

Section contents

These are the topics covered in this section:

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Scope

This section will guide the user through the process of planning the ac power requirements of a **TotalPlant** Solution (TPS) System installation that uses Local Control Network (LCN), Data Hiway (DH), Process Manager (PM), Advanced Process Manager (APM), High-Performance Process Manager (HPM), and Logic Manager (LM) equipment.

Introduction

The TPS system must be supplied with ac power that is within certain limits, and the system electrical equipment must be properly grounded. System performance and safety is directly related to the quality of the power and grounding system supplied by the user. Honeywell provides that portion of the electrical power and grounding equipment contained within the system cabinets, consoles, and peripheral devices.

Analyze before planning

When planning the power system, carefully analyze the present requirements, together with future planned expansions and anticipate unplanned expansion. It is always easier and less expensive to initially install power systems to serve these future needs, than to retrofit cabling trays, new distribution panels, etc., at a later date. Technical assistance is available from the Honeywell Project Engineer or Project Manager. The electrical power requirements for each system component are listed in the *Local Control Network Planning*, *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Planning*, *Logic Manager Planning*, and *Data Hiway Planning* manuals. To determine the total KVA and Btu/hr, use the installation specifications in the manuals.

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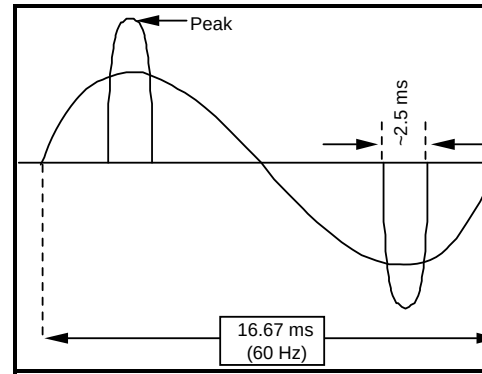
3.1 Overview, Continued

Background

Note that the electrical power (amperage) requirements provided in the *Local Control Network Planning*, *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Planning*, *Logic Manager Planning*, and *Data Hiway Planning* manuals for each of the TPS system components are given in terms of RMS, Peak, and Watts. It is important to understand how these values were measured and the measuring device used.

Most ac-type volt/ampmeters use a rectification and averaging technique to measure current, but are scaled to indicate RMS, based on an assumed sinusoidal input.

For TPS, the majority of the loads are of the rectifier/capacitor-input type, where current flows on the peaks of the line voltage, as illustrated:



Higher peak currents

The current drawn by the input is far from sinusoidal and contains large amounts of odd harmonics. As a result, the peak currents are much higher than those of a linear (resistive or inductive) load.

This relationship between the peak value and the RMS or effective value of the current is indicated by the "crest" factor, which is 1.41 for a resistive load (sinusoidal current).

Large errors

Using an average reading instrument that is scaled for sinusoidal RMS values to measure this type of load can result in errors of 50% or more; that is, the reading is half of the RMS effective value of the current.

The RMS (effective) value of the current determines the heat generated as a result of resistive losses in the TPS system. The peak current, however, determines the voltage losses in the power source and the distribution system; therefore, when the current waveform is distorted and the source impedance is not low enough, there is an appreciable amount of distortion. System fuse or circuit breaker ratings, however, should be based on RMS values (taking into account possible inrush current).

3.2 AC Power System Design

Overview

Strive to design the power distribution system for no more than 5% power loss.

- Transformer wire losses $\leq 3\%$
- Distribution system $\leq 2\%$

Guidelines for power system components

Breaker sizing and wire sizing should be done according to the lowest acceptable voltage (use maximum current listed in specification table) because the constant current outputs will draw more input current when the line voltage drops to the lowest acceptable voltage.

Circuit Breakers:	<u>Inrush current</u>—Breakers are selected on the basis of RMS current. • LCN systems have soft start circuitry to limit turn-on current to the rated full load running current. • PM, APM, and HPM systems do have soft start circuitry, but the inrush current is 35 amps. • Data Hiway systems have CVT input transformers that have a one cycle current inrush of up to 30 times the running current. <u>Guideline</u>—Startup philosophy will determine the selection of delay curve-type of breaker. After a total power failure and power returns, all power supplies will start simultaneously and can cause a significant breaker overload. • Breakers with only CVT power supplies should have the 25x delay curve. • Breakers with 50% CVT loads can have medium (6x) delay curves. • Breakers for PM, APM, and HPM products should have medium (6x) delay curves. <u>Sizing</u>—Use the typical RMS currents from the planning manuals for sizing breakers. ATTENTION The typical RMS values do not include the increase in current caused by low voltages (102/204 Vac.) • Each cabinet must have its own dedicated breaker so that a single cabinet (or power strip in an equipment cabinet) can be shut off separately. See Figure 3-3. • Because each cabinet or power strip is limited to 15 amperes by the breaker in the cabinet, a larger breaker in the distribution box is not called for. • The purpose of these breakers in the distribution panel is twofold: – Protect the power wiring from damage. – Allow for complete shut off of power for service to the station or cabinet.
-------------------	---

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3.2 AC Power System Design, Continued

Guidelines for power system components, continued

Observe the following wiring rules.

Distribution Wiring:	Use size 12 AWG (2.5 mm²) wire from the Universal Station or cabinet to the distribution panel if • the distance from the distribution panel to the Universal Station or cabinet is 75 feet (22 meters) or less • the total current in the cabinet is 10 amps or less If the distance is greater than 75 feet or if the current is larger than 10 amps, use Table 3-1, Table 3-2, and Table 3-3. ATTENTION For wire sizes larger than 12 AWG (2.5 mm²), special steps must be taken to reduce the wire size to 12 AWG as it enters the Universal Station, LCN equipment cabinet, or PM/APM/HPM cabinet (the power terminal will not accept wire larger than 12 AWG or 2.5 mm²).
----------------------	--

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3.2 AC Power System Design, Continued

Guidelines for power system components, continued

Transformers should be sized using typical RMS values.

Transformer Sizing:	When a transformer is used to directly supply ac power to the TPS equipment (LCN, PM, APM, HPM, LM, or personal computer-based systems) an Underwriter's Laboratory (UL) rated transformer with a K factor of 9 or better is recommended with 5% taps to provide an additional 5% output voltage on a contingency basis. The resistance rating of the transformer should be 3% or better and the total impedance should be less than 5%. Low temperature-rise models or oversized units may be used to get the equivalent source resistance and to protect against additional temperature rise due to the harmonic eddy current content that is defined by the K = 9 rating. The 3% resistance is required to minimize the "flat-top" voltage effect. The K factor-rated transformers are made by many vendors and have a premium cost factor of from 10% (for transformers in the 15 KVA range) to 100% (for 500 KVA). K = 9 transformers will have longer life. Transformers for use in systems having only Data Hiway equipment can be normal-rated transformers. Systems having a mix of Data Hiway and LCN, PM, APM, HPM, LM, and personal computer equipment should have transformers with a K factor = 9.
UPS Systems:	Note: The following procedure is for sizing Uninterruptible Power Systems. 1. Add up the individual typical RMS currents for the TPS equipment chosen. 2. Multiply the total currents by 120 or 240 volts. 3. Check that the UPS equipment can handle a crest factor of at least 3.0. ATTENTION These steps do not account for any overhead from other equipment or for possible expansions. These values are typical and do not include any margins.

Continued on next page

3.2 AC Power System Design, Continued

Wire size table

Select the wire size from Table 3-1 according to load current, length, and percentage voltage drop allowed. Table 3-1 determines the wire size based on the RMS current.

Table 3-1 Copper Wire Size Based on RMS Current Requirement for 120 Vac

Amp s	1% (Voltage Drop)						2% (Voltage Drop)						KVA
	Cable Feet						Cable Feet						
	25	50	75	100	200	500	25	50	75	100	200	500	
5	12	12	12	10	8	4	12	12	12	12	10	6	0.6
10	12	10	8	8	4	1	12	12	12	10	8	4	1.2
15	12	8	6	6	2	2/0	12	12	10	8	6	2	1.8
20	10	8	6	4	2	3/0	12	10	8	8	4	1	2.4
25	10	6	4	4	1	4/0	12	10	8	6	4	0	3.0
30	86	4	2	0	—		10	8	6	6	6	2/0	3.6
40	84	2	2	2/0	—		88	6	4	2	3/0		4.8
50	64	2	1	3/0	—		86	4	4	1	4/0		6.0
60	62	1	0	4/0	—		66	4	2	0	—		7.2
80	42	0	2/0	—	—		44	2	2	2/0	—		9.6
100	21	2/0	3/0	—	—		22	2	1	3/0	—		12
125	10	3/0	4/0	—	—		11	1	0	4/0	—		15

Continued on next page

3.2 AC Power System Design, Continued

Wire size table (IEC)

Select the wire size from Table 3-2 according to load current and length in meters for a 2% maximum voltage drop for 240 Vac power.

Table 3-2 Maximum Cable Length in Meters for 240 Vac

Cable Size in mm ²	Current in Amps				
	5	10	15	20	30
1,5	38	19	—	—	—
2,5	63	32	21	—	—
4	100	50	33.3	25	—
6	150	75	50	37.5	25
10	252	126	84	63	42
16	400	200	133	100	66.6

Wire size table (IEC)

Select the wire size from Table 3-3 according to load current and length in meters for a 2% maximum voltage drop for 120 Vac power.

For a 1% maximum voltage drop, allow half of the cable length in Table 3-3.

Table 3-3 Maximum Cable Length in Meters for 120 Vac

Cable Size in mm ²	Current in Amps				
	5	10	15	20	30
1,5	19	9.5	—	—	—
2,5	31.5	16	10.5	—	—
4	50	25	16.6	12.5	—
6	75	37.5	25	18.7	12.5
10	126	63	42	32	21
16	200	100	66	50	33.3

3.3 AC Primary Power Source

Guidelines

Figure 3-1 illustrates a typical quality primary power source layout for a TPS system, including an Uninterruptible Power Supply (UPS) for backup. The following guidelines must be followed:

- **If ac power is not securely referenced to local safety ground**, the ac power for the TPS system equipment must be supplied by a dedicated power transformer located within 18.3 m (60 ft.) of the system distribution panel or dedicated branch circuits.

ATTENTION

ATTENTION—The term, dedicated, means that the transformer secondary and panel breaker(s) are exclusively used to power the TPS System and for no other purpose.

This is because of various power line disturbances inherent in an industrial environment, which can adversely affect system performance.

- Where multiple process control/computer systems are required, care must be exercised to separate the electrical power requirements of each system.
- The power transformer must accept single- or 3-phase 120/240/480 Vac $\pm 10\%$ (nominal power-line rating) at its primary and deliver 120/240 Vac nominal at its secondary, and must operate in a frequency range of 47-62 Hz for 50 or 60 Hz. Local nominal power line voltage can vary; typical values include 110/220, 115/230, and 120/240 Vac. Contact your local utility for actual values. The operating voltage range of the TPS System is 102-132 Vac. (Refer to the appropriate planning manuals for the voltage range of each system device.)

Three-Phase power

Special considerations apply if your ac power is 3-phase with no neutral. They are

- A dedicated transformer is required where the secondary must be a 4-wire system in a Y configuration.
- The secondary neutral must be wired in accordance with *NEC Article 250-26*, single-phase systems, or IEC standards. All other aspects of single-phase power and ground philosophies, as outlined in this manual, must be followed.
- A balanced load must be employed.
- Neutral returns are not to be daisy chained from device-to-device.
- See subsection 3.2 for wire sizing (the neutral wire must be two wire sizes larger because of the currents from the odd harmonics on the line).

Continued on next page

3.3 AC Primary Power Source, Continued

Frequency variations

Frequency variations are more critical because some peripherals operate within 0.5 Hz of their nominal rating (e.g., the matrix printer). Although power companies deliver ac power within fairly tight constraints (e.g., ± 0.5 Hz), a power outage or brownout may cause the more frequency-sensitive devices to reset; carefully read the subsection entitled *Power Failure Considerations*. Check the equipment specifications in the appropriate planning manuals to identify the frequency-sensitive devices. In addition, the input ac power to the system must not contain more than 6.0% Total Harmonic Distortion (THD), including the power transformer.

This is true of older equipment still in use today.

Continued on next page

3.3 AC Primary Power Source, Continued

Illustration

The following three figures represent an example of a typical ac power entry, distribution system, and Uninterruptible Power System (UPS).

- The three figures represent one drawing of an ac power system.
- The letters A, B, and C represent "connections" between the three figures.

Figure 3-1 Typical Three Phase Power Entry

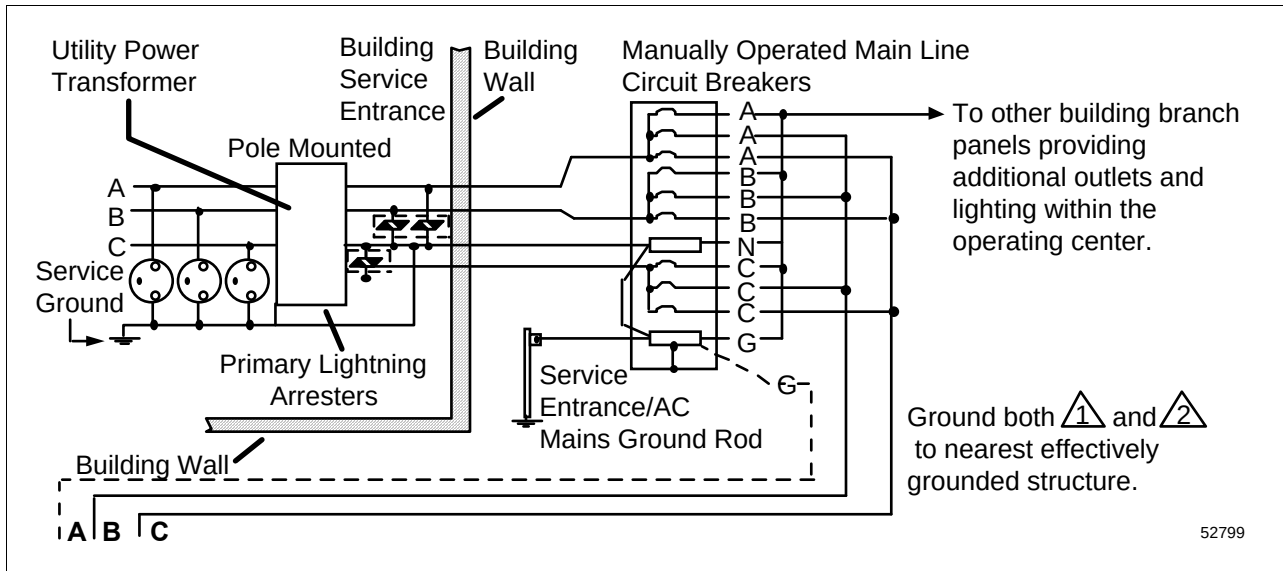
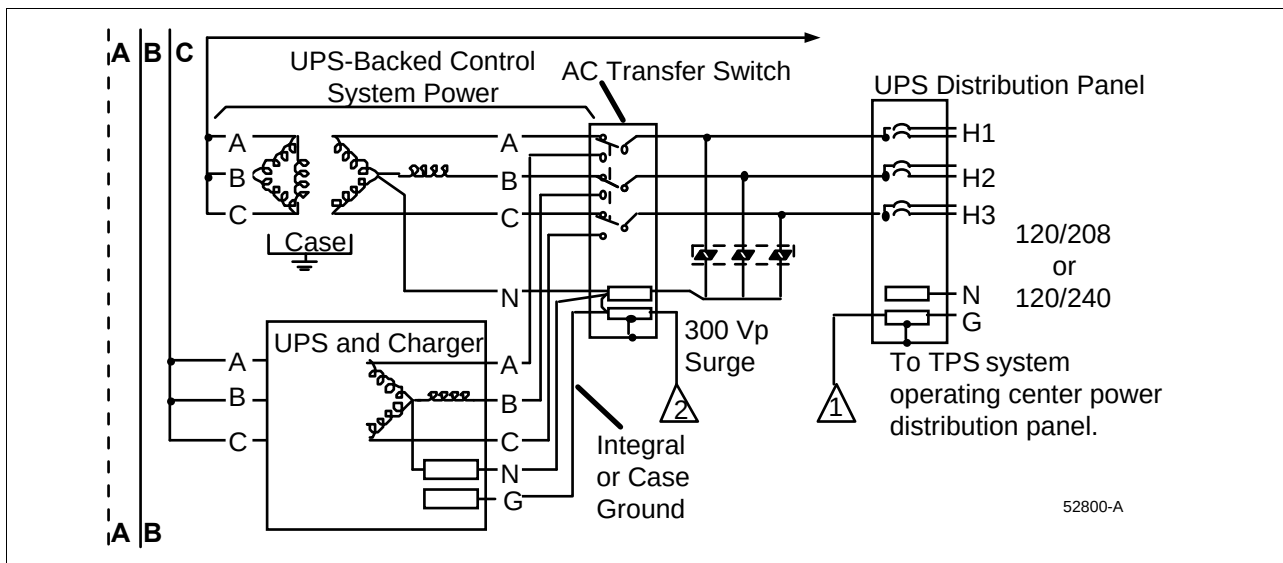


Figure 3-2 Typical Uninterrupted Power System (UPS) Distribution

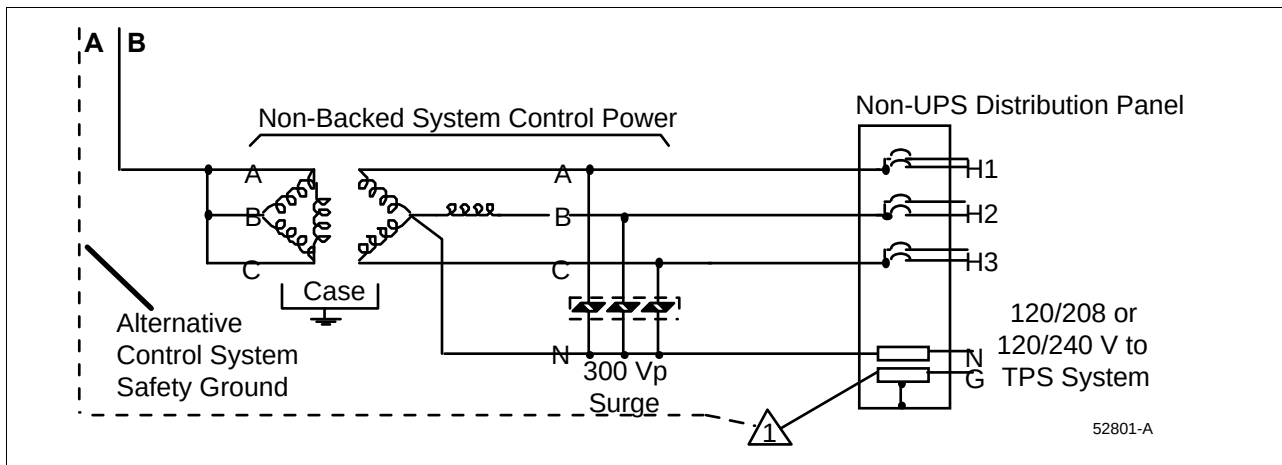


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3.3 AC Primary Power Source, Continued

Illustration

Figure 3-3 Non-UPS Power Distribution



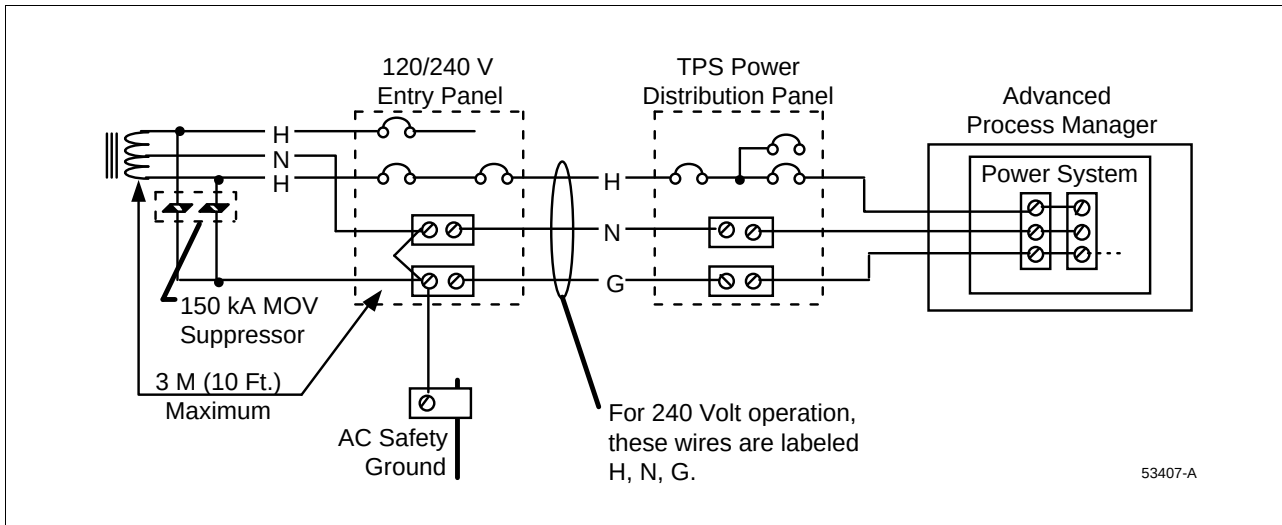
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3.3 AC Primary Power Source, Continued

Illustration

Figure 3-4 shows the details of a typical ac power system with ac safety ground stake connections. Also shown are the MOV suppressor connections put in for lightning protection.

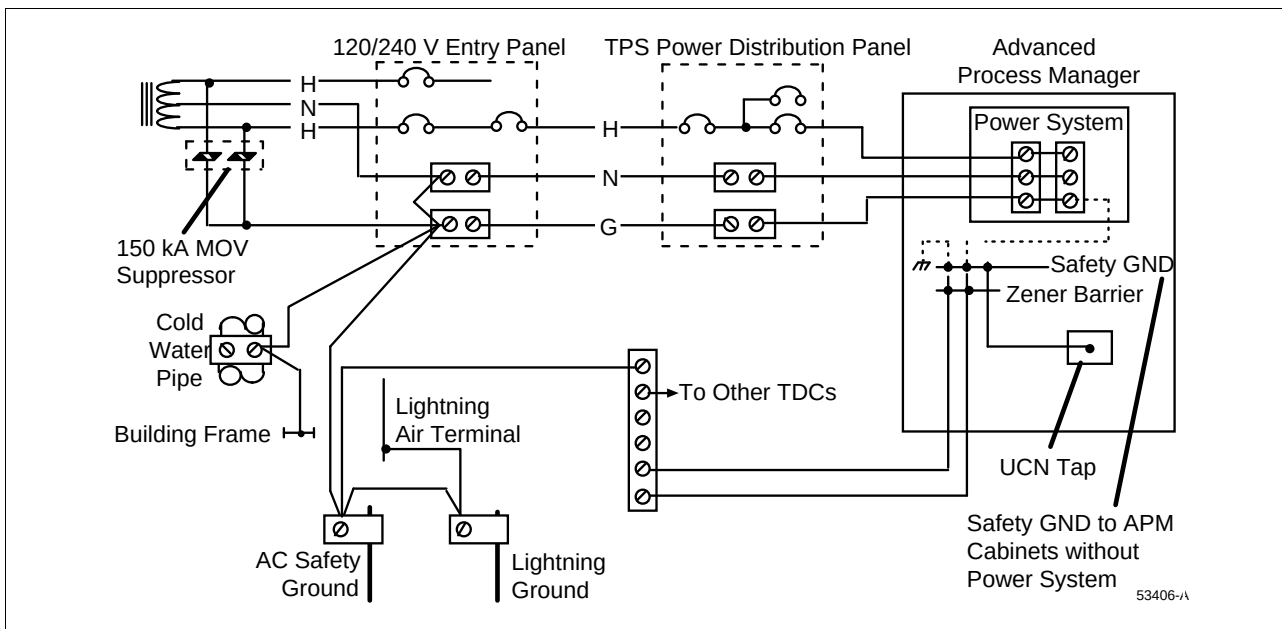
Figure 3-4 Typical PM/APM/HPM AC Power System and System Grounding



Illustration

Figure 3-5 shows the detail of a typical ground system with safety ground, lightning ground, and star plates.

Figure 3-5 Typical PM/APM/HPM Grounding System

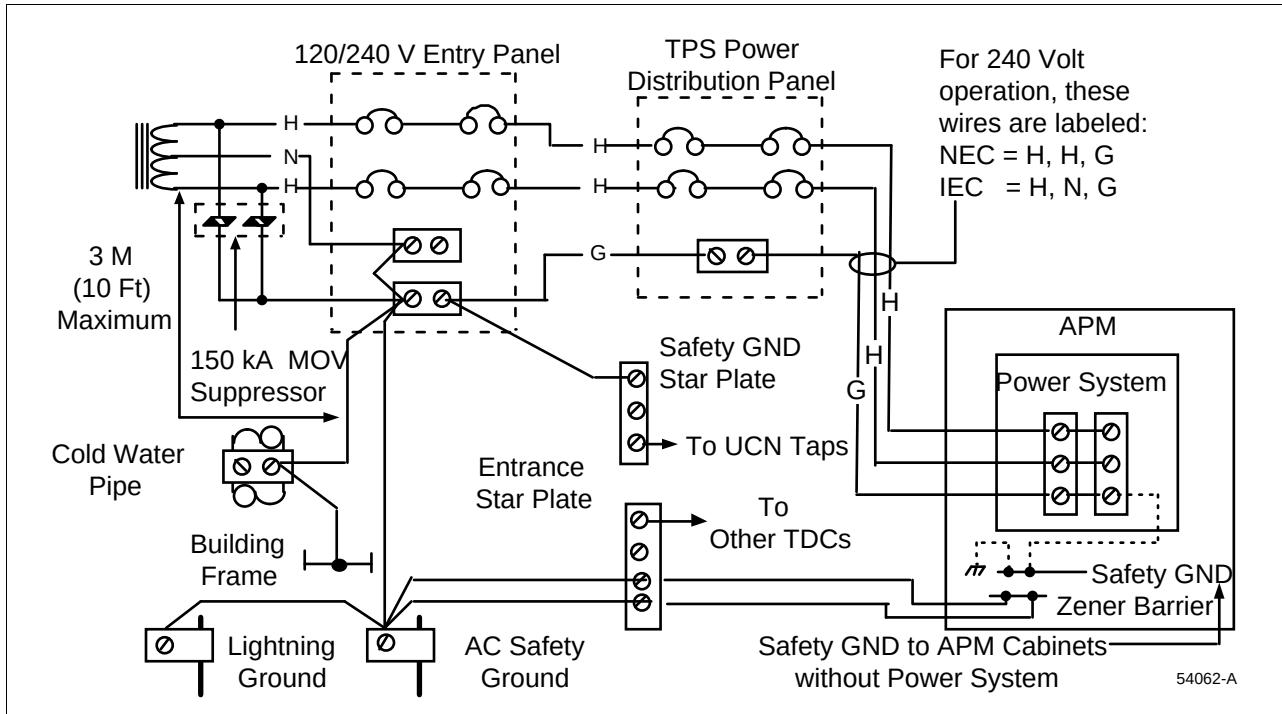


Continued on next page

3.3 AC Primary Power Source, Continued

Illustration

Figure 3-6 Typical PM/APM/HPM AC Power System and System Grounding for IEC 240 Vac



Continued on next page

3.3 AC Primary Power Source, Continued

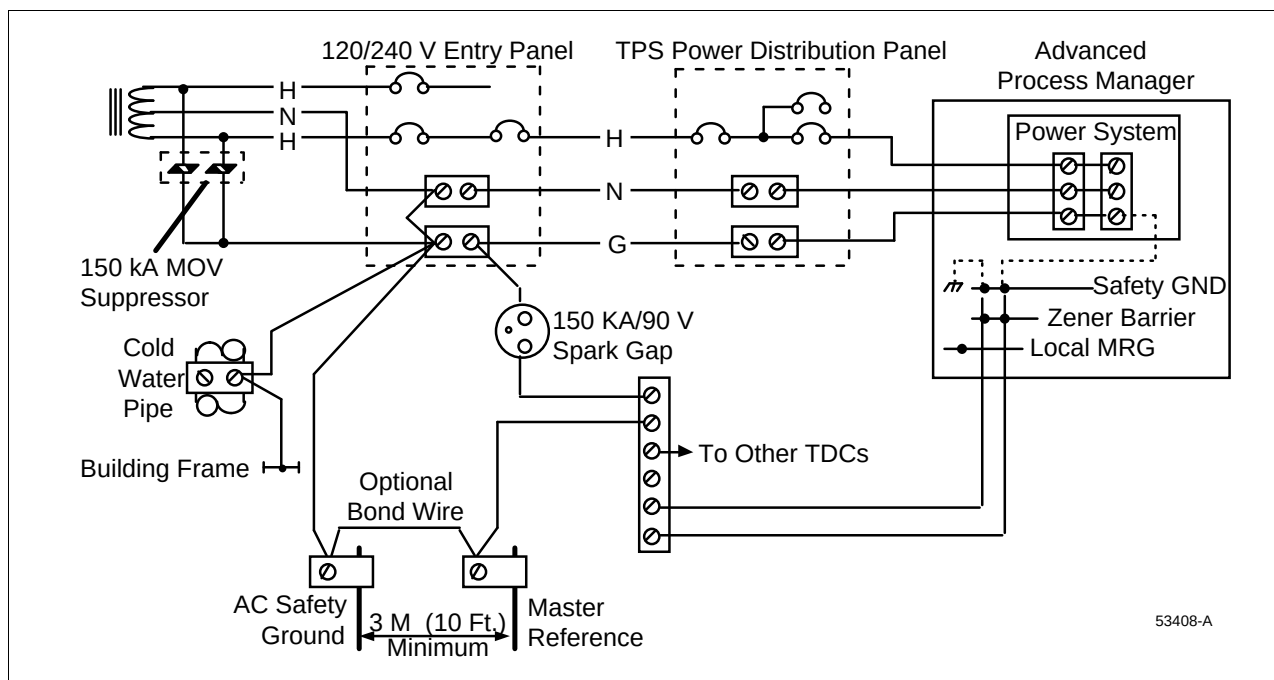
ATTENTION

ATTENTION—Systems that were installed with a Master Reference Ground (MRG) are acceptable as long as code is not violated. **The are no changes to the application of Data Hiway system usage of the MRG.**

Data Hiway MRG

TPS systems that have Data Hiways must have a Master Reference Ground (MRG) as a part of their grounding system as shown in Figure 3-7.

Figure 3-7 Typical PM/APM/HPM AC Power System and System Grounding



3.4 Multiple Power Sources

UPS as backup power

As shown in Figures 3-1, 3-2, and 3-3 an Uninterruptible Power System (UPS) is used to provide primary power. Local power can be used for backup power in case primary power fails. Several UPSs can be used for redundant operation of critical console equipment; however, each must have its own power transfer switch. Normally, the UPS is the primary feed and the local power is the backup feed.

For nonredundant multiple UPS installations, each UPS has its own power distribution box and switchover panel (if required). If there are non-TPS loads on the UPS, they require separate transformers and distribution panels.

A situation that may arise involves support equipment that is not TPS-related, but is to be included in a Universal Station. Examples are telephones, radios, chargers, programmable controllers, custom switch and annunciator panels, and related displays. Such equipment must not be powered from the TPS-dedicated transformer distribution box.

PM/APM/HPM with separate power sources

As shown in Figure 3-5, the PM, APM, or HPM can be simultaneously powered by two separate power sources:

- Line power could be power source #1 and a UPS could be power source #2.
 - The in-plant generator could be power source #1 and local power could be power source #2.
-

3.5 TPS AC Power Distribution Guidelines

Conduit

Metal conduits are highly recommended for all power cable runs. The conduit must be grounded at the power distribution box and at the system-equipment power entry box. The case of the equipment power entry box and branch-circuit distribution panel must be grounded (bonded) to building steel.

The IEC allows cable trays for power cables when this is desirable.

ATTENTION

- **ATTENTION**—To avoid electrical noise problems in the system, observe the power cable(s) to signal cable(s) separation rules discussed in the *Data Hiway Planning*, *Process Manager/Advanced Process Manager Planning*, and *High-Performance Process Manager Planning* manuals.
- **All ac power wiring must be 3-conductor, with a fire-resistant jacket**
 - Hot
 - Neutral
 - Safety Ground

Distribution box sizing

The size or capacity of the system power distribution box or dedicated branch circuit(s) depends on the size of the system. The panel must contain a safety ground (equipment) bus bar to which the ground (green wire or yellow green) from the system cabinets, consoles, and device receptacles can be connected. The safety ground bus bar must then be connected to the ac safety ground rod through an insulated cable specified by local code requirements. Refer to the subsection, Cabinet Ground Connections, in this publication.

Power entry boxes

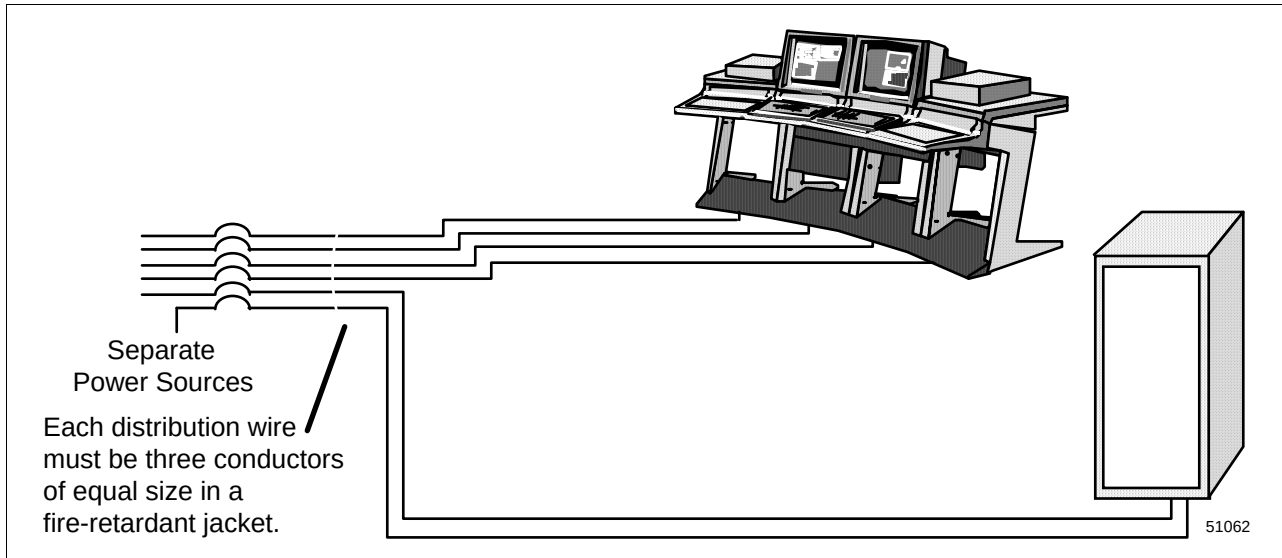
Each TPS product has its own power entry scheme. Refer to the following subsections for the specific product.

Continued on next page

3.5 TPS AC Power Distribution Guidelines, Continued

Illustration

Figure 3-8 LCN AC Power Distribution



Continued on next page

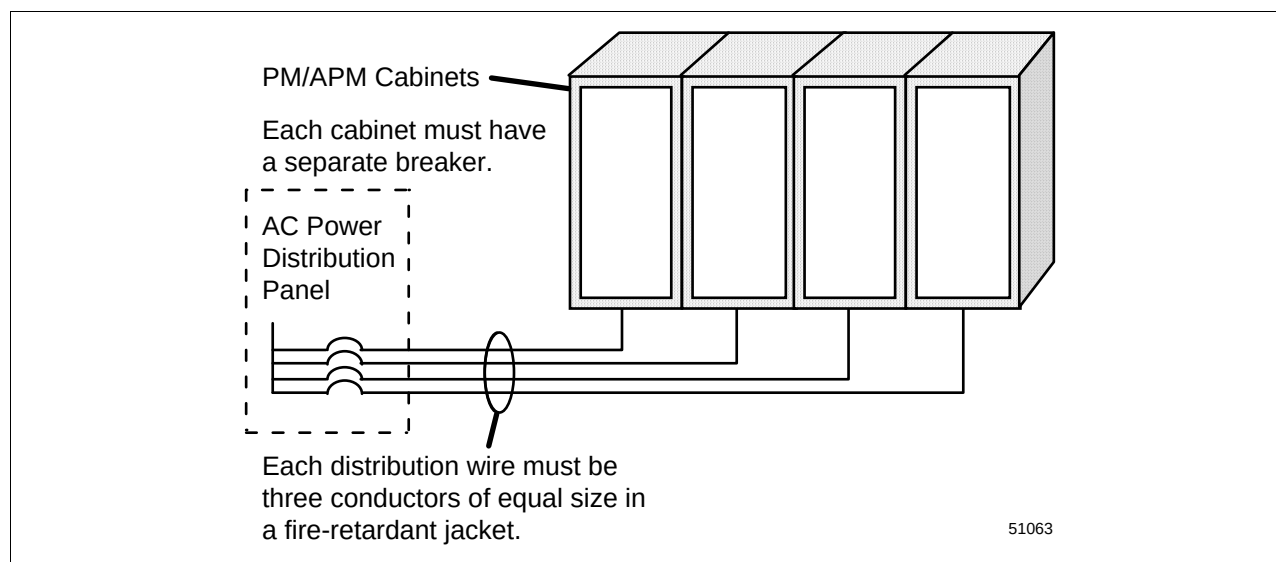
3.5 TPS AC Power Distribution Guidelines, Continued

PM, APM, and HPM equipment

Process Manager (PM), Advanced Process Manager (APM), and High-Performance Process Manager (HPM) equipment cabinets do not have ac breaker boxes. The ac power connects directly to the system power distribution panel.

Illustration

Figure 3-9 PM/APM/HPM AC Power Distribution



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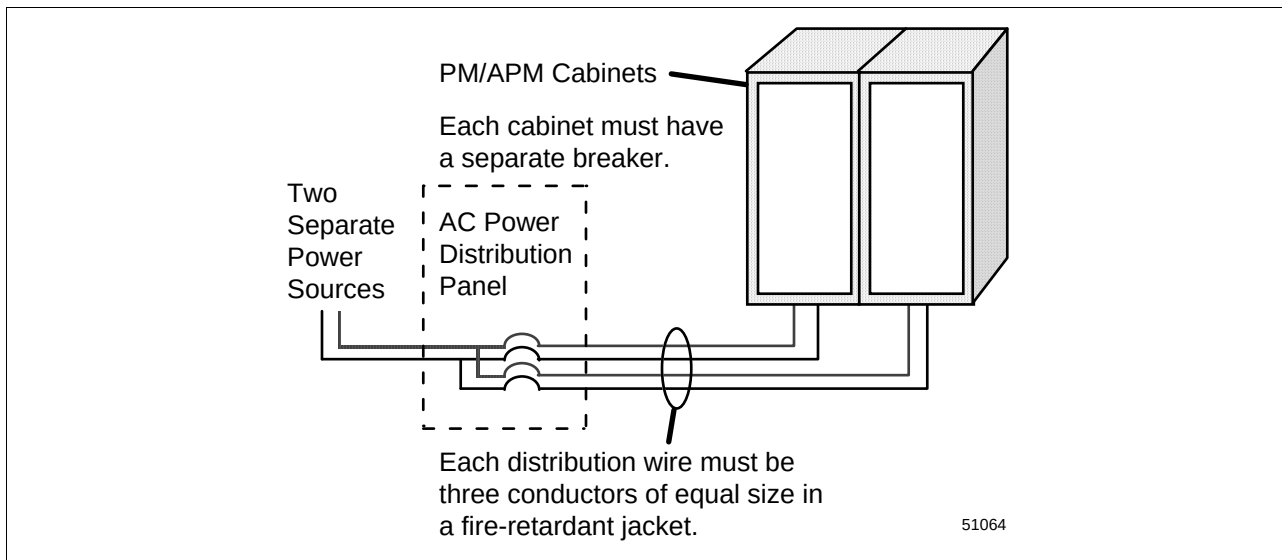
3.5 TPS AC Power Distribution Guidelines, Continued

PM, APM, and HPM equipment, continued

PM, APM, and HPM equipment cabinets can be connected to two separated power sources. Each source must be separately wired. This allows for automatic backup of the power input. **Thus, an individual 15 A breaker is required for each power source.**

Illustration

Figure 3-10 PM/APM/HPM AC Power Distribution—Separate Power Sources



3.6 LCN AC Power Entry

Cabinet breakers

Both the LCN console and cabinet power entry boxes are equipped with a ganged, circuit breaker.

- LCN consoles
 - with the 115 V option, both the hot and neutral lines are switched
 - with the 230 V option, both hot lines are switched
 - For LCN cabinets
 - the hot lines are switched
 - the neutral line is not switched
-

Power entry box

Each Universal Station cabinet is equipped with a power entry box as shown in Figure 3-6 and Figure 3-7. Universal Station modules, gateways, and peripherals can plug into the NEMA L5-15R receptacles (6) provided on the power entry (P/E) box.

The equipment cabinets include a similar power entry box, except that 10 receptacles are provided on a vertical strip. Cabinets can be furnished with a redundant power entry box, mounted in the cabinet, for six additional receptacles. All of the power cabling and wiring to TB1 in each power entry box is provided by the customer.

Distribution wiring

Each TPS cabinet and each power strip in an equipment cabinet **must be on its own branch circuit**. This isolation is desirable in the event of a failure that requires shutdown for repairs. More than one unit on a branch would require shutdown of more equipment than necessary.

Continued on next page

3.6 LCN AC Power Entry, Continued

Illustrations

The NEC wire color coding and the IEC color coding are different as shown in Figure 3-6 and 3-7.

Figure 3-11 Universal Station Power Entry Box with NEC Wire Color Codes

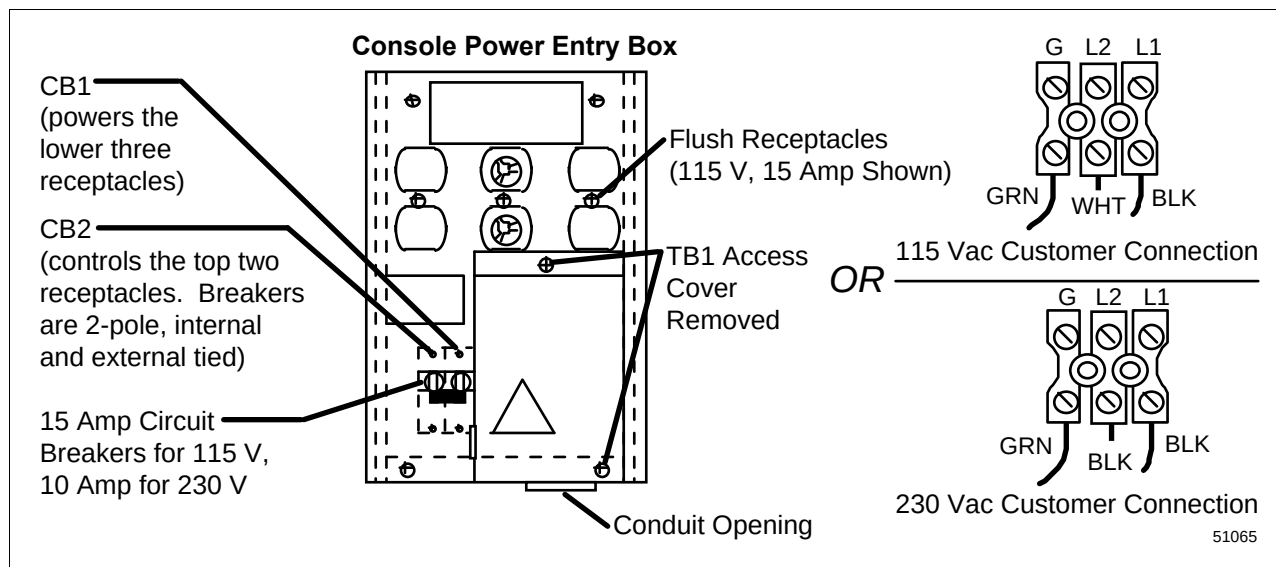
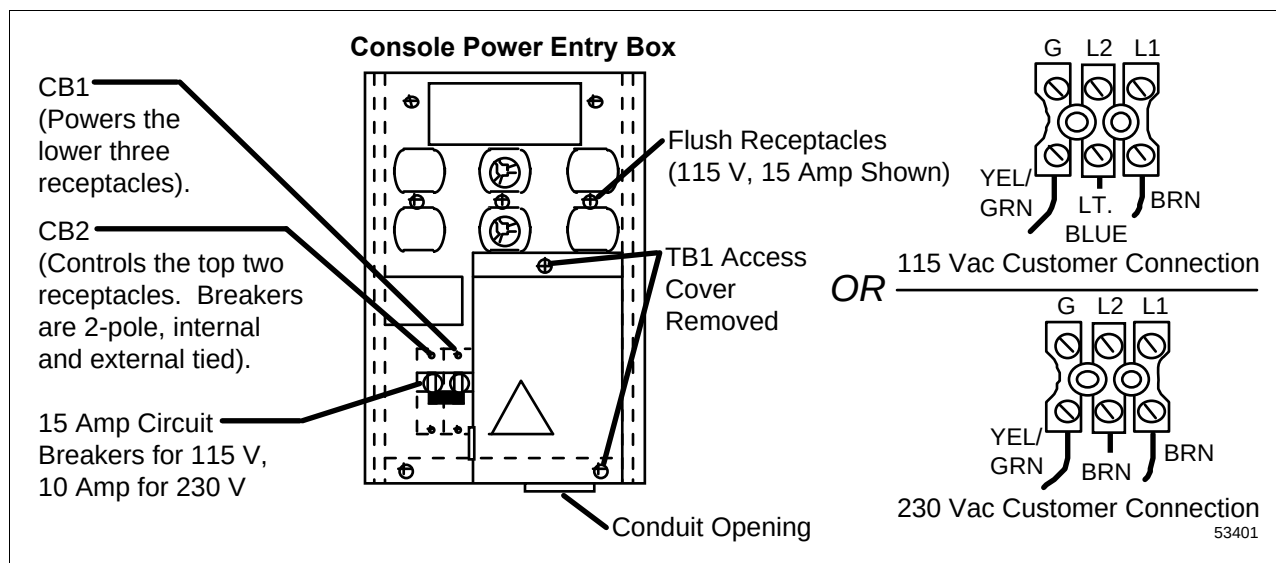


Figure 3-12 Universal Station Power Entry Box with IEC Color Codes



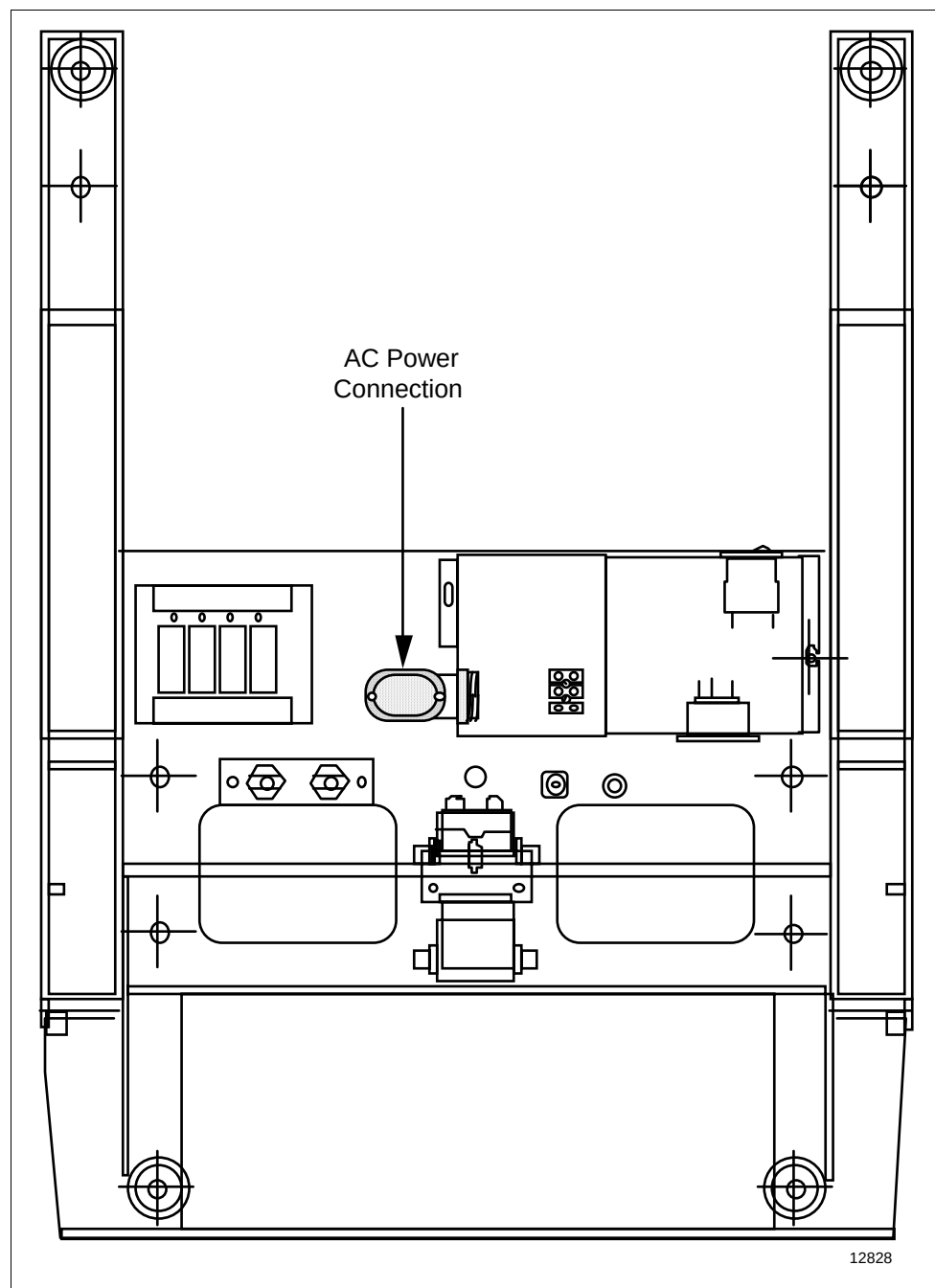
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3.6 LCN AC Power Entry, Continued

LCN ergonomic furniture

Ergonomic Universal Station furniture has a power entry box located in the bottom of the station under the footrest cover. Figure 3-13 shows power entry from directly below the station. For rear entry of ac power, see *LCN System Installation*.

Figure 3-13 Universal Station (Ergonomic) Power Entry Box



Continued on next page

3.6 LCN AC Power Entry, Continued

LCN equipment cabinet power entry

There is an additional (optional) power entry box available in equipment cabinets. This allows any redundant module to be connected to a separate power source.

The NEC wire color-coding and the IEC color-coding are different as shown in Figure 3-14 and 3-15.

Figure 3-14 LCN Equipment Cabinet Power Entry Box with NEC Wire Color Codes

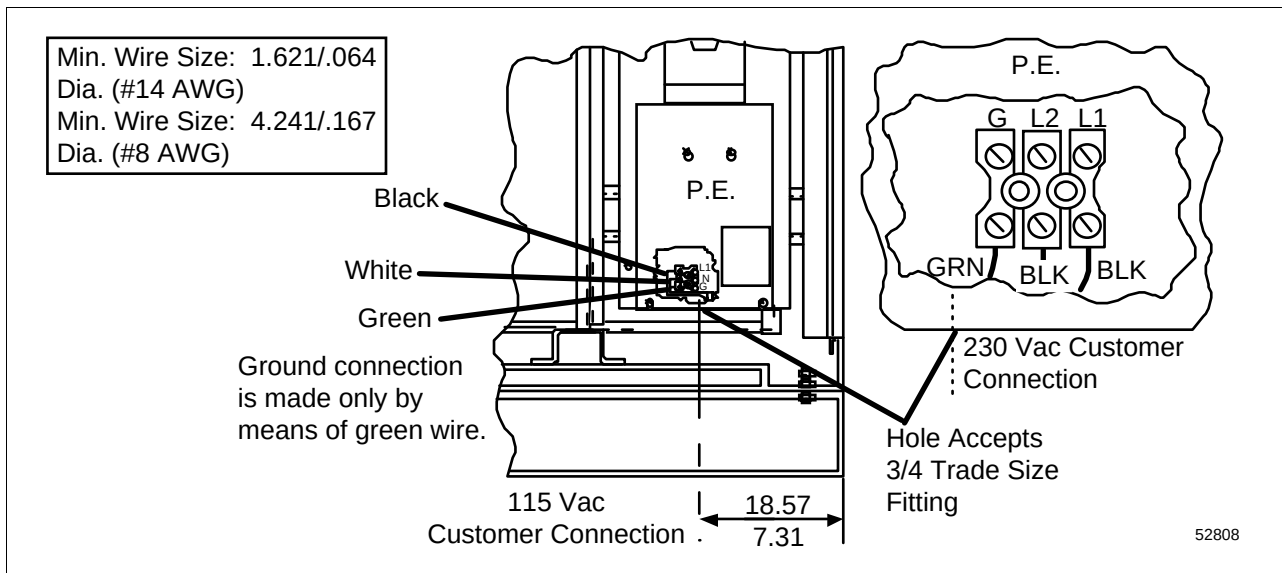
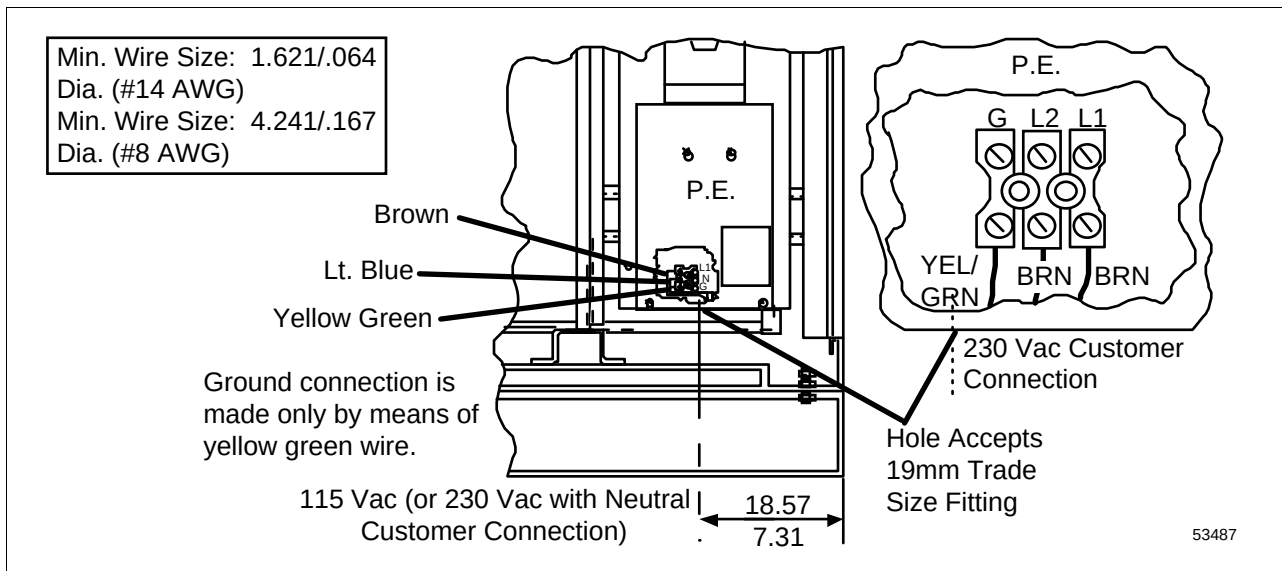


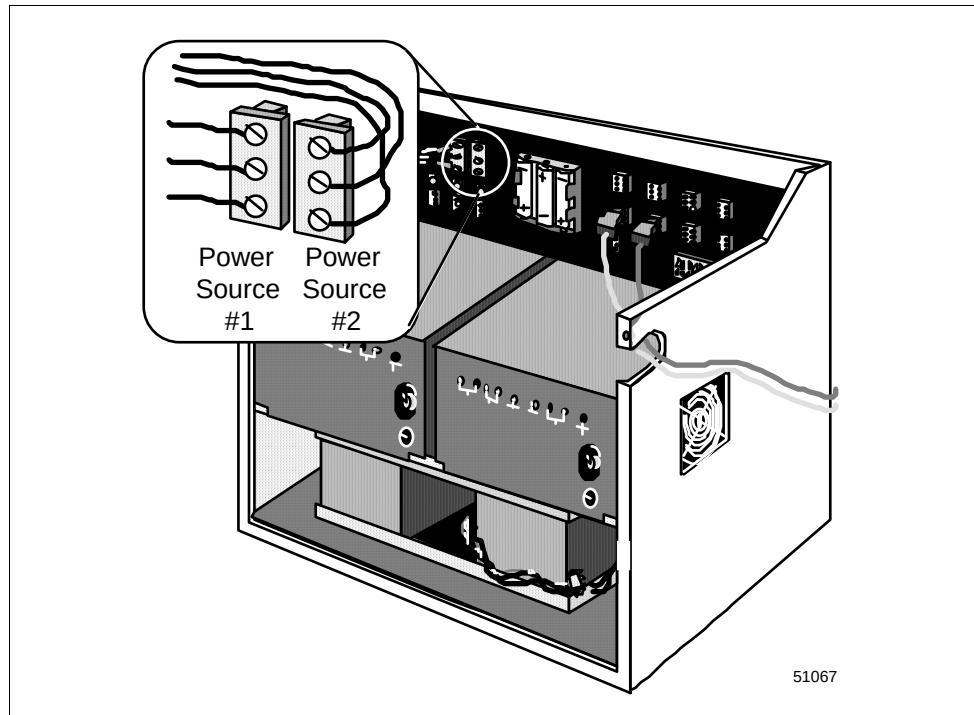
Figure 3-15 LCN Equipment Cabinet Power Entry Box with IEC Wire Color Codes



3.7 PM/APM/HPM Power Entry

Illustration

Figure 3-16 PM/APM/HPM Power Supply Power Entry Wiring for Separate Power



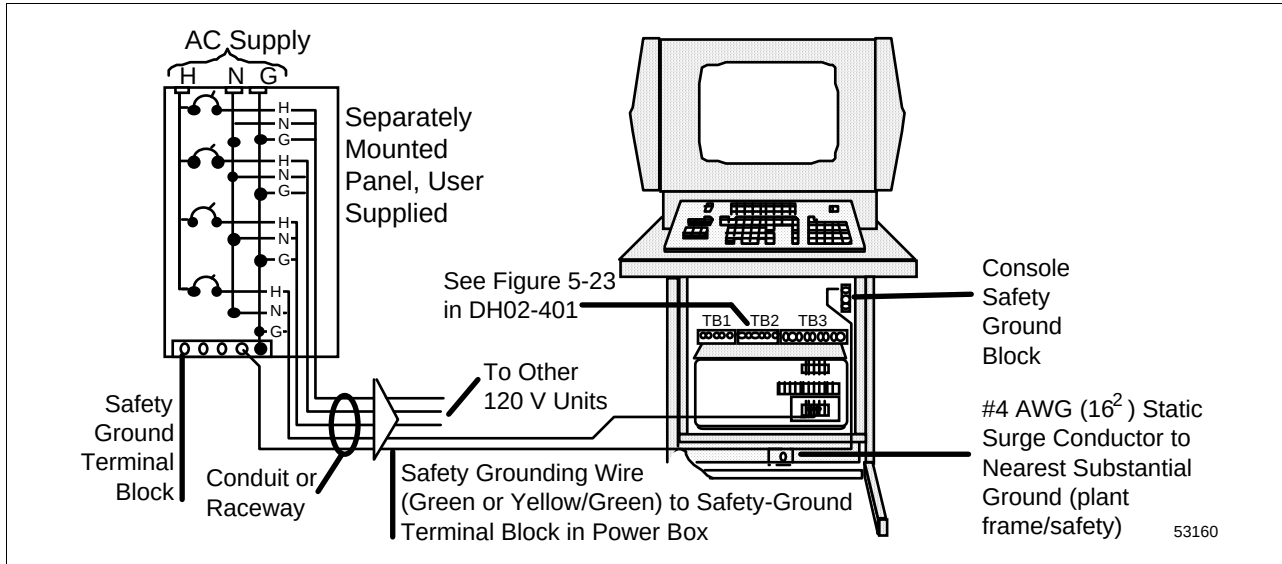
3.8 Data Hiway AC Power Entry

Data Hiway equipment

Data Hiway (DH) equipment has a 15 A breaker box mounted in the cabinets.

Data Hiway ac power Refer to the *Data Hiway Planning* manual for more detailed guidelines.

Figure 3-17 Basic Operator Station with Remote AC Panel

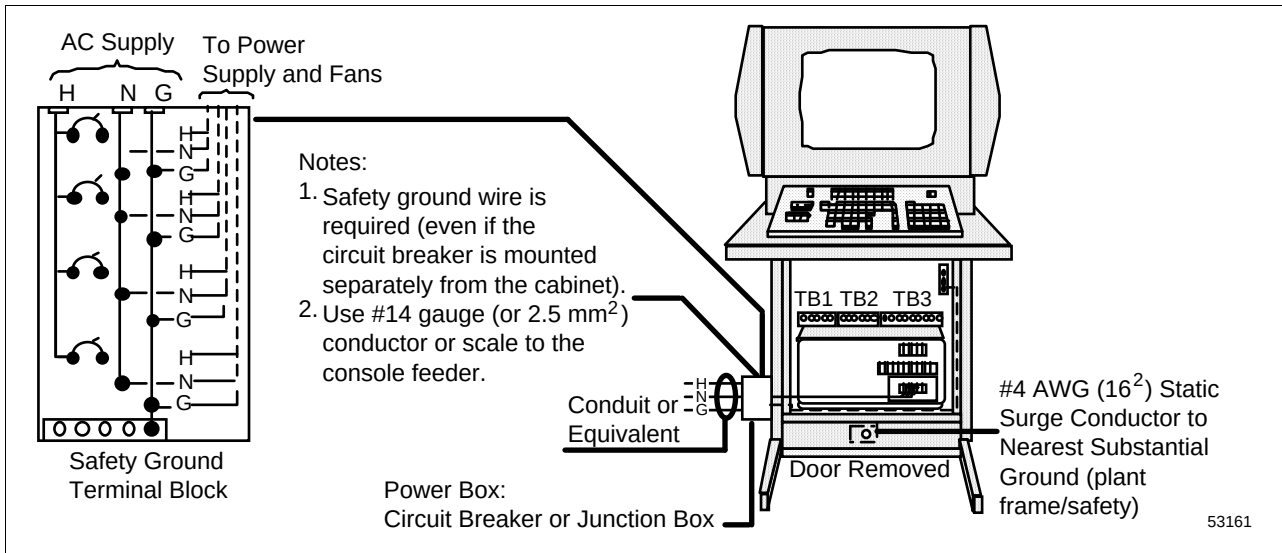


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3.8 Data Hiway AC Power Entry, Continued

Data Hiway ac power, continued Refer to the *Data Hiway Planning* manual for more detailed guidelines.

Figure 3-18 Console with Integral Breaker Panel

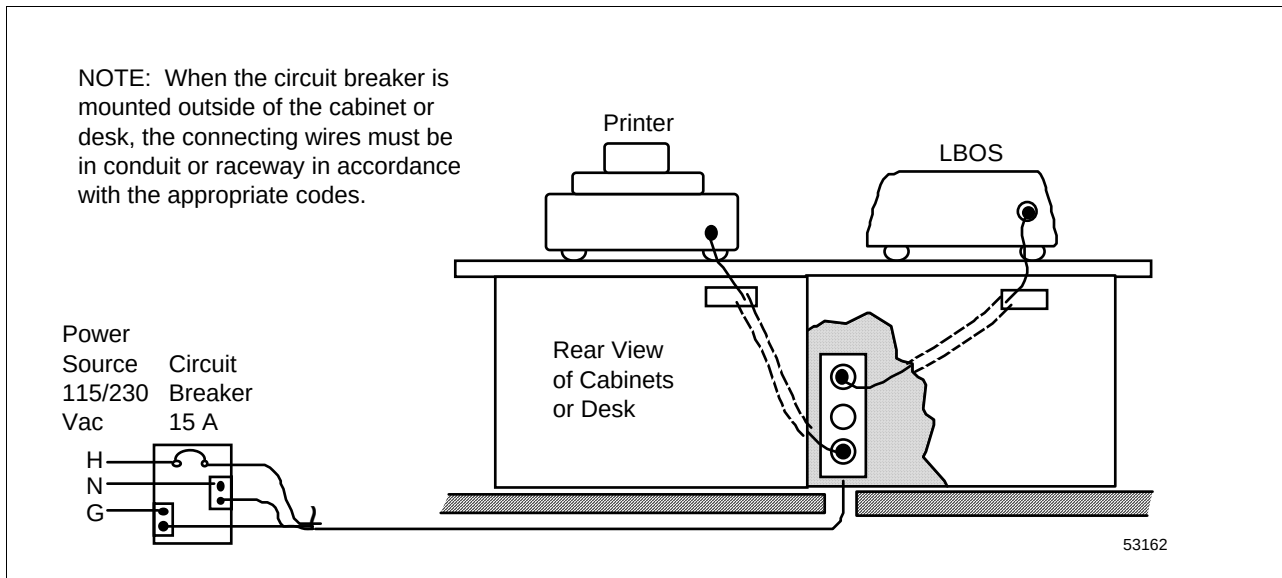


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3.8 Data Hiway AC Power Entry, Continued

Data Hiway ac power, continued Refer to the *Data Hiway Planning* manual for more detailed guidelines.

Figure 3-19 AC Wiring to LBOS Console

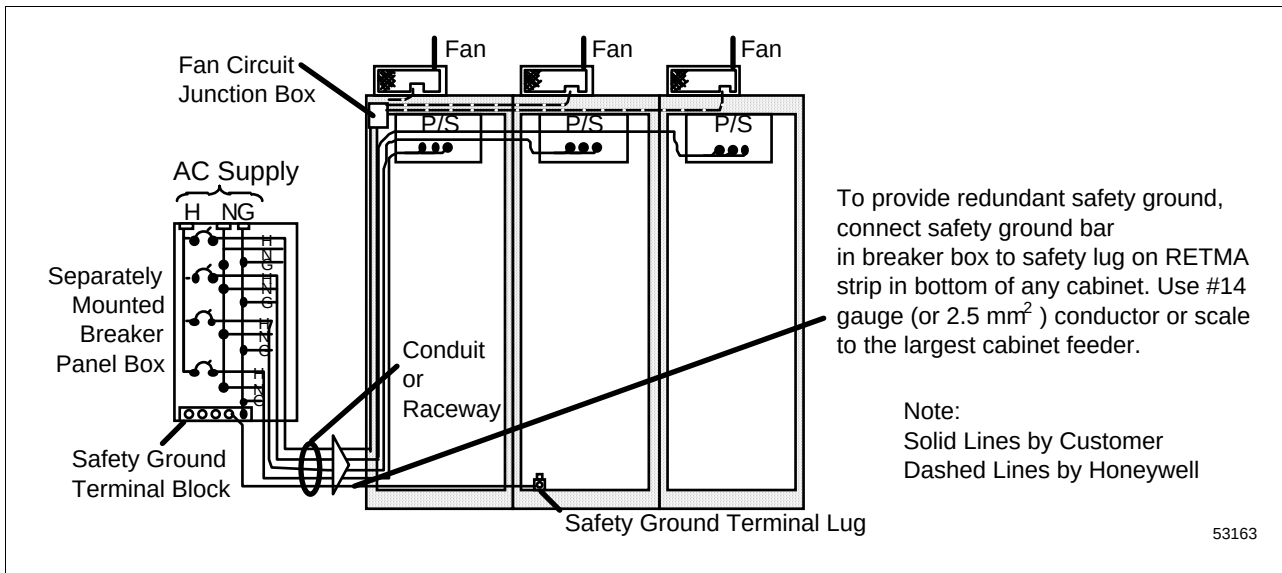


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3.8 Data Hiway AC Power Entry, Continued

Data Hiway ac power, continued Refer to the *Data Hiway Planning* manual for more detailed guidelines.

Figure 3-20 Optional Routing of AC Power Wiring from Separate Breaker Panel

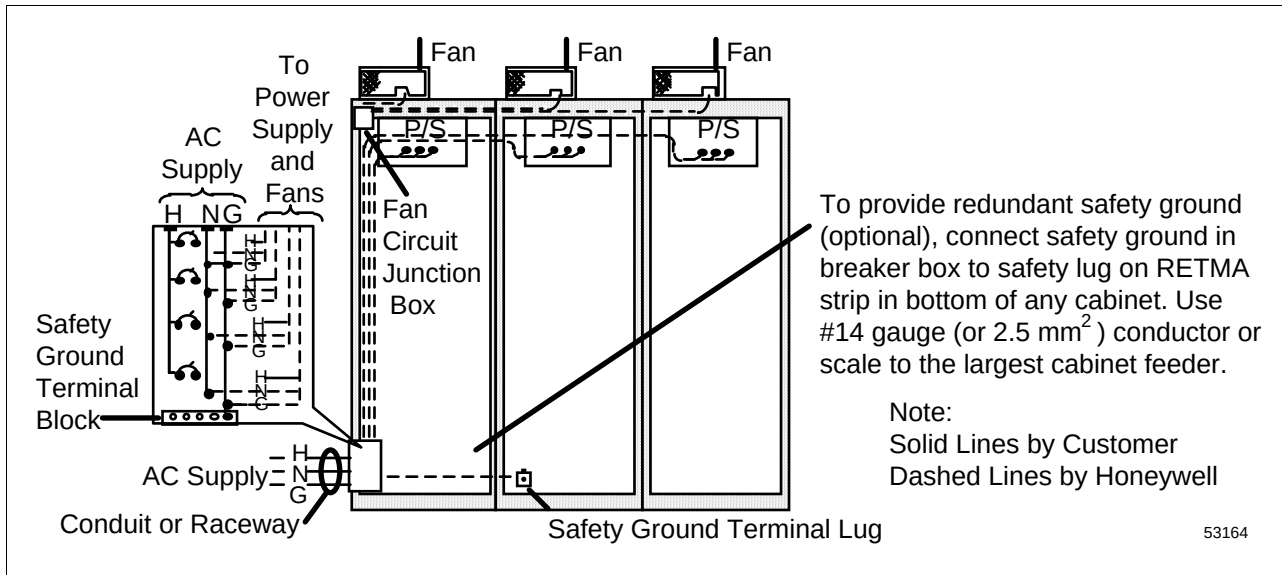


Continued on next page

3.8 Data Hiway AC Power Entry, Continued

Data Hiway ac power, continued Refer to the *Data Hiway Planning* manual for more detailed guidelines.

Figure 3-21 Routing of AC Power Wiring from Cabinet-Mounted Breaker Box

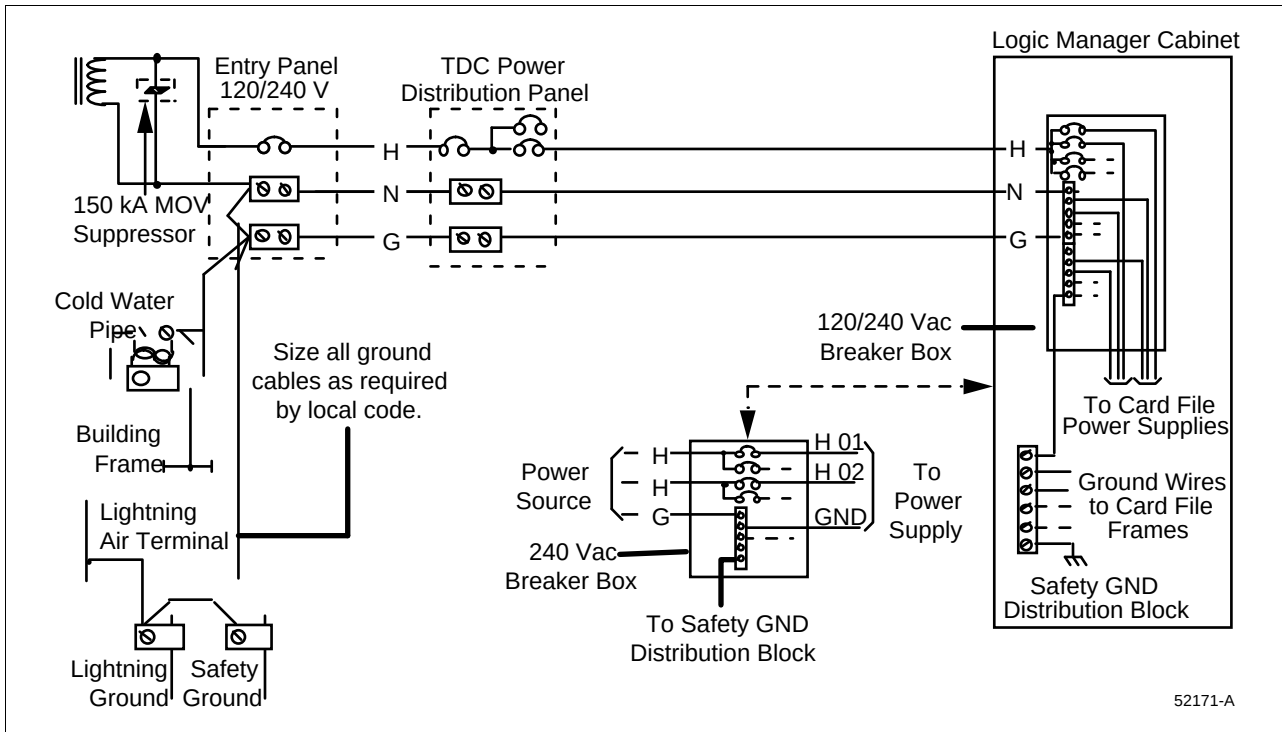


3.9 Logic Manager AC Power Entry

Description

Logic Manager cabinets are equipped with a separate 15 amp breaker for each Logic Manager (up to four). Refer to the *Logic Manager Planning* manual for more detailed guidelines.

Figure 3-22 Logic Manager Cabinet AC Power Wiring



3.10 Floating AC Power

Scope

Floating in-plant ac power systems are in use in several European countries.

Why?

The reasoning behind such an arrangement is that a single earth fault does not cause power loss to any device connected to the power system. Earth fault detectors signal the presence of such a fault.

Risks

Experience has shown that the floating power does not affect TPS operation. The risk is that if high voltages related to ground penetrate the power system, those high voltages can damage power supplies.

ATTENTION

ATTENTION—It is the responsibility of the user to prevent such overvoltages. When a local UPS provides the power and the entire power distribution is inside a single building, surge suppression by means of MOVs provide adequate protection.

- **All connections to external power lines should be made through isolation transformers.**
 - **For best security, those transformers should have a grounded static screen between windings.**
-

Section 4 – Grounding for CE Compliant Systems

4.1 Overview



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

Section contents

These are the topics covered in this section:

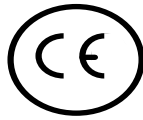
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ATTENTION

ATTENTION—This section applies to CE Compliant **TotalPlant** Solution (TPS) system installations. Any equipment designated as "CE Compliant" complies with the European Union EMC and Health and Safety Directives.

This section describes single point grounding to Building Safety Ground. For systems using Master Reference Ground, see Section 5.



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4.2 System Grounds

Required grounds

Grounding of electronics can make the difference between a system that is unusable and one that performs well. The required grounds, their connections, and design are discussed below.

TPS Systems require the use of three ground systems.

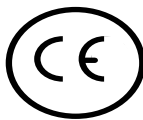
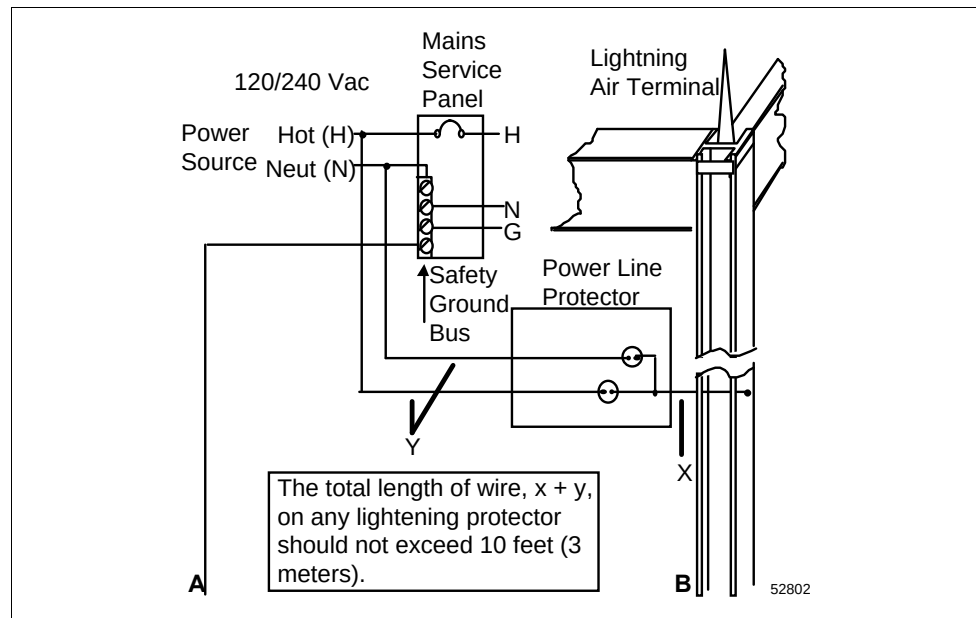
- Safety Ground
- Lightning Ground
- Master Reference Ground (**for pre-existent TPS systems only**)

Safety Ground

AC Safety Ground interconnects all metallic structures and enclosures in an area. This minimizes the voltage that people can come into contact with. AC Safety Ground is used to ground all power systems for lightning. It is also used to ground UCN coaxial cable shields at every tap.

For each building power entry, an independent AC Safety Ground system with an electrode cable is preferred. Examples of such independent ground systems with a driven electrode is shown in Figure 4-1 and Figure 4-2.

Figure 4-1 TPS Ground System—Upper View



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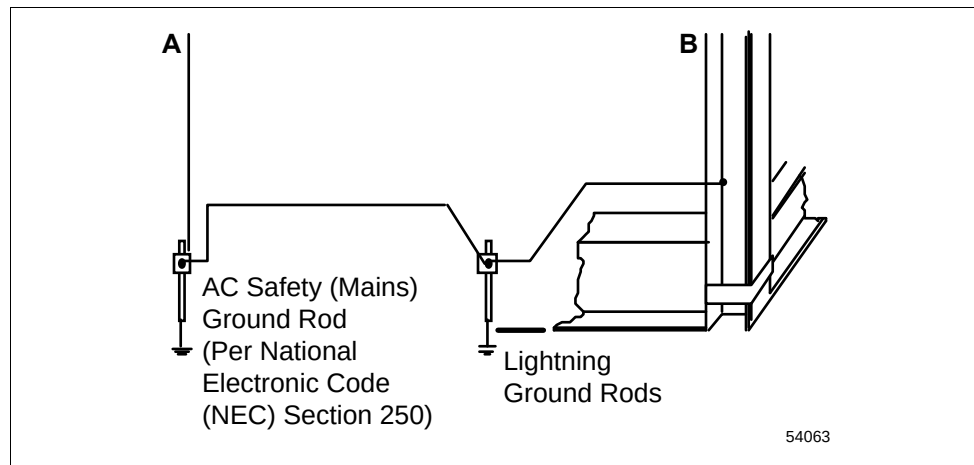
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4.2 System Grounds, Continued

Safety Ground, continued

Figure 4-2 TPS Ground System—Lower View

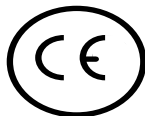


Safety grounding requires electrode systems that provide grounding resistances between 0.1 and 5.0 ohms. The required performance varies with the circumstances described as follows:

- When a facility has neither zener barriers nor lightning problems, AC Safety Ground has only to meet the minimum grounding requirements specified in the local electrical code. Usually, such codes require a grounding resistance of 5 ohms maximum to true earth.
- When a facility uses zener barriers for intrinsic safety, AC Safety Ground should be less than 0.1 ohms to true earth. Refer to the Lightning Ground discussion that follows.

IEC standards require:

- Zener ground and equipment logic ground must be connected together.
- Zener ground and logic ground must have a resistance to safety ground of less than 1.0 ohm.
- Data Hiway systems MRG must be less than 1.0 ohm to safety ground.



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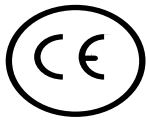
4.2 System Grounds, Continued

Lightning Ground

The purpose of lightning ground is to safely disperse a lightning charge that may be picked up by the facility metallic structure and electrical system. Lightning ground is frequently integrated with AC Safety Ground as shown in Figure 4-2 through Figure 4-4.

Lightning grounds must conform to applicable codes such as NFPA No. 78 (ANSI), IEEE Std 142-1972, Code LP1-175, and other local codes. When a facility is subject to lightning, each lightning rod should be grounded at the base with a maximum resistance of 0.1 ohms to earth.

As discussed in the lightning section in this manual, the TPS lightning protection philosophy is that each lightning rod should be grounded with a resistance to earth of less than 0.1 ohms. A 100 kA lightning strike into a 0.1 ohm ground generates 10 kV. Experience has shown that 10 kV does not cause sparking inside of wiring trays or across terminal blocks. Without sparking and the Electromagnetic Interference (EMI) it generates, TPS systems perform well during a lightning strike.



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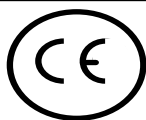
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4.2 System Grounds, Continued

Master Reference Ground (pre-existing systems only)

Master Reference Ground (MRG) that is an isolated ground electrode array that references Honeywell TPS Data Hiway electronics and the process signals is shown added to AC Safety ground system in Figure 4-3. An isolated MRG assures stable process signals to the Data Hiway connected devices by divorcing signal ground from AC Safety Ground. An MRG installation can have the following properties:

- An MRG is an isolated ground electrode system reserved for Honeywell TPS System and **can be used only in pre-existing TDC 3000^X systems.**
- An MRG usually has a maximum resistance through the earth to AC Safety Ground of 5 ohms.
- When the MRG is grounding zener barriers, it must have a maximum resistance through the earth to AC Safety Ground of less than 1.0 ohm.
- Use a minimum separation of 3 m (10 feet) between the MRG electrodes and any grounded metal such as pipes or a grounding grid or mat.
- MRG electrodes usually perform well when placed immediately next to building footings with electrically floating reinforcing bar.
- No TPS electronics is to be further than 152.4 cable-meters (500 feet) from its MRG electrode nailhead.
- An MRG electrode conductor is typically 11.8 mm (4/0 AWG), is 13 m (50 feet) long, and terminates at a star plate upon entry into a building.
- An MRG entry may require a spark gap to AC Safety Ground to control arcing during a lightning strike. See the lightning section in this manual for more information.
- Large TPS System can use MRG star plates that feed other MRG star plates.
- An MRG star plate is insulated from the building structure and concrete, and is capable of accepting up to three 5.3 mm (4 AWG) local Master Reference Ground conductors from each TPS cabinet or complex in the TPS system.
- Radial wiring to a star plate is preferred to daisy chaining from cabinet-to-cabinet.
- A system can have as many MRGs distributed through the process as is convenient.



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Continued on next page

4.2 System Grounds, Continued

**Master Reference
Ground (pre-existing
systems only),
continued**

-
- There is no need to connect MRGs together.
 - A building can have several independent MRGs.
 - There is no limit to the amount of equipment that can be placed on one MRG.

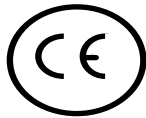
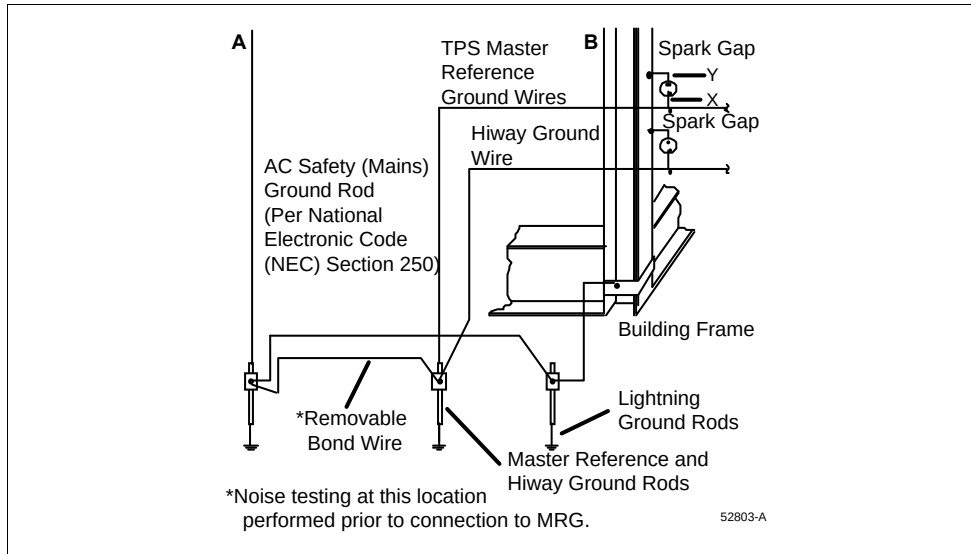
The design and testing of ground electrodes is discussed in the following subsection.

4.3 Alternate Grounding Method

Scope

In the event that there are concerns about the grounding system not meeting a regulating body's requirements, such as NEC, OSHA or local codes, it is permissible to add a direct wire, 4/0 AWG (120 mm² or larger), between the AC Safety Ground and the Master Reference Ground at their grounding electrodes according to codes as shown in Figure 4-3.

Figure 4-3 TPS Ground System—Lower View



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4.3 Alternate Grounding Method, Continued

Description

Bonding of the Master Reference Ground (MRG) to AC Safety Ground is created by connecting to the installation's AC Safety Ground as shown in Figure 4-3.

- The MRG connection must be made at the electrode nailhead. Moving just a small distance away adds wiring inductance, preventing proper sinking of noise.
- The AC Safety Ground connection must be made at the connection point previously tested for an acceptable noise level by the procedure described later in this subsection.
- The connection must be disconnectable for periodic testing or troubleshooting noise problems.
- The bonding connection between the MRG nailhead and AC Safety Ground must be made with a green, or yellow/green, insulated copper wire that is 4/0 AWG (120 mm²) or larger. The attachment points for the bonding connection are critical.

MRG noise test

A suitable automatic tester for measurement of the ground system noise is manufactured by PowerCet, Santa Clara, CA, at telephone number (408) 988-1346. Use the model PowerVisa 440 tester. The test requires a reference ground. Testing for noise is discussed in more detail in the *Basic System Checkout Checkout* manual, SY-09-03.

Testing must

- take place over a period of at least one week before it can be considered suitable for operation
- continue at least 6 months or more until operational confidence is high
- never indicate a measured impulse noise exceeding 10 volts peak



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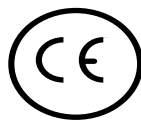
4.3 Alternate Grounding Method, Continued

Problems

Problems with this connection are most likely caused by electrical impulse noise from electrically operated equipment that is sharing the AC Safety Ground.

Most bonded connections pass the tests just described, but not all. Therefore, a method must be incorporated that will permit removing the connection between MRG and Safety Ground.

A given connection between MRG and AC Safety Ground can be quiet because the details of plant construction have made it so. These details are important, but not always understood. Excessive noise usually announces itself by causing “ghost alarms” on the operating system; therefore, the connection can fail at any time. The use of a permanently installed noise monitor is preferred over the alternative of using control system performance as an alarm.



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4.4 Implementation of Alternate Grounding

Guidelines

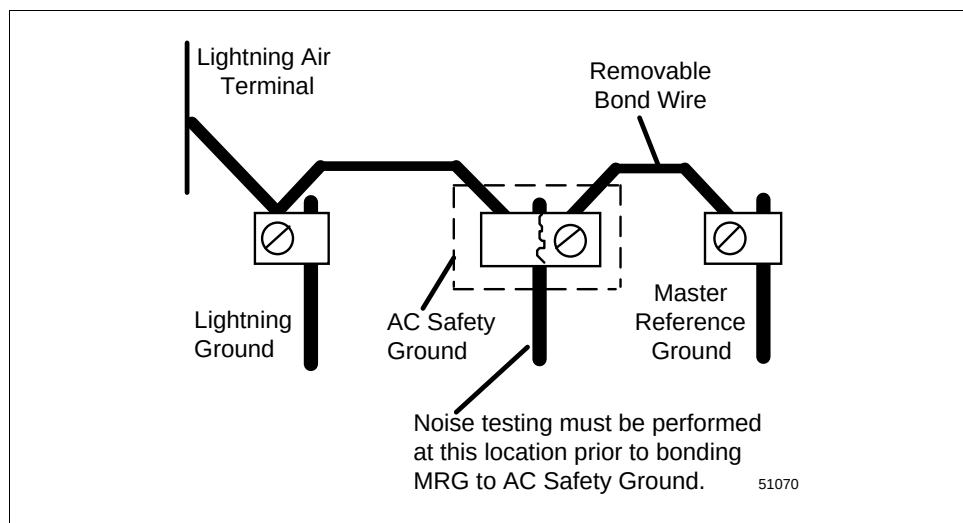
ATTENTION

With reference to Figures 4-2 and 4-3, the following guidelines are to be implemented before bonding Master Reference Ground to AC Safety Ground:

ATTENTION—All conditions for the Master Reference Ground (MRG) and AC Safety Ground (resistance, isolation, lightning protection, etc.) are to be individually met before the connection between the MRG and AC Safety Ground is made as shown in Figure 4-4; otherwise, the system will not have a fall back position if the connection is removed.

- The MRG bonding connection is to be made with a suitable heavy conductor, insulated copper, (4 AWG or larger) directly from the MRG nailhead to the noise-tested AC Safety Ground connection point. A conductor is desirable so it can be opened for troubleshooting. In the event that conditions develop on the AC Safety Ground that cause problems on the MRG, the connection can be broken until the AC Safety Ground is corrected.
- The AC Safety Ground may introduce a level of noise on the MRG without causing system degradation if the total is always under 10 volts peak. A detailed test, described in "Facility AC Safety Ground Verification" in the *Process Manager/Advanced Process Manager Checkout* or *High-Performance Process Manager Checkout* manual, is to be performed at the AC Safety Ground connection point. If the test fails, another AC Safety Ground connection point must be selected and tested.

Figure 4-4 AC Safety Ground/MRG Bonding



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4.5 Electrode Systems

Establishing System

Any grounding system (safety, lightning, MRG, etc.) for a site is designed by (1) measuring the soil resistivity, (2) determining the best electrode configuration for the soil, and (3) installing and interconnecting the electrodes.

Soil Test

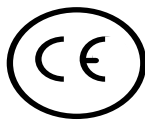
The only way to get electrical information about the soil at a given site is to drive electrodes down to the water table and measure resistance. The electrical measurement is done with a commercial tester, such as one made by James G. Biddle Co. in Plymouth Meeting, Pennsylvania at telephone number (215) 646-9200. Use model 250220 or 250241.

A single 3-meter (10-foot) electrode might have a contact resistance of, typically, 5 ohms to earth.

Electrode Material

The preferred electrode usually is 25 mm (1 inch) diameter stainless steel rod, in 2- to 3-meter (8- to 10-foot) lengths. The rod can be driven through soft rock and push smaller stones aside. Weld the rod sections together as the hammering progresses.

Avoid using electrodes with copper when making large arrays. They can turn nearby buried steel into sacrificial electrodes. Avoid using rods with protective platings or coatings. Welding the electrode generates unhealthy fumes and also destroys the coating on the rod.



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4.6 Electrode Arrangements

Single driven electrode

The most economical electrode system consists of one long electrode driven into the water table. A single rod going down 45 m (150 feet) is not uncommon. Deep grounds are desirable because of their electrical stability with the changing seasons.

Electrode Array

When soil conditions do not permit a single deep electrode, shorter electrodes can be used in arrays.

The most efficient form of an array is created by placing the rods in a circle or in a straight line. The rod separation should equal the rod length. Therefore, 3-meter (10-foot) rods would be driven into the soil at 3-meter (10-foot) separations.

The relationship of separating rods at a distance equal to the rod length comes from each rod having a circle of influence out to a diameter equal to the rod length. Placing a rod inside of the circle of influence of another rod gains little in system grounding performance.

Ground rods can be placed in various arrays such as circles, triangles, squares, and straight lines. The straight line configuration is the most flexible because both ends are open for adding rods, if needed. On the other hand, filling in a circle or any other closed array improves performance very little, optimistically 2:1.

Array performance

The performance of circular and in-line arrays can be estimated using test results from a single isolated rod. For example, assume a test rod resistance of 10 ohms. To achieve 5 ohms, one would think that two rods in parallel would be acceptable. It actually requires four. When rods are separated at distances equal to their length, the circles of influence overlap such that each added rod has only a 50% independent circular contact with the soil.

The rods are commonly connected together using cadmium welding and 11.8 mm (4/0 AWG) green insulated copper cable. Encase the ends of all rods and make all connections inside a reinforced concrete pipe with an 18 mm (3/4 inch) thick steel cover.



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4.6 Electrode Arrangements, Continued

Drilled electrode

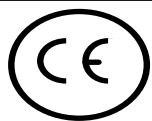
When soil conditions or space do not allow the use of driven rods, a 10 cm (4-inch) or larger well can be drilled or pounded into the water table. Specify that drilling continues until two aquifers (rivers) are struck if reasonable for the area.

The well requires a casing down to bed rock, or it may require casing for the full depth if there is no bed rock in the area. Casing can be perforated steel or plastic. Leave the top 9 meters (30 feet) solid and nonperforated to prevent surface water from draining into the well and contaminating the substrata aquifer.

For the grounding electrode, use a stainless steel rod centering it inside and extending it to the full depth of the well. Then fill the well with concrete mix, coke breeze, or other ashes. These materials slowly release ions that improve grounding over time. Do not use strong chemicals. They may cause water contamination.

The stainless steel rod must comply with the specification contained in IEC standards or NEC 250-83-C, paragraphs 1, 2, and 3.

Do not omit the 150 kA spark gap shown in Figure 4-3, because the well may have to sink all lightning in the area.



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4.7 Cabinet Ground Connections

Overview

The TPS System has products that represent different technologies, and therefore different grounding schemes.

- LCN Five-Slot Modules
- LCN Dual Node Modules
- Process Manager, Advanced Process Manager, and High-Performance Process Manager
- Logic Manager
- Data Hiway
- New TPS equipment delivered after 1/1/96.

New Module grounding

All Five-Slot Modules, Ten-Slot Modules, and Dual Node Modules have the same grounding rules.

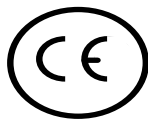
- Logic ground is connected to chassis ground on the back of the module using serrated washers.
- By bolting the module into stations or cabinets the chassis ground to safety ground connection is made.
- **If a new module is added to a pre-existing system, the ground rules listed below must be followed.**

LCN Five-Slot Module (existing prior to 1/1/96)

Five-Slot Modules are grounded according to the type of node present in the module.

- Universal Stations have the logic ground terminal connected to chassis ground.
- All other LCN nodes have the logic ground connected to MRG.

Refer to the *Local Control Network Planning* manual for additional grounding information.



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4.7 Cabinet Ground Connections, Continued

LCN Dual Node Module (existing prior to 1/1/96)

Dual Node Modules require grounding of both nodes in the module according to the type of node present in the module.

- Hiway Gateways (HG) have the logic ground connected to MRG.
- All other node types have the logic terminal connected to chassis ground.
- Redundant gateways (HGs, NIMs, and PLCGs) must be located in separated Dual Node Modules.

Refer to the *Local Control Network Planning* manual for additional grounding information.

PM/APM/HPM (existing prior to 1/1/96)

PM, APM, or HPM FTAs (that are not grounded at the process end) connect the shields for the process-connected wiring as follows:

- Analog input FTAs and Analog output FTAs are connected to MRG
- 3-30 Vdc digital FTAs are connected to MRG
- 30-200 Vdc digital FTAs that have safety grounds or shields connect them to Safety Ground
- Ac digital FTAs that have safety grounds or shields connect them to Safety Ground

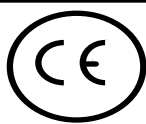
The cabinets have a separate bus bar for safety ground, MRG, and an optional Zener Barrier ground bus bar located near the bottom of the cabinet. Each bus bar is connected to the appropriate system star plate (one for safety ground, one for MRG, and one for zener ground). Refer to the *Process Manager/Advanced Process Manager Planning* or *High-Performance Process Manager Planning* manual for additional details about grounding the PM, APM, or HPM.

Logic Manager

The Logic Manager does not require any special grounding other than what is in the ac power cord. Refer to the *Logic Manager Planning* manual for additional details about grounding the Logic Manager.

Data Hiway

Data Hiway products have all grounds connected to the MRG vertical bus bar in each cabinet. These bus bars are horizontally connected to a local reference bar (if present) mounted to the top of the cabinets. This bus bar is then connected to MRG with a 5 mm (#4 AWG or larger) wire. Refer to the *Data Hiway Planning* manual for additional details about grounding of Data Hiway products.



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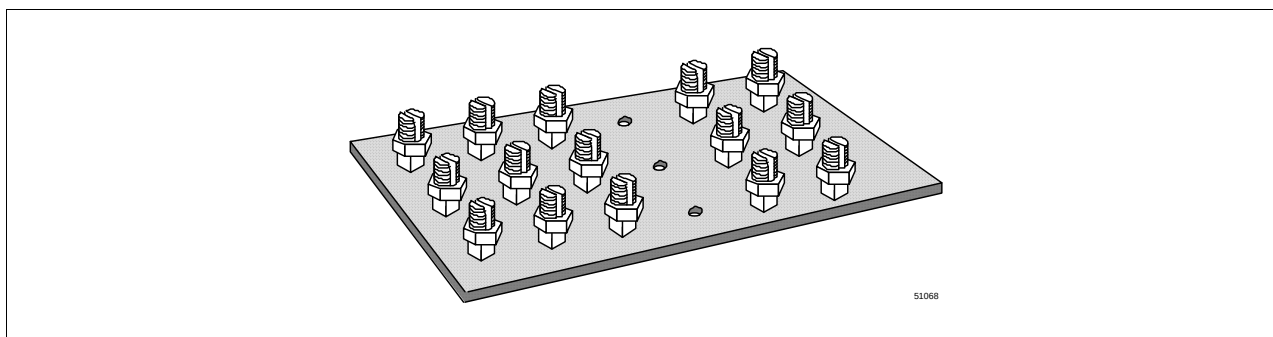
4.8 Star Plate

Description

A star plate is a copper or brass plate that provides for the connection of ground wires at a common point. The required attributes of a star plate are

- **must be completely physically isolated from all conductive material except those ground wires that are to be connected.**
- heavy enough (thick enough) to resist damage from being stepped on or hit a heavy blow. A minimum of 6.35 mm (1/4 inch) thick.
- have more fastening studs (or stud holes) on the plate than the grounding system wiring requires. **Star plates CANNOT be replaced with any of the attached system hardware in operation.**
- each fastening lug must be capable of retaining a single 5.2 mm (#4/0 gauge) insulated ground wire. **One, and only one, wire can be terminated in each lug.**
- be located where it can be readily inspected.
- must be kept dry.

Figure 4-5 Example of a Star Plate

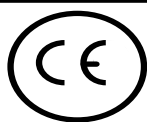


Application rules

Star plates can be used when

- The total distance from the stake to the farthest end point to be grounded is less than 152.4 meters (500 feet). Refer to the MRG grounding requirements shown in Figure 4-3 and described in subsection 4.2.
- each lug has one, and only one, #4/0 copper wire terminated in it.
- **Only ground wires from Honeywell TPS equipment are to be connected to the star plate.**

Star plates for safety ground systems, MRG grounds systems, and zener barrier grounds systems must be kept separate until they reach the respective electrode nailhead.



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4.9 System Grounding

Overview

The TPS ground system is a combination of

- Local Control Network grounds
- Process Manager, Advanced Process Manager, and High-Performance Process Manager grounds
- Data Hiway grounds
- IPC 620 system grounds
- DEC system grounds.

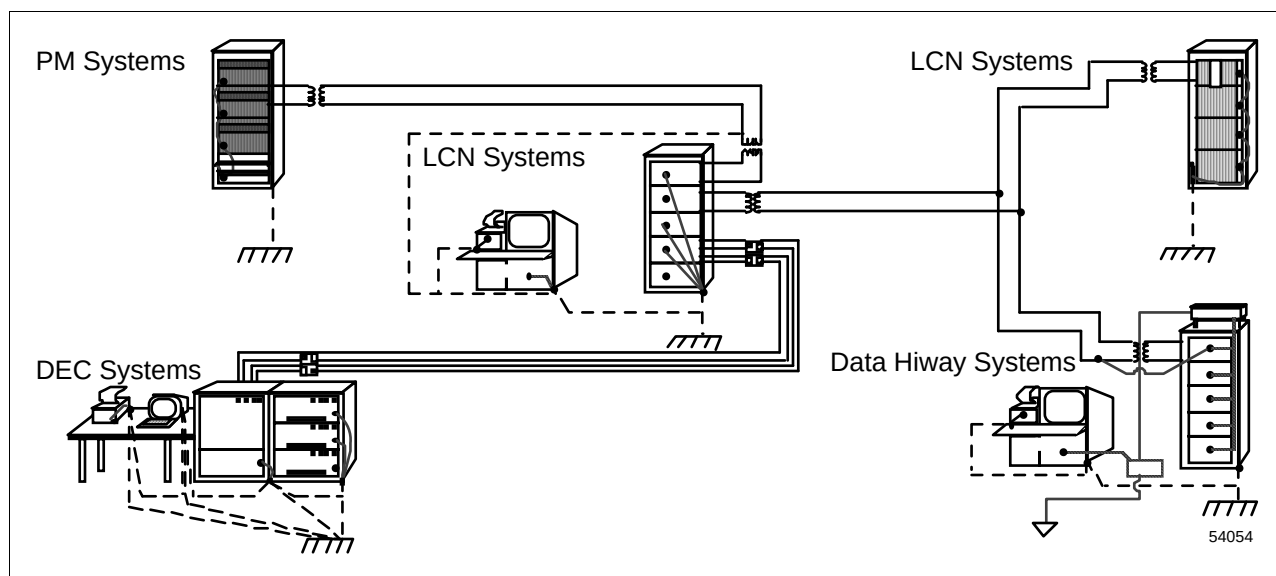
Individual ground systems

Each system has a unique grounding scheme local to itself (star plate) that is directly connected to the ground stake for

- AC Safety ground
- Master Reference ground (LCN, PM, APM, HPM, and Data Hiway)
- Zener Barrier ground (PM, APM, and HPM).

Illustrative overview Figure 4-6 is an illustrative overview of the TPS system grounding.

Figure 4-6 TPS Grounding Systems Overview



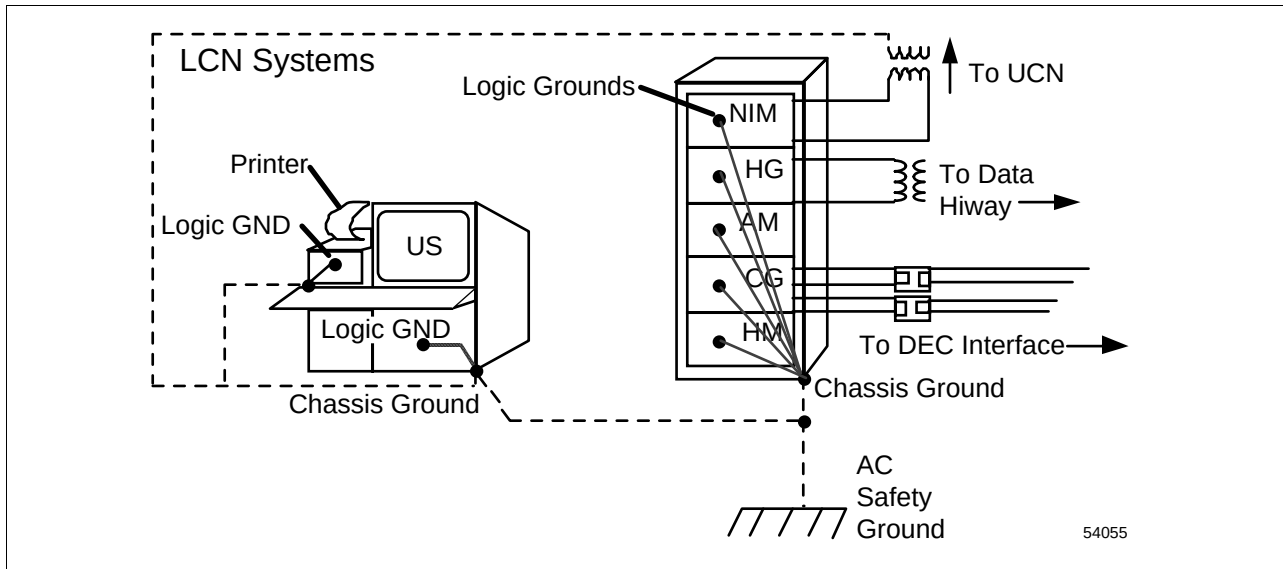
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4.10 LCN Grounds

Scope

The LCN module grounding scheme is shown in Figure 4-7. Each part of the overall TPS system is individually grounded and communications between each one and the LCN is transformer coupled.

Figure 4-7 LCN Cabinet Grounding



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4.11 LCN Node Grounding Methods

ATTENTION

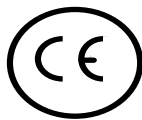
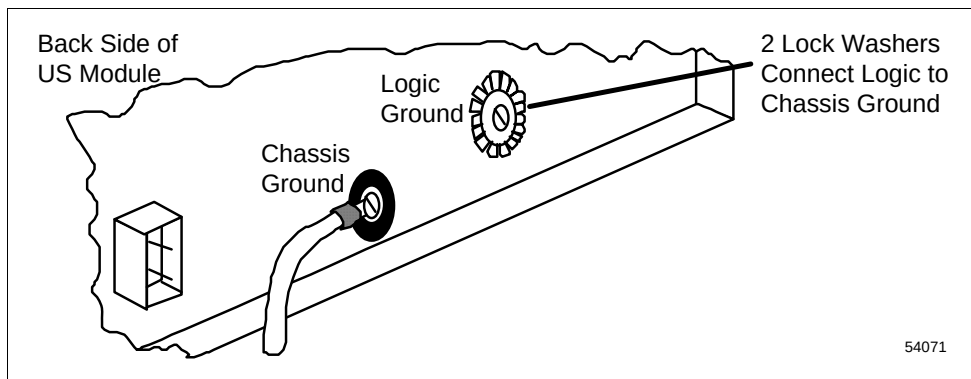
ATTENTION—All modules get their chassis ground connection through the mounting hardware.

Scope

There are two different grounding methods in use in the LCN. The method of grounding the nodes is unique for each module type.

- Five/Ten Slot Module
 - All nodes in a Five/Ten-Slot Module have the logic ground terminal connected to chassis ground (AC Safety ground) by means of a special serrated grounding washer that connects the terminal to the module chassis as illustrated in Figure 4-8.

Figure 4-8 Five/Ten-Slot Module Chassis Ground Connections for USs



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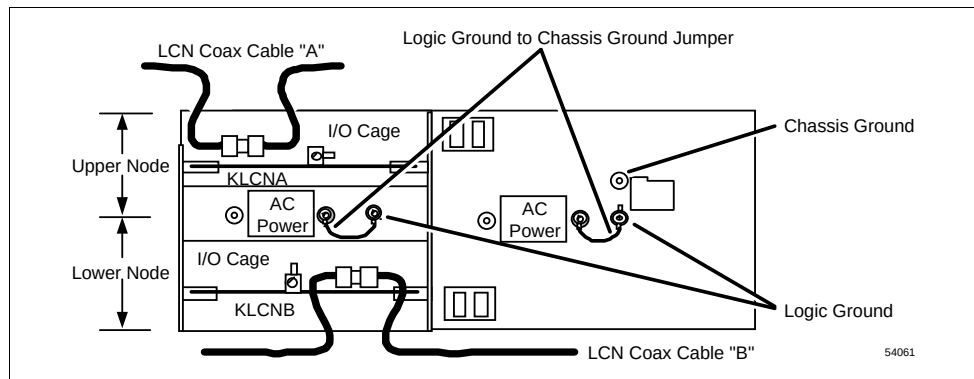
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4.11 LCN Node Grounding Methods, Continued

Scope, continued

- Dual Node Module
 - All node types in a Dual Node Module have the logic ground terminal wired to a chassis screw (chassis ground/AC Safety Ground) as illustrated in Figure 4-9.

Figure 4-9 Dual Node Module Ground Wiring



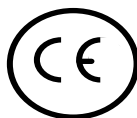
Grounding of console furniture

The Universal Station and Universal Station^X furniture has a Safety Ground post. This applies to both classic and ergonomic furniture styles. The following describes the three grounds:

- Chassis Ground
 - The chassis is grounded to the cabinet through module mounting screws. No wiring to the ground lug from a module is required.
- Logic Ground
 - Logic ground to chassis ground—logic ground is grounded to chassis ground directly by the serrated washer located on the logic ground post on the rear of the module.
- Safety Ground
 - Ground post is mounted to the station sheet metal and can be wired to the Safety Ground in the power panel or to building steel with an insulated #4/0 AWG (100.2 mm²) wire.

ATTENTION

ATTENTION—The power cord has a safety ground connection. This second Safety Ground connection is required by some codes.



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4.11 LCN Node Grounding Methods, Continued

LCN coax shield ground

Each of the coax cable systems (Cable A and Cable B) must be grounded at one, and only one, point. See Figures 4-10 and 4-11 for details.

- Five-Slot Modules—The Clock Source Repeater (CS/R) board that provides the clock, grounds the shield automatically.

If you have Five-Slot Modules with all K2LCN boards installed, it is necessary (mandatory) to have two (one for each cable) CS/R boards installed in empty slots.

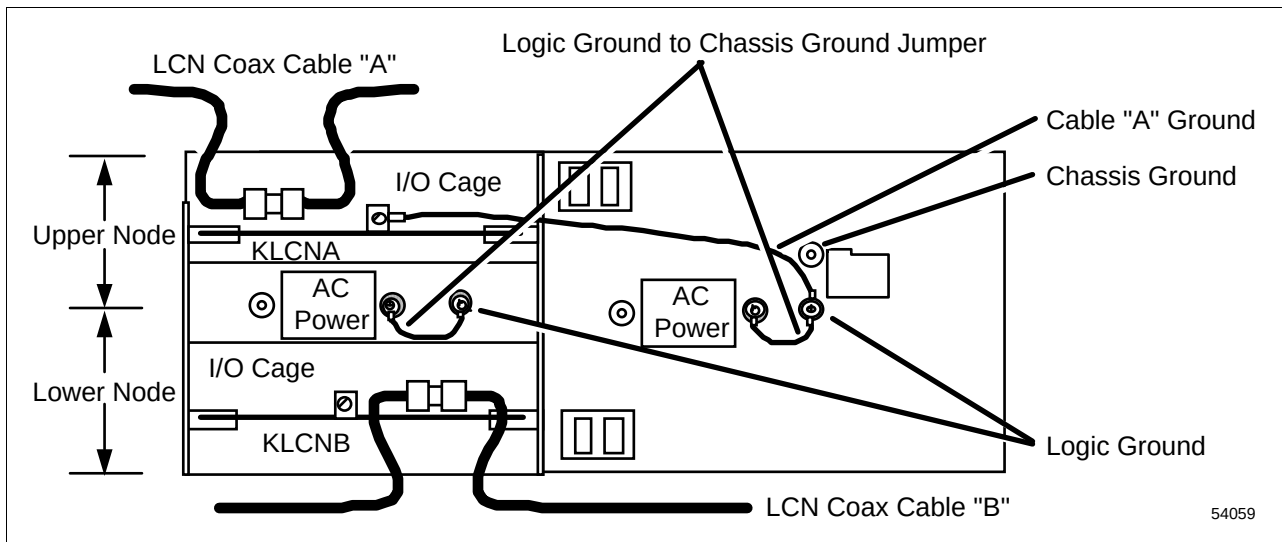
- Dual node modules—
 - Place a wire between the screw terminal on the CLCNA/CLCNB board and the logic ground terminal on the rear of the module.
 - Cable A must be grounded on a different dual node module than Cable B.

Upper Dual Node Module ground

Figure 4-10 illustrates the upper Dual Node Module chassis ground connections.

Figure 4-10

Upper Dual Node Module Chassis Ground Connections



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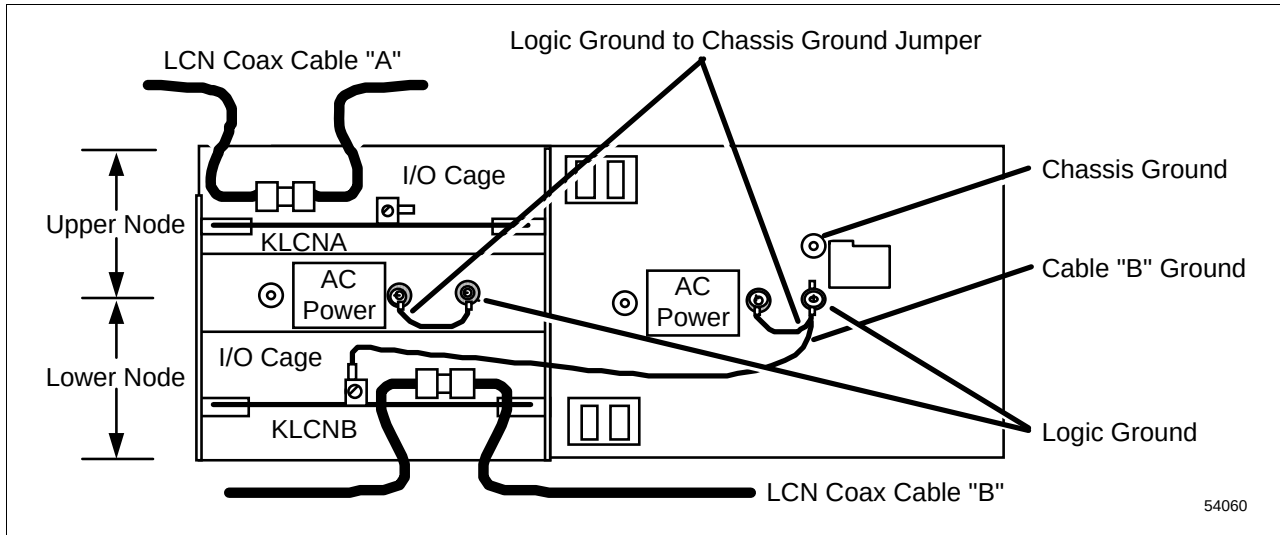
4.11 LCN Node Grounding Methods, Continued

Lower Dual Node Module ground

Figure 4-10 illustrates the lower Dual Node Module chassis ground connections.

Figure 4-11

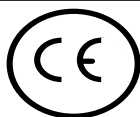
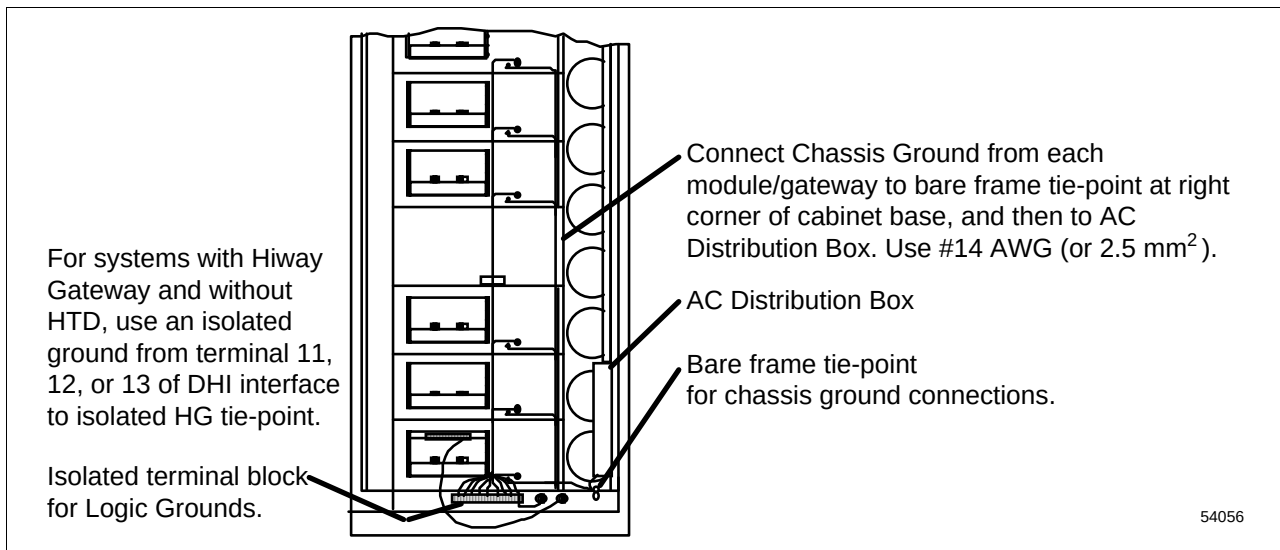
Lower Dual Node Module Chassis Ground Connection



LCN utility cabinet ground connections

Figure 4-12 illustrates the utility cabinet ground connections.

Figure 4-12 LCN Utility Cabinet Ground Connections



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4.12 PM/APM/HPM Grounds

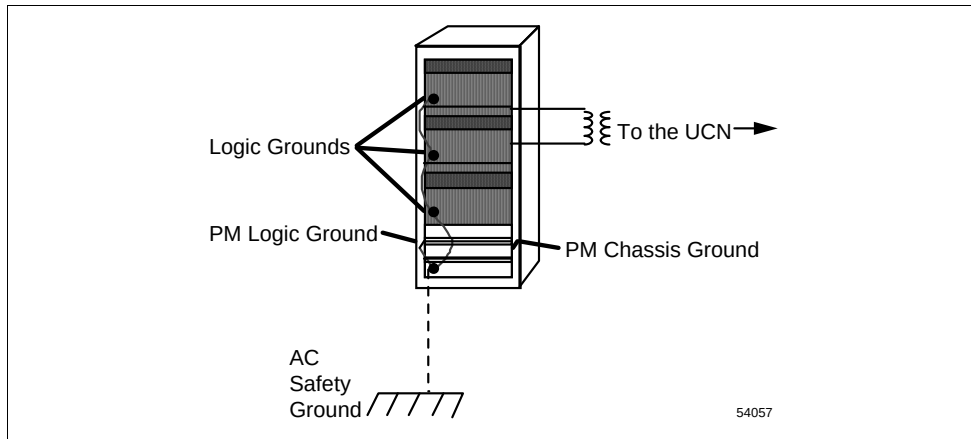
Scope

The PM, APM, or HPM is grounded as shown in Figure 4-13 and Figure 4-14. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer coupled.

PM/APM/HPM cabinet grounding

Figure 4-13 illustrates PM, APM, and HPM cabinet grounding.

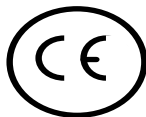
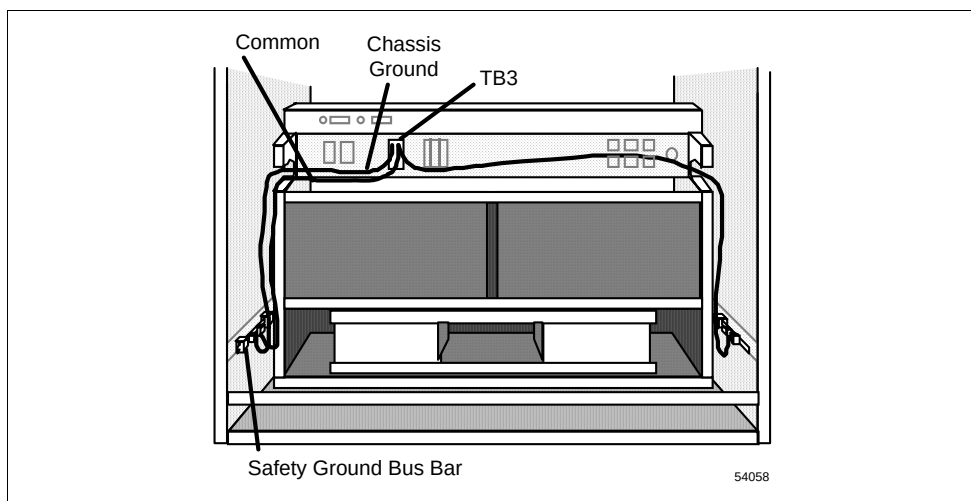
Figure 4-13 PM/APM/HPM Cabinet Grounding



PM/APM/HPM cabinet ground wiring

Figure 4-14 illustrates PM, APM, and HPM cabinet ground wiring.

Figure 4-14 PM/APM/HPM Cabinet Ground Wiring (Factory Provided)



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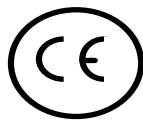
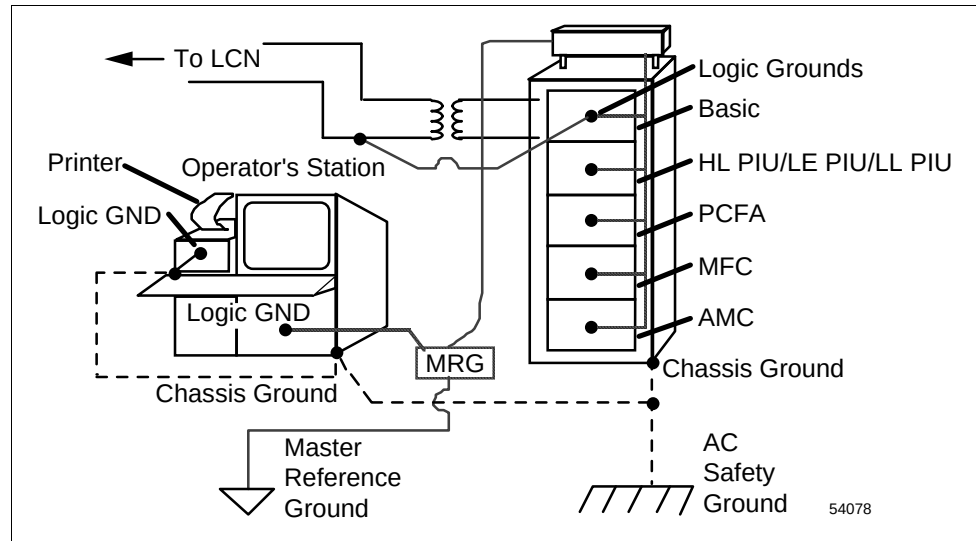
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4.13 Data Hiway Grounds

Scope

The Data Hiway is grounded as shown in Figure 4-15. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer coupled.

Figure 4-15 Data Hiway Cabinet Grounding



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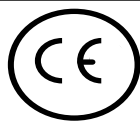
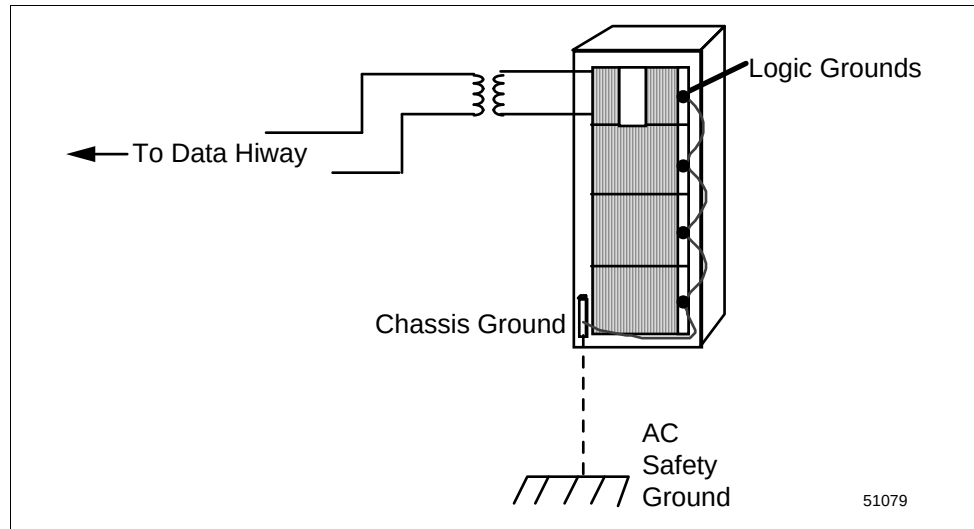
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4.14 620 Logic Controller Grounds

Scope

The 620 Logic Controller is grounded as shown in Figure 4-16. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer coupled.

Figure 4-16 620 Logic Controller Cabinet Grounding



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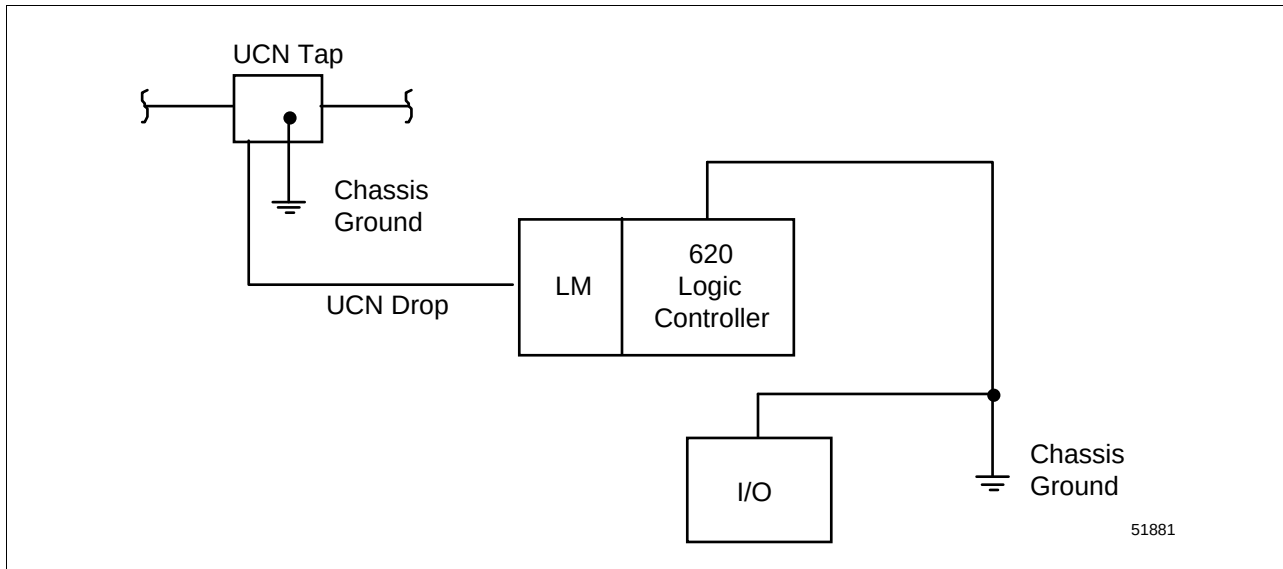
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4.15 Logic Manager Grounds

Scope

The Logic Manager is grounded as shown in Figure 4-17. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer coupled.

Figure 4-17 Logic Manager Cabinet Grounding



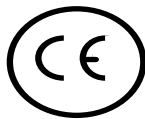
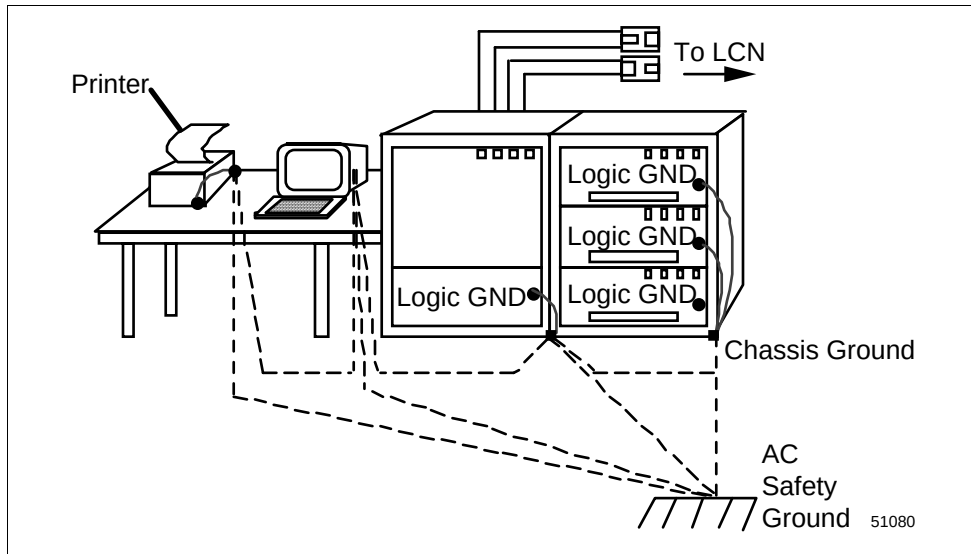
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4.16 DEC System Grounds

Scope

The DEC system is grounded as shown in Figure 4-18. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer coupled.

Figure 4-18 DEC System Cabinet Grounding



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Section 5 – Grounding for Systems Using Master Reference Ground

5.1 Overview



SHOCK HAZARD

Troubleshooting, opening this equipment and removing any panels or covers will expose the **user** to the risk of a shock hazard. There are no **user** serviceable parts inside this equipment. Refer all servicing only to **qualified service personnel**.

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ATTENTION

ATTENTION—This section applies to non-CE Compliant **TotalPlant** Solution (TPS) System installations.



For Non-CE Compliant equipment Installations only

5.2 System Grounds

Required grounds

Grounding of electronics can make the difference between a system that is unusable and one that performs well. The required grounds, their connections, and design are discussed below.

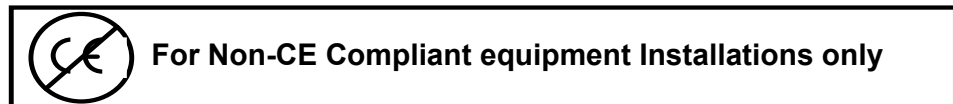
A TPS System requires the use three ground systems.

- Safety Ground
- Lightning Ground
- Master Reference Ground

AC Safety Ground

AC Safety Ground interconnects all metallic structures and enclosures in an area. This minimizes the voltage that people can come into contact with. AC Safety Ground is used to ground all power systems for lightning. It is also used to ground the UCN coaxial cable shields at every tap.

For each building power entry, two independent AC Safety Ground systems with two electrode cables are preferred. Examples of such independent ground systems would be a driven electrode and a metal water pipe as shown in Figure 5-1 and Figure 5-2.



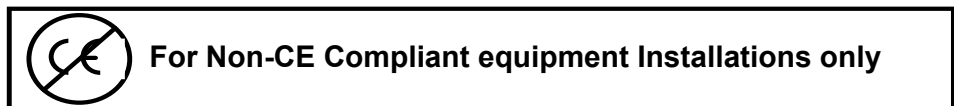
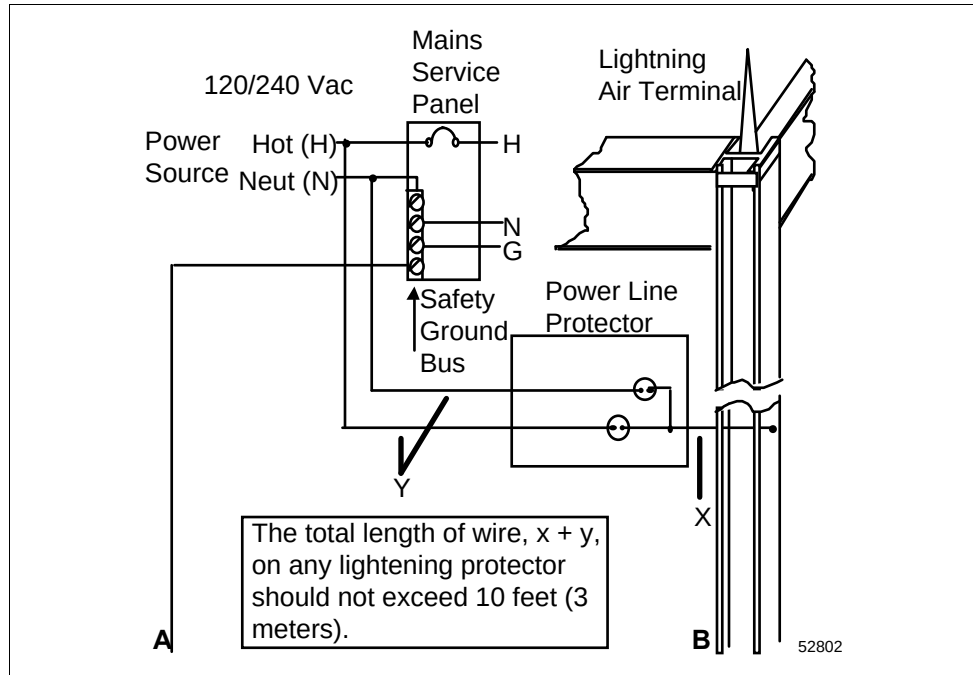
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5.2 System Grounds, Continued

TPS ground system – upper view

Figure 5-1 illustrates an upper view of the TPS ground system.

Figure 5-1 TPS Ground System—Upper View



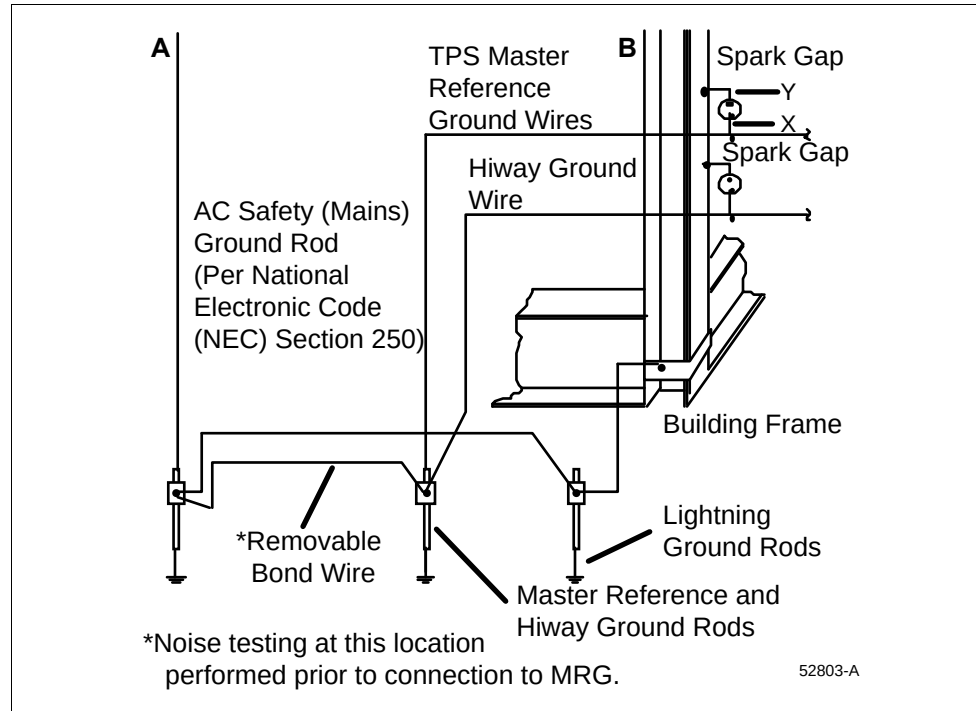
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5.2 System Grounds, Continued

TPS ground system – lower view

Figure 5-2 illustrates an lower view of the TPS ground system.

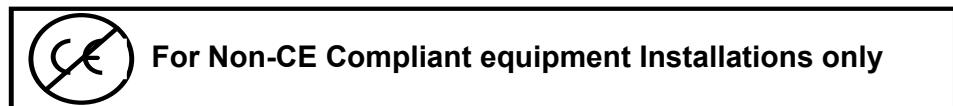
Figure 5-2 TPS Ground System—Lower View



Electrode system requirements

Safety grounding requires electrode systems that provide grounding resistances between 0.1 and 5.0 ohms. The required performance varies with the circumstances described as follows:

- When a facility has neither zener barriers nor lightning problems, AC Safety Ground has only to meet the minimum grounding requirements specified in the local electrical code. Usually, such codes require a grounding resistance of 5 ohms maximum to true earth.
- When a facility uses zener barriers for intrinsic safety, AC Safety Ground should be less than 0.1 ohms and MRG less than 1.0 ohms to true earth. Refer to the Lightning Ground discussion that follows.



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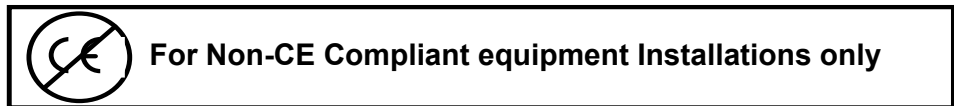
5.2 System Grounds, Continued

Lightning Ground

The purpose of lightning ground is to safely disperse a lightning charge that may be picked up by the facility metallic structure and electrical system. Lightning ground is frequently integrated with AC Safety Ground as shown in Figure 5-1 and Figure 5-2.

Lightning grounds must conform to applicable codes such as NFPA No. 78 (ANSI), IEEE Std 142-1972, Code LP1-175, and other local codes. When a facility is subject to lightning, each lightning rod should be grounded at the base with a maximum resistance of 0.1 ohms to earth.

As discussed in the lightning section in this manual, the TPS lightning protection philosophy is that each lightning rod should be grounded with a resistance to earth of less than 0.1 ohms. A 100 kA lightning strike into a 0.1 ohm ground generates 10 kV. Experience has shown that 10 kV does not cause sparking inside of wiring trays or across terminal blocks. Without sparking and the Electromagnetic Interference (EMI) it generates, TPS Systems perform well during a lightning strike.



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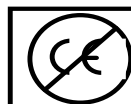
5.2 System Grounds, Continued

Master Reference Ground properties

Master Reference Ground (MRG) is an isolated ground electrode array that references Honeywell TPS electronics and the process signals as shown in Figure 5-2.

An isolated MRG assures stable process signals to the Process Manager, Advanced Process Manager, or High-Performance Process Manager by divorcing signal ground from AC Safety Ground. An MRG installation can have the following properties:

- An MRG is an isolated ground electrode system reserved for Honeywell TPS Systems.
- An MRG usually has a maximum resistance through the earth to AC Safety Ground of 5 ohms.
- When the MRG is grounding zener barriers, it must have a maximum resistance through the earth to AC Safety Ground of less than 1.0 ohm.
- Use a minimum separation of 3 m (10 feet) between the MRG electrodes and any grounded metal such as pipes or a grounding grid or mat.
- MRG electrodes usually perform well when placed immediately next to building footings with electrically floating reinforcing bar.
- No TPS electronics is to be further than 152.4 cable-meters (500 feet) from its MRG electrode nailhead.
- An MRG electrode conductor is typically 11.8 mm (4/0 AWG), is 13 m (50 feet) long, and terminates at a star plate upon entry into a building.
- An MRG entry may require a spark gap to AC Safety Ground to control arcing during a lightning strike. See the lightning section in this manual for more information.
- Large TPS Systems can use MRG star plates that feed other MRG star plates.
- An MRG star plate is insulated from the building structure and concrete, and is capable of accepting up to three 5.3 mm (4 AWG) local Master Reference Ground conductors from each TPS cabinet or complex in the TPS System .
- Radial wiring to a star plate is preferred to daisy chaining from cabinet-to-cabinet.



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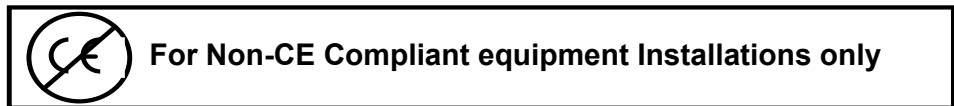
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5.2 System Grounds, Continued

Master Reference Ground properties, continued

- A system can have as many MRGs distributed through the process as is convenient.
- There is no need to connect MRGs together.
- A building can have several independent MRGs.
- A building can have several independent MRGs.
- There is no limit to the amount of equipment that can be placed on one MRG.

The design and testing of ground electrodes is discussed in the following subsection.



5.3 Alternate Grounding Method

Scope

In the event that there are concerns about the grounding system not meeting a regulating body's requirements, such as NEC, OSHA, or local codes, it is permissible to add a direct wire, 4/0 AWG or larger, between the AC Safety Ground and the Master Reference Ground at their grounding electrodes according to codes as shown in Figure 5-2.

Description

Bonding Master Reference Ground (MRG) to AC Safety Ground is done by connecting to the site's AC Safety Ground as shown in Figure 5-1.

- The MRG connection must be made at the electrode nailhead. Moving just a small distance away adds wiring inductance, preventing proper sinking of noise.
 - The AC Safety Ground connection must be made at the connection point previously tested for an acceptable noise level by the procedure described later in this subsection.
 - The connection must be disconnectable for periodic testing or troubleshooting noise problems.
 - The bonding connection between the MRG nailhead and AC Safety Ground must be made with a green, insulated copper wire that is 4 AWG or larger. The attachment points for the bonding connection are critical.
-

MRG noise test

A suitable automatic tester for measurement of the ground system noise is manufactured by PowerCet, Santa Clara, CA, at telephone number (408) 988-1346. Use the model PowerVisa 440 tester. The test requires a reference ground. Testing for noise is discussed in more detail in the *Process Manager/Advanced Process Manager Checkout* or *High-Performance Process Manager Checkout* manual.

Testing must

- take place over a period of at least one week before it can be considered suitable for operation
 - continue at least 6 months or more until operational confidence is high
 - never indicate a measured impulse noise exceeding 10 volts peak
-



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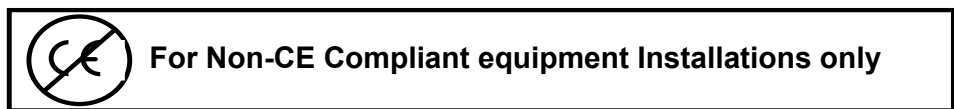
5.3 Alternate Grounding Method, Continued

Problems

Problems with this connection are most likely caused by electrical impulse noise from electrically operated equipment that is sharing the AC Safety Ground.

Most bonded connections pass the tests just described, but not all; therefore, a method must be incorporated that will permit removing the connection between MRG and Safety Ground.

A given connection between MRG and AC Safety Ground can be quiet because the details of plant construction have made it so. These details are important, but not always understood. Excessive noise usually announces itself by causing “ghost alarms” on the operating system; therefore, the connection can fail at any time. The use of a permanently installed noise monitor is preferred over the alternative of using control system performance as an alarm.



5.4 Implementation of Alternate Grounding

Guidelines

With reference to Figure 5-2, the following guidelines are to be implemented before bonding Master Reference Ground to AC Safety Ground:

- The MRG bonding connection is to be made with a suitable heavy conductor, insulated copper, (4 AWG or larger) directly from the MRG nailhead to the noise-tested AC Safety Ground connection point. A conductor is desirable so it can be opened for troubleshooting. In the event that conditions develop on the AC Safety Ground that cause problems on the MRG, the connection can be broken until the AC Safety Ground is corrected.
- The AC Safety Ground may introduce a level of noise on the MRG without causing system degradation if the total is always under 10 volts peak. A detailed test, described in "Facility AC Safety Ground Verification" of the *Process Manager/Advanced Process Manager Checkout* or *High-Performance Process Manager Checkout* manual, is to be performed at the AC Safety Ground connection point. If the test fails, another AC Safety Ground connection point must be selected and tested.

ATTENTION

ATTENTION—All conditions for the Master Reference Ground (MRG) and AC Safety Ground (resistance, isolation, lightning protection, etc.) are to be individually met before the connection between the MRG and AC Safety Ground is made as shown in Figure 5-3; otherwise, the system will not have a fall back position if the connection is removed.



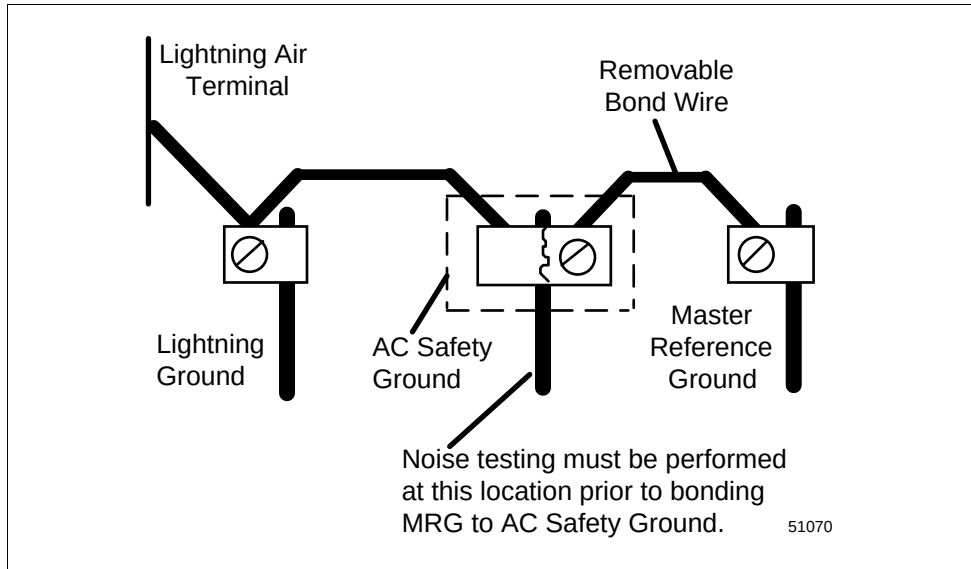
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5.4 Implementation of Alternate Grounding, Continued

Illustration

Figure 5-3 AC Safety Ground/MRG Bonding



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5.5 Electrode Systems

Establishing System

Any grounding system (safety, lightning, MRG, etc.) for a site is designed by (1) measuring the soil resistivity, (2) determining the best electrode configuration for the soil, and (3) installing and interconnecting the electrodes.

Soil Test

The only way to get electrical information about the soil at a given site is to drive electrodes down to the water table and measure resistance. The electrical measurement is done with a commercial tester, such as one made by James G. Biddle Co. in Plymouth Meeting, Pennsylvania at telephone number (215) 646-9200. Use model 250220 or 250241.

A single 3-meter (10-foot) electrode might have a contact resistance of, typically, 5 ohms to earth.

Electrode Material

The preferred electrode usually is 25 mm (1 inch) diameter stainless steel rod, in 2- to 3-meter (8- to 10-foot) lengths. The rod can be driven through soft rock and push smaller stones aside. Weld the rod sections together as the hammering progresses.

Avoid using electrodes with copper when making large arrays. They can turn nearby buried steel into sacrificial electrodes. Avoid using rods with protective platings or coatings. Welding the electrode generates unhealthy fumes and also destroys the coating on the rod.



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5.6 Electrode Arrangements

Single driven electrode

The most economical electrode system consists of one long electrode driven into the water table. A single rod going down 45 m (150 feet) is not uncommon. Deep grounds are desirable because of their electrical stability with the changing seasons.

Electrode Array

When soil conditions do not permit a single deep electrode, shorter electrodes can be used in arrays.

The most efficient form of an array is created by placing the rods in a circle or in a straight line. The rod separation should equal the rod length; therefore, 3-meter (10-foot) rods would be driven into the soil at 3-meter (10-foot) separations.

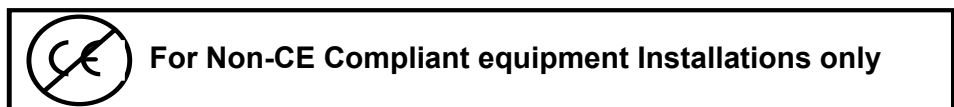
The relationship of separating rods at a distance equal to the rod length comes from each rod having a circle of influence out to a diameter equal to the rod length. Placing a rod inside of the circle of influence of another rod gains little in system grounding performance.

Ground rods can be placed in various arrays such as circles, triangles, squares, and straight lines. The straight line configuration is the most flexible because both ends are open for adding rods, if needed. On the other hand, filling in a circle or any other closed array improves performance very little, optimistically 2:1.

Array performance

The performance of circular and in-line arrays can be estimated by using test results from a single isolated rod. For example, assume a test rod resistance of 10 ohms. To achieve 5 ohms, one would think that two rods in parallel would be acceptable. It actually requires four. When rods are separated at distances equal to their length, the circles of influence overlap such that each added rod has only a 50% independent circular contact with the soil.

The rods are commonly connected together using cadmium welding and 11.8 mm (4/0 AWG) green insulated copper cable. Encase the ends of all rods and make all connections inside a reinforced concrete pipe with an 18 mm (3/4 inch) thick steel cover.



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5.6 Electrode Arrangements, Continued

Drilled electrode

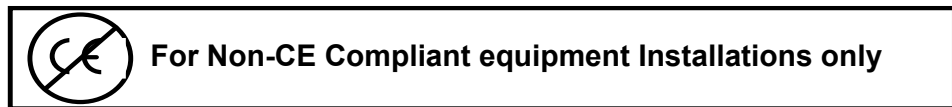
When soil conditions or space do not allow the use of driven rods, a 10 cm (4-inch) or larger well can be drilled or pounded into the water table. Specify that drilling continues until two aquifers (rivers) are struck if reasonable for the area.

The well requires a casing down to bed rock, or it may require casing for the full depth if there is no bed rock in the area. Casing can be perforated steel or plastic. Leave the top 9 meters (30 feet) solid and nonperforated to prevent surface water from draining into the well and contaminating the substrata aquifer.

For the grounding electrode, use a stainless steel rod centering it inside and extending it to the full depth of the well. Then fill the well with concrete mix, coke breeze, or other ashes. These materials slowly release ions that improve grounding over time. Do not use strong chemicals. They may cause water contamination.

The stainless steel rod must comply with the specification contained in NEC 250-83-C, paragraphs 1, 2, and 3.

Do not omit the 150 kA spark gap shown in Figure 5-2, because the well may have to sink all lightning in the area.



5.7 Cabinet Ground Connections

Overview

The TPS System has products that represent different technologies, and therefore different grounding schemes.

- LCN Five-Slot Modules
- LCN Dual Node Modules
- Process Manager, Advanced Process Manager, and High-Performance Process Manager
- Logic Manager
- Data Hiway

LCN Five-Slot Module

Five-Slot Modules are grounded according to the type of node present in the module.

- Universal Stations have the logic ground terminal connected to chassis ground.
- All other LCN nodes have the logic ground connected to MRG.

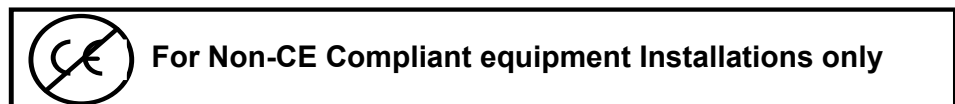
Refer to the *Local Control Network Planning* manual for additional details about grounding the Logic Manager.

LCN Dual Node Module

Dual Node Modules require grounding of both nodes in the module according to the type of node present in the module.

- Hiway Gateways (HG) have the logic ground connected to MRG.
- All other node types have the logic terminal connected to chassis ground.
- Redundant gateways (HGs, NIMs, and PLCGs) must be located in separated Dual Node Modules.

Refer to the *Local Control Network Planning* manual for additional details about grounding the Logic Manager.



Continued on next page

5.7 Cabinet Ground Connections, Continued

PM/APM/HPM

PM/APM/HPM FTAs (that are not grounded at the process end) connect the shields for the process-connected wiring as follows:

- Analog input FTAs and Analog output FTAs are connected to MRG
- 3-30 Vdc digital FTAs are connected to MRG
- 30-200 Vdc digital FTAs that have safety grounds or shields connect them to Safety Ground
- Ac digital FTAs that have safety grounds or shields connect them to Safety Ground

The cabinets have a separate bus bar for safety ground, MRG, and an optional Zener Barrier ground bus bar located near the bottom of the cabinet. Each bus bar is connected to the appropriate system star plate (one for safety ground, one for MRG, and one for zener ground). Refer to the *Process Manager/Advanced Process Manager Planning* or *High-Performance Process Manager* manual for additional details about grounding the PM, APM, or HPM.

Logic Manager

The Logic Manager does not require any special grounding other than what is in the ac power cord. Refer to the *Logic Manager Planning* manual for additional details about grounding the Logic Manager.

Data Hiway

Data Hiway products have all grounds connected to the MRG vertical bus bar in each cabinet. These bus bars are horizontally connected to a local reference bar (if present) mounted to the top of the cabinets. This bus bar is then connected to MRG with a 5 mm (#4 AWG or larger) wire. Refer to the *Data Hiway Planning* manual for additional details about grounding of Data Hiway products.



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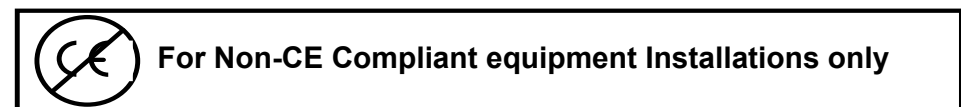
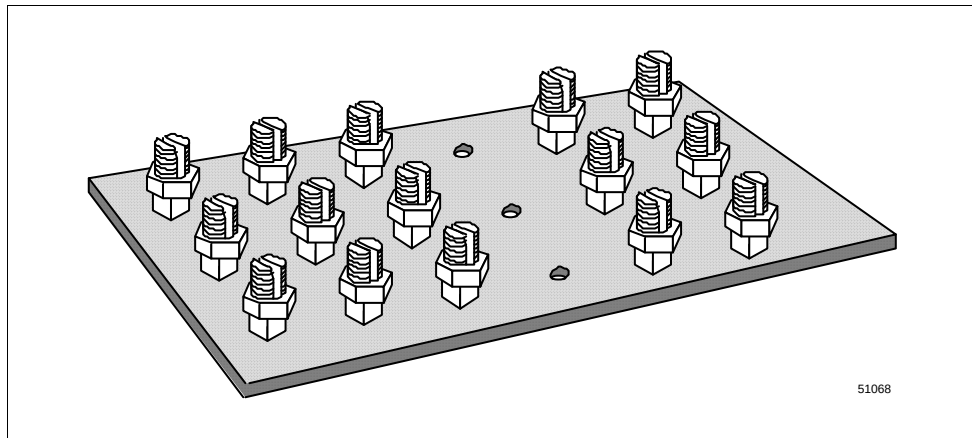
5.8 Star Plate

Description

A star plate is a copper or brass plate that provides for the connection of ground wires at a common point. The required attributes of a star plate are

- **must be completely physically isolated from all conductive material except those ground wires that are to be connected.**
- heavy enough (thick enough) to resist damage from being stepped on or hit a heavy blow. A minimum of 6.35 mm (1/4 inch) thick.
- have more fastening studs (or stud holes) on the plate than the grounding system wiring requires. **Star plates CANNOT be replaced with any of the attached system hardware in operation.**
- each fastening lug must be capable of retaining a single 5.2 mm (#4/0 gauge) insulated ground wire. **One, and only one, wire can be terminated in each lug.**
- be located where it can be readily inspected.
- must be kept dry.

Figure 5-4 Example of a Star Plate



Continued on next page

5.8 Star Plate, Continued

Application rules

Star plates can be used when

- The total distance from the stake to the farthest end point to be grounded is less than 152.4 meters (500 feet). Refer to the MRG grounding requirements shown in Figure 5-2 and described in subsection 5.2.
- each lug has one, and only one, #4/0 copper wire terminated in it.
- **Only ground wires from Honeywell TPS equipment are to be connected to the star plate.**

Star plates for safety ground systems, MRG grounds systems, and zener barrier grounds systems must be kept separate until they reach the respective electrode nailhead.



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5.9 System Grounding

Overview

The TPS ground system is a combination of

- Local Control Network (LCN) grounds
- Process Manager, Advanced Process Manager, and High-Performance Process Manager grounds
- Data Hiway grounds
- IPC 620 system grounds
- DEC system grounds.

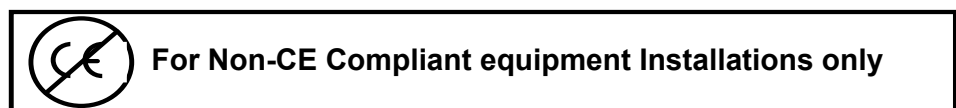
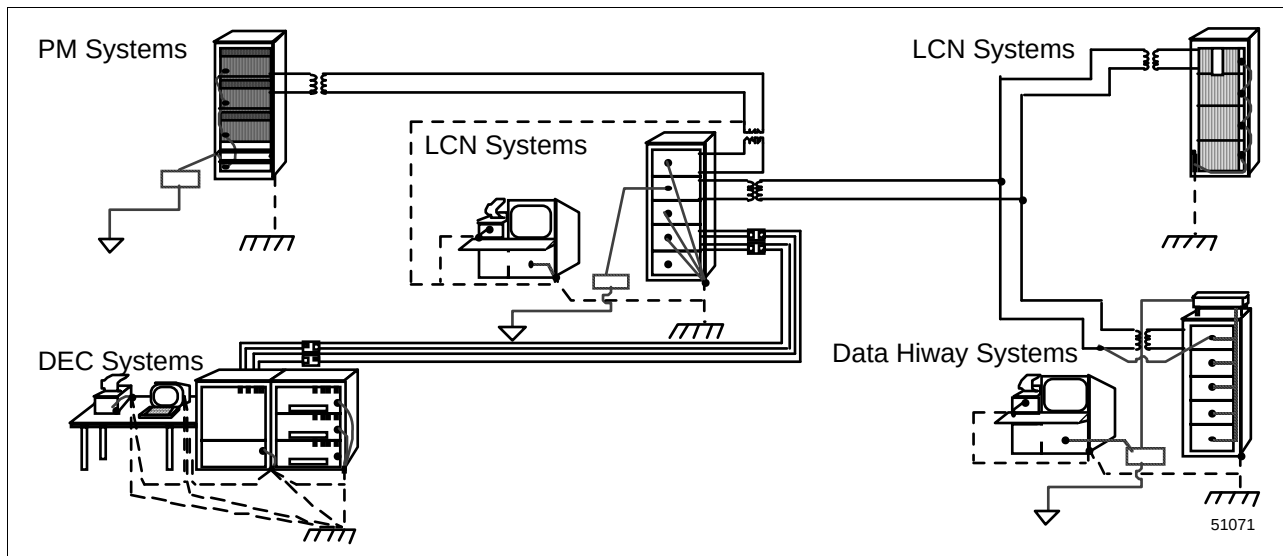
Individual ground systems

Each system has a unique grounding scheme local to itself (star plate) that is directly connected to the ground stake for

- AC Safety ground
- Master Reference ground (LCN, PM, APM, HPM, and Data Hiway)
- Zener Barrier ground (PM, APM, and HPM).

Illustrative overview Figure 5-5 is an illustrative overview of the TPS system grounding.

Figure 5-5 TPS Grounding Systems Overview



5.10 LCN Grounds

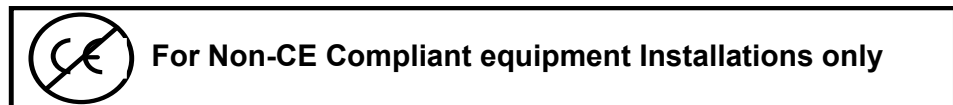
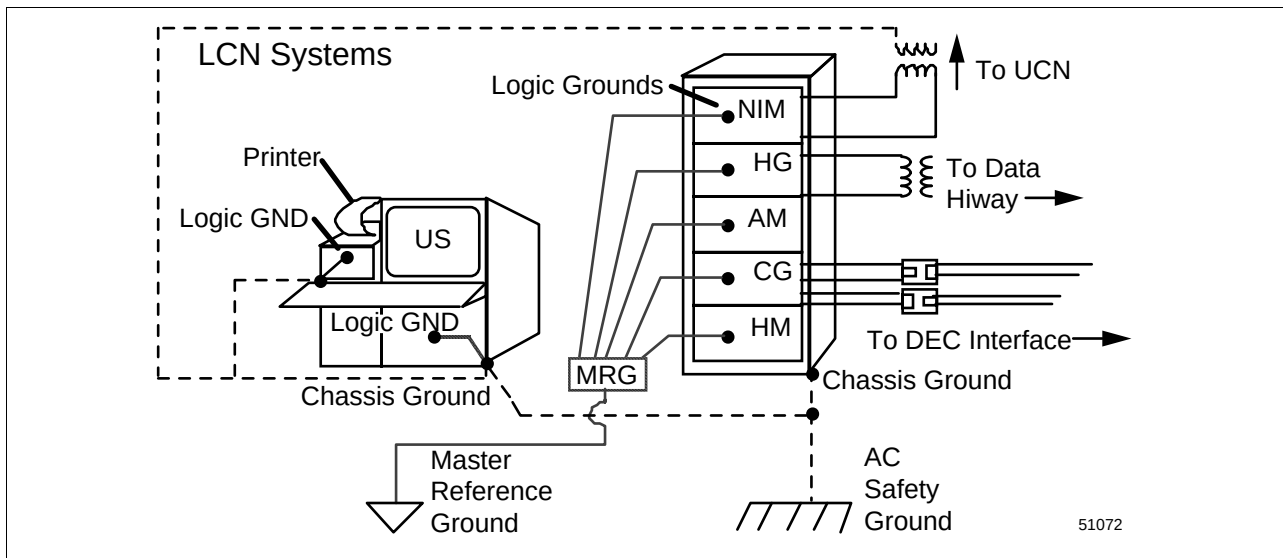
Scope

The LCN module grounding scheme is shown in Figure 5-6 and Figure 5-7. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer coupled.

LCN cabinet grounding Five/Ten-Slot Module

Figure 5-6 is an illustration of LCN cabinet grounding for the Five/Ten-Slot Module.

Figure 5-6 LCN Cabinet Grounding—Five/Ten-Slot Module



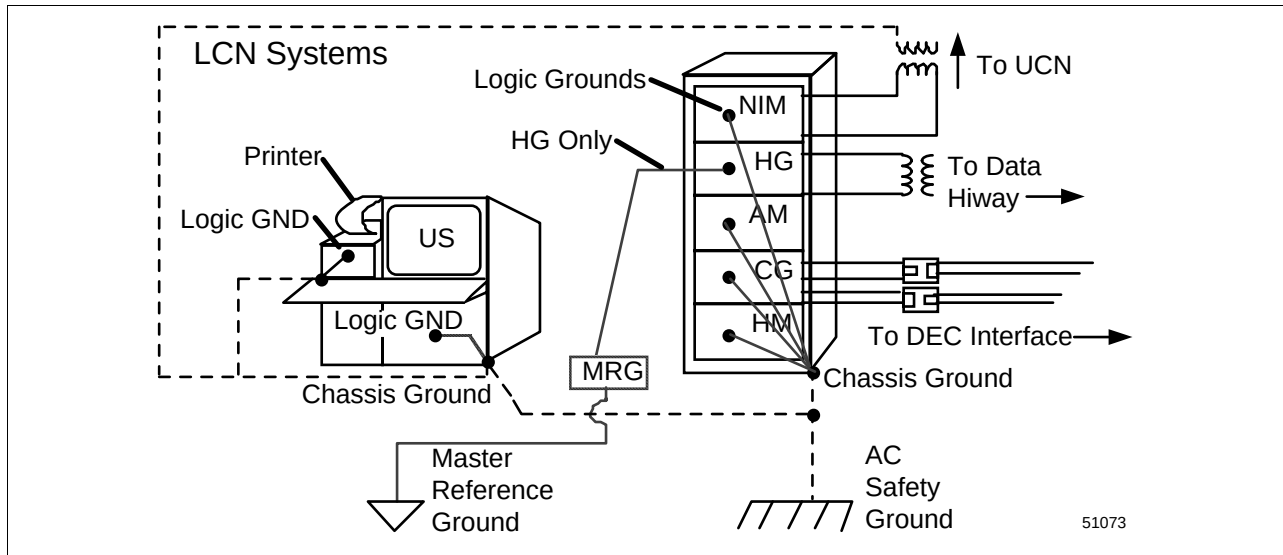
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5.10 LCN Grounds, Continued

LCN cabinet grounding Dual Node Module

Figure 5-7 is an illustration of LCN cabinet grounding for the Dual Node Module.

Figure 5-7 LCN Cabinet Grounding—Dual Node Module



5.11 LCN Node Grounding Methods

ATTENTION

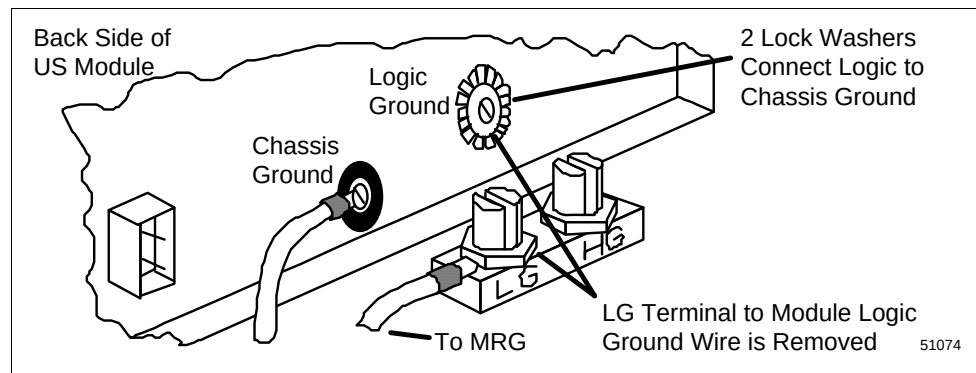
ATTENTION—All modules get their chassis ground connection through the mounting hardware.

Scope

There are two different grounding methods in use in the LCN. The method of grounding the nodes is unique for each module type.

- Five/Ten Slot Module
 - Universal Station nodes in a Five/Ten-Slot Module have the logic ground terminal connected to chassis ground (AC Safety ground) by means of a special serrated grounding washer that connects the terminal to the module chassis as illustrated in Figure 5-8.
 - All other node types in a Five/Ten-Slot Module have the logic ground terminal wired to Master Reference Ground (MRG).

Figure 5-8 Five/Ten-Slot Module Chassis Ground Connections for USs



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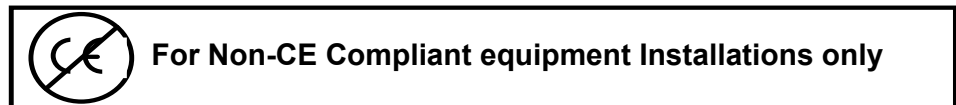
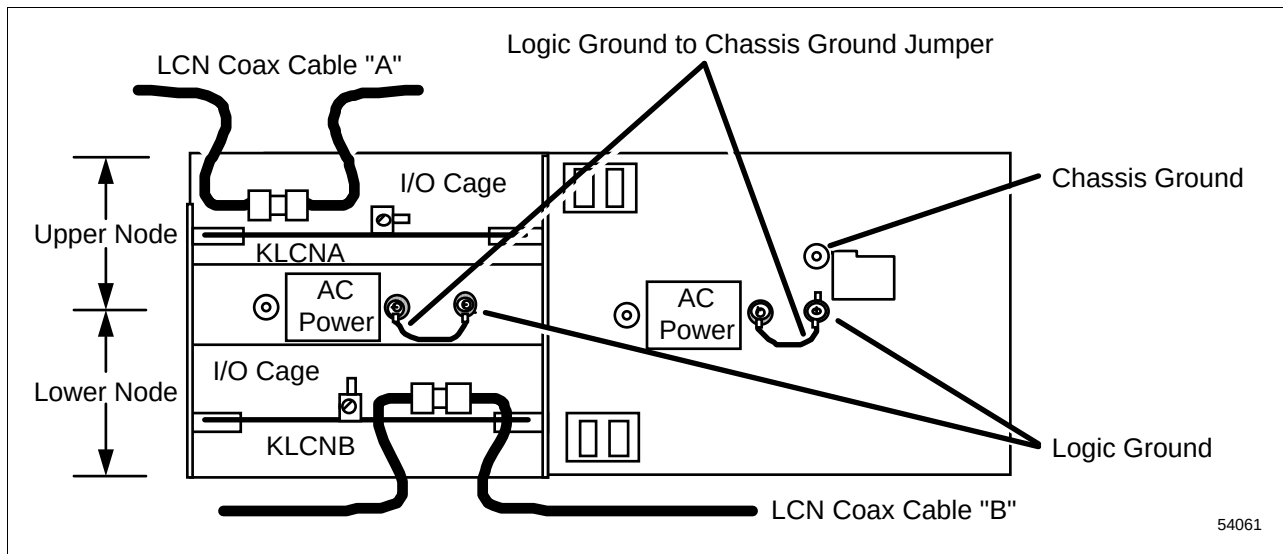
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5.11 LCN Node Grounding Methods, Continued

Scope, continued

- Dual Node Module
 - Hiway Gateway nodes in a Dual Node Module have the logic ground
 - All other node types in a Dual Node Module have the logic ground terminal wired to a chassis screw (chassis ground/AC Safety Ground) as illustrated in Figure 5-9.

Figure 5-9 Dual Node Module Ground Wiring



Continued on next page

5.11 LCN Node Grounding Methods, Continued

Grounding of console furniture

The Universal Station and Universal Station^X furniture has two collection-point ground lugs for in-station grounds (one for chassis ground and one for MRG). There is also a third post for Safety Ground. This applies to both classic and ergonomic furniture styles.

The following describes the three grounds:

- Chassis Ground
 - The chassis is grounded to the cabinet through module mounting screws. No wiring to the ground lug from a module is required.
- Logic Ground
 - Logic ground to chassis ground—Logic ground is grounded to chassis ground directly by the serrated washer located on the logic ground post on the rear of the module.
 - Logic ground to MRG— Individual ground wires from each logic ground terminal on each module is individually connected to the Master Reference Ground (MRG) stake with an insulated #4/0 AWG wire.
- Safety Ground
 - A ground post is mounted on the station sheet metal and can be wired to the Safety Ground in the power panel or to building steel with an insulated #4/0 AWG wire.

ATTENTION

ATTENTION—The power cord has a safety ground connection. This second Safety Ground connection is required by some codes.



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5.11 LCN Node Grounding Methods, Continued

LCN coax shield ground

Each of the coax cable systems (Cable A and Cable B) must be grounded at one, and only one, point. See Figures 5-10 and 5-11 for details.

- Five-Slot Modules—The Clock Source Repeater (CS/R) board that provides the clock, grounds the shield automatically.

If you have Five-Slot Modules with all K2LCN boards installed, it is necessary (mandatory) to have two (one for each cable) CS/R boards installed in empty slots.

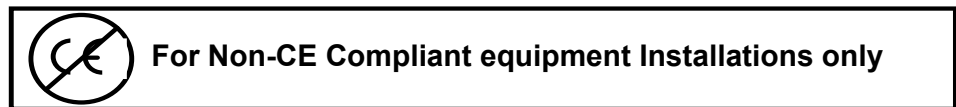
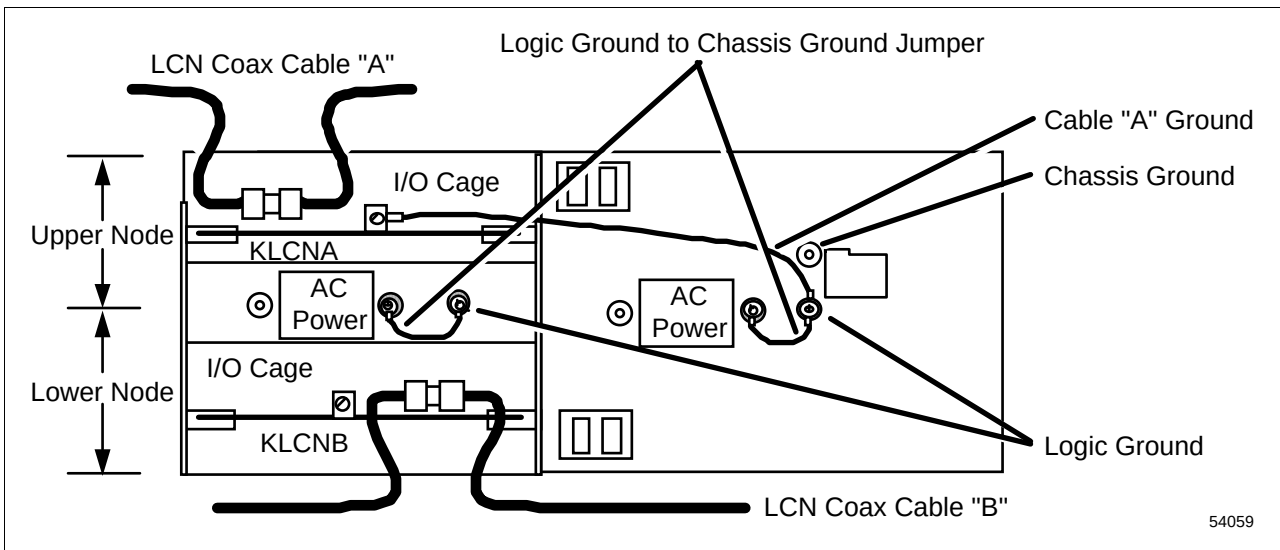
- Dual node modules—
 - Place a wire between the screw terminal on the KLCNA/KLCNB board and the logic ground terminal on the rear of the module.
 - Cable A must be grounded on a different dual node module than Cable B.

Upper Dual Node Module ground

Figure 5-10 illustrates the upper Dual Node Module chassis ground connections.

Figure 5-10

Upper Dual Node Module Chassis Ground Connection



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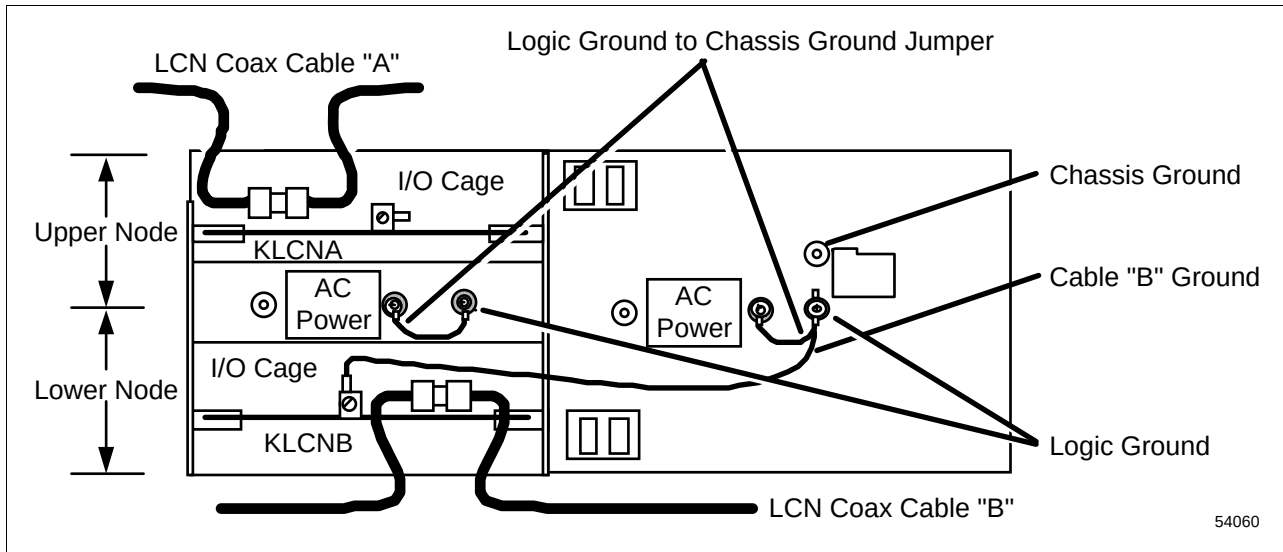
5.11 LCN Node Grounding Methods, Continued

Lower Dual Node Module ground

Figure 5-11 illustrates the lower Dual Node Module chassis ground connections.

Figure 5-11

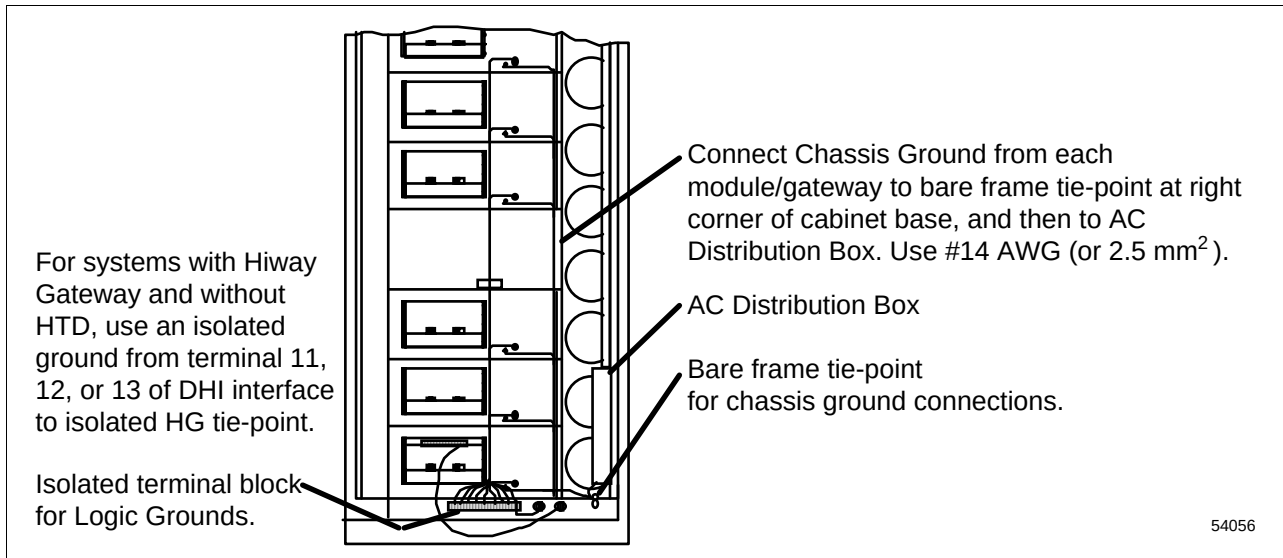
Lower Dual Node Module Chassis Ground Connection



LCN utility cabinet ground connections

Figure 5-12 illustrates the utility cabinet ground connections. Do not confuse the Hiway Gateway module logic ground with the Data Hiway ground.

Figure 5-12 LCN Utility Cabinet Ground Connections



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5.12 PM/APM/HPM Grounds

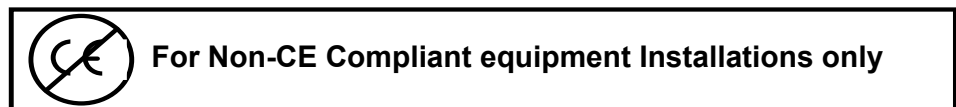
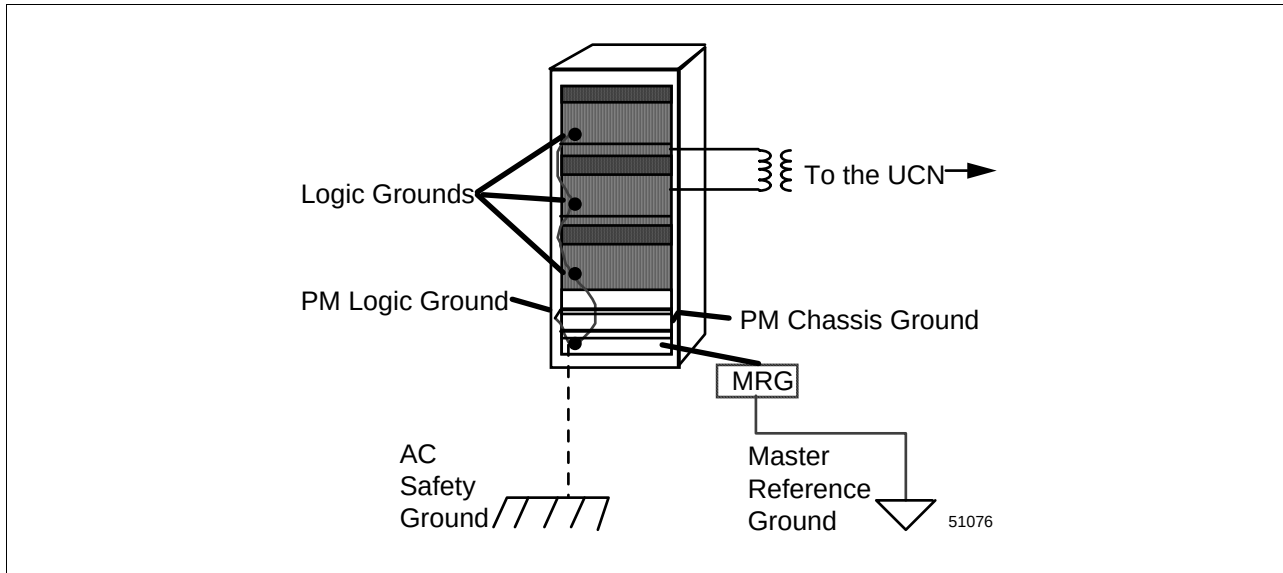
Scope

The PM, APM, and HPM is grounded as shown in Figures 5-13 and 5-14. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer-coupled.

PM/APM/HPM cabinet grounding

Figure 5-13 illustrates PM, APM, or HPM cabinet grounding.

Figure 5-13 PM/APM/HPM Cabinet Grounding

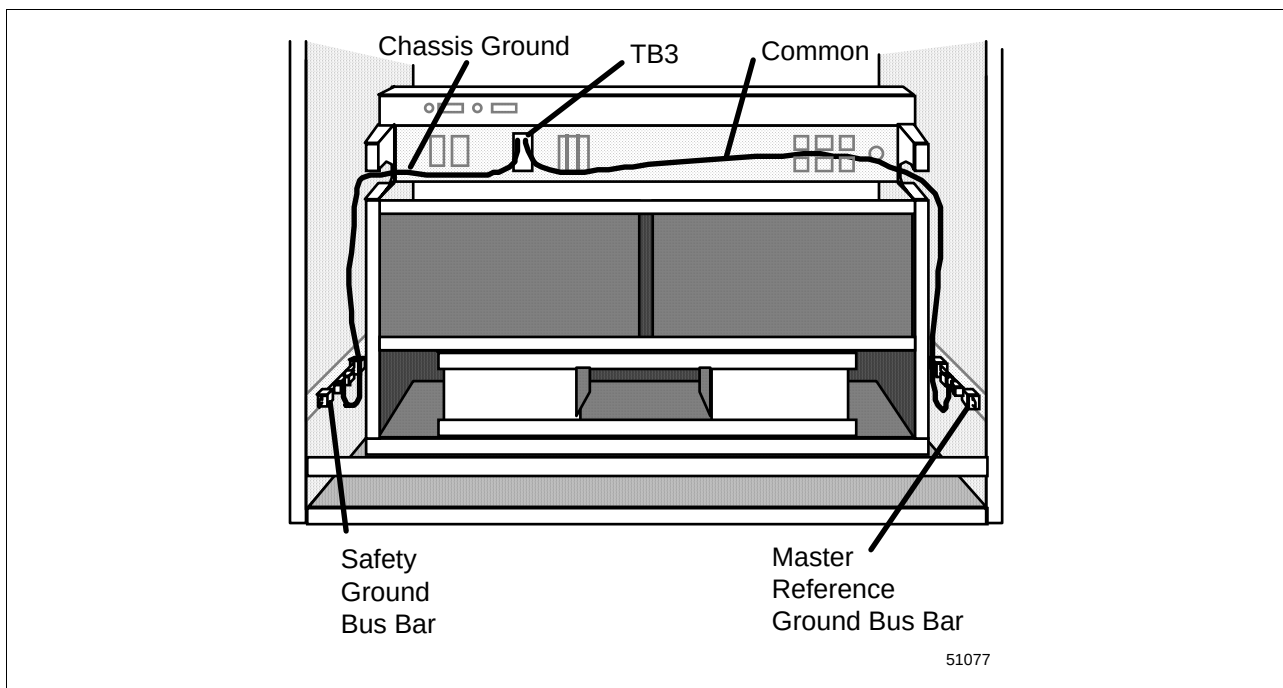


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5.12 PM/APM/HPM Grounds, Continued

PM/APM/HPM cabinet ground wiring Figure 5-14 illustrates PM, APM, and HPM cabinet ground wiring.

Figure 5-14 PM/APM/HPM Cabinet Ground Wiring (Factory Provided)



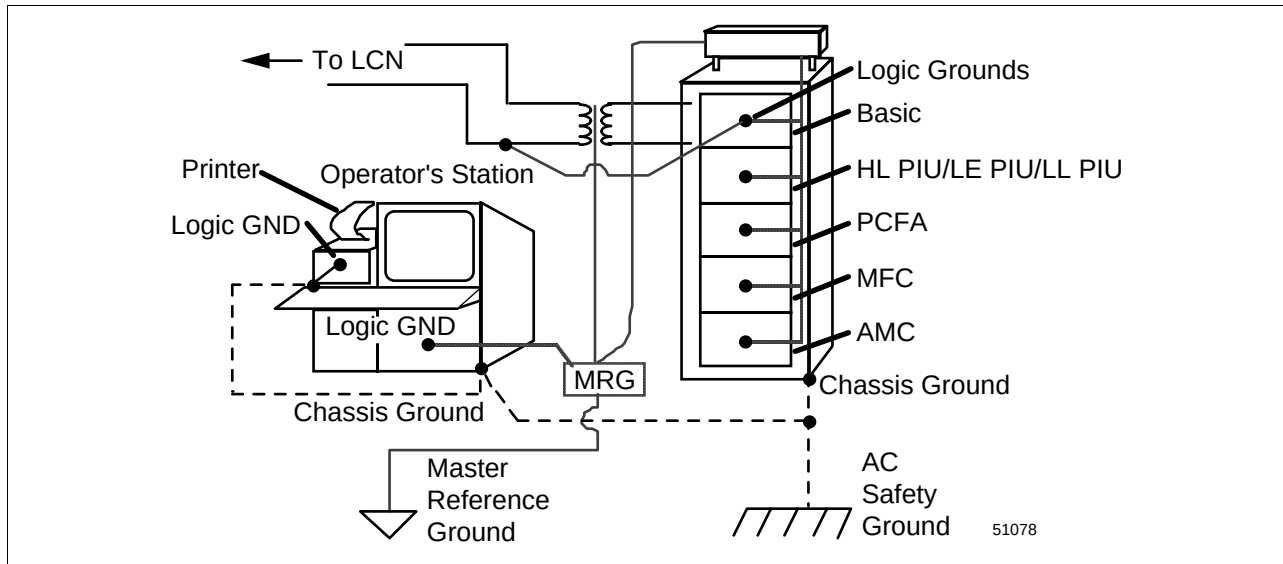
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5.13 Data Hiway Grounds

Scope

The Data Hiway is grounded as shown in Figure 5-15. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer-coupled.

Figure 5-15 Data Hiway Cabinet Grounding

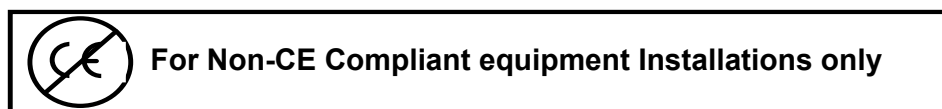
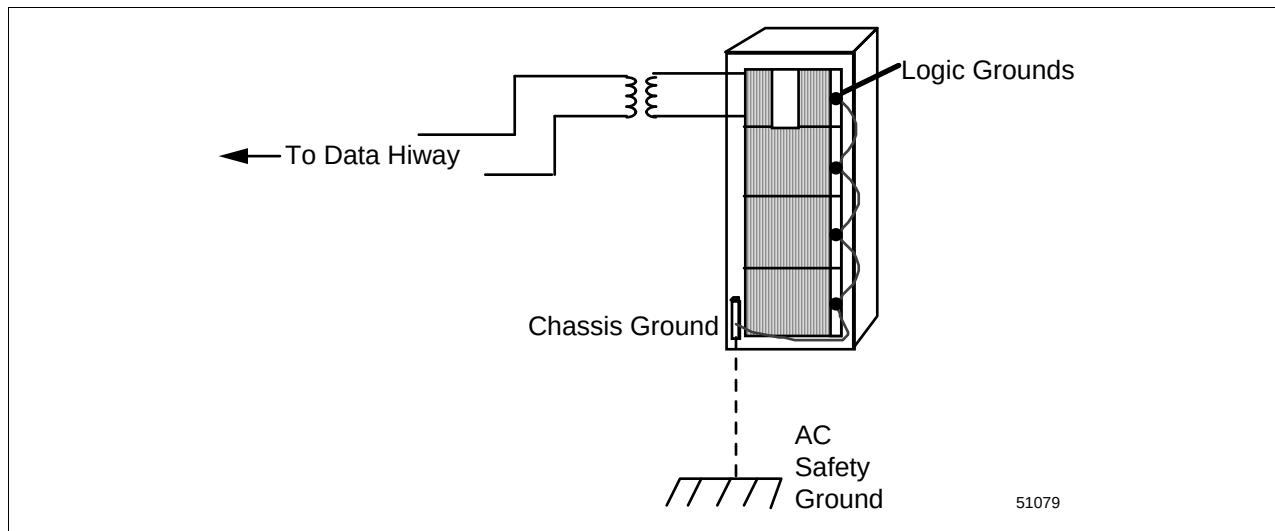


5.14 620 Logic Controller Grounds

Scope

The 620 Logic Controller is grounded as shown in Figure 5-16. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer-coupled.

Figure 5-16 620 Logic Controller Cabinet Grounding

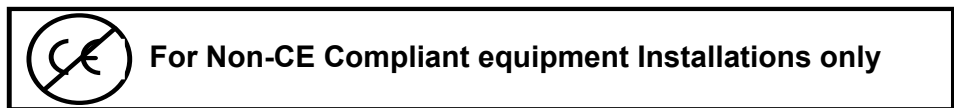
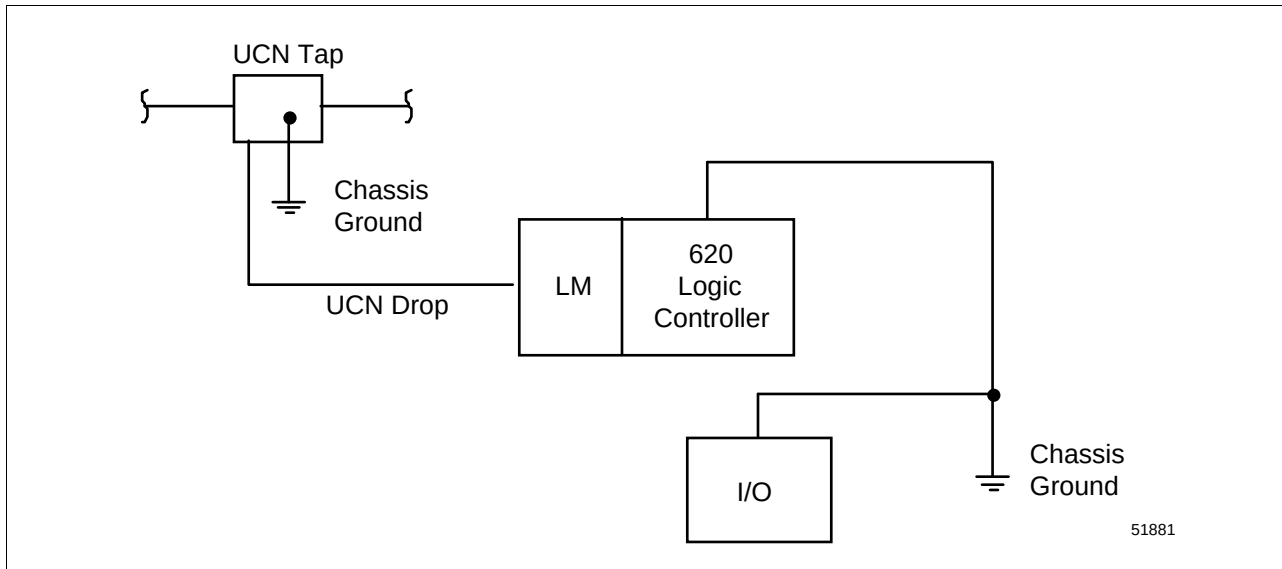


5.15 Logic Manager Grounds

Scope

The Logic Manager is grounded as shown in Figure 5-17. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer-coupled.

Figure 5-17 Logic Manager Cabinet Grounding

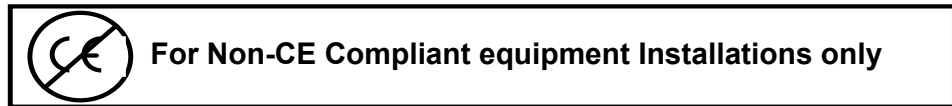
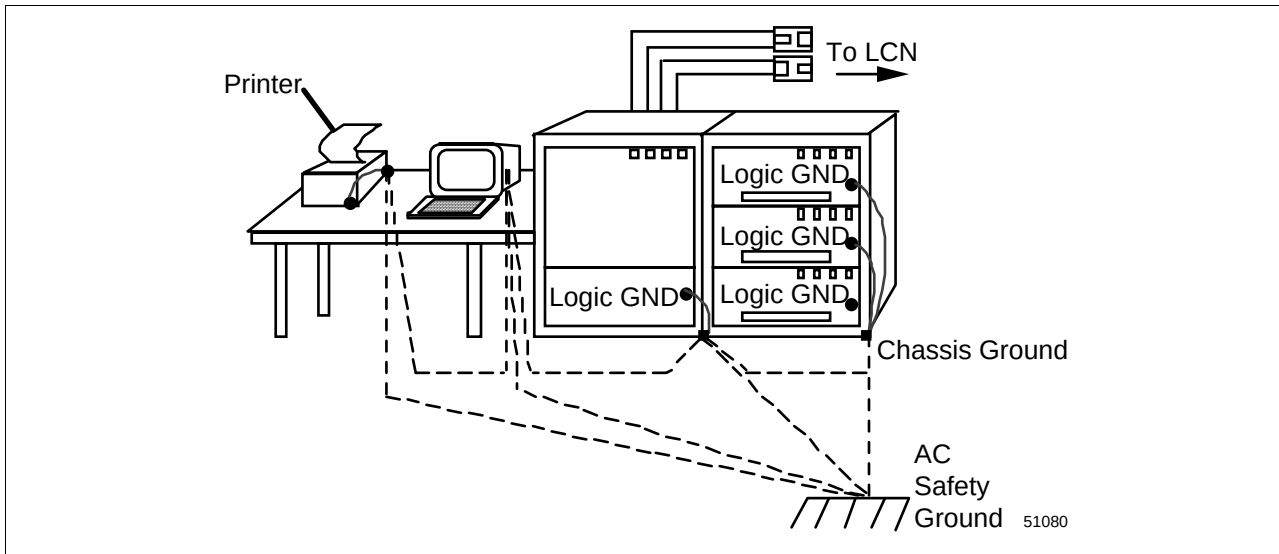


5.16 DEC System Grounds

Scope

The DEC system is grounded as shown in Figure 5-18. Each part of the overall TPS System is individually grounded and communications between each one and the LCN is transformer-coupled.

Figure 5-18 DEC System Cabinet Grounding



Section 6 – Electrical Noise Protection

6.1 Overview

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6.2 Routing and Separation Techniques

ATTENTION

ATTENTION—The following is applicable only when nonshielded wiring is present.

Overview

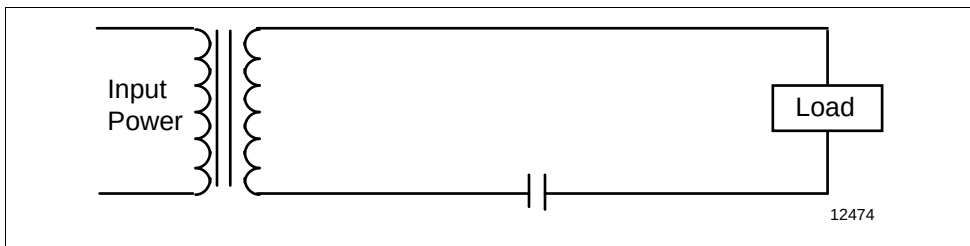
An unwanted signal that obscures or modifies the information content of the system is called noise. The coaxial cables should not be installed in areas with excessive electrical noise.

The most common source of noise is the high frequency component of the magnetic field around power and digital signal lines. On/off-switching causes high frequency changes of the magnetic field and a transformer-like coupling of noise onto the system cables. This problem becomes extremely important when the power potential is in the kilovolt range. The following guidelines suggest ways to reduce the effect of the noise signals. These guidelines consider only the normal switching operations. (Fuses, circuit breakers and other safety devices are not considered switching devices.)

Open parallel power lines

When you directly switch the loads on open parallel power lines, install the coax cable at least 15 m (50 ft.)/kV from the power lines.

Figure 6-1 Open Parallel Power Lines



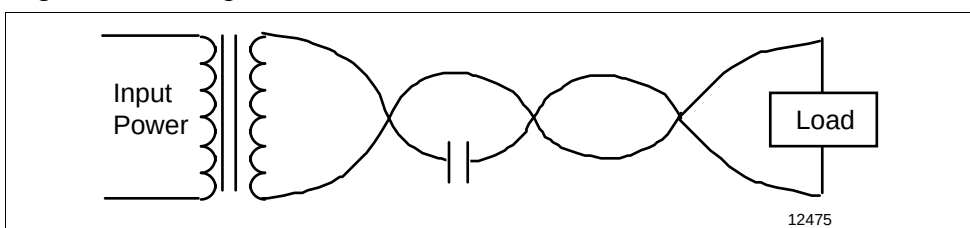
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6.2 Routing and Separation Techniques, Continued

Open twisted power lines

When you directly switch the load on open twisted power lines (1 turn/1.5 m or 5 ft.), install the coax cable at least 3 m (10 ft.)/kV from the power lines.

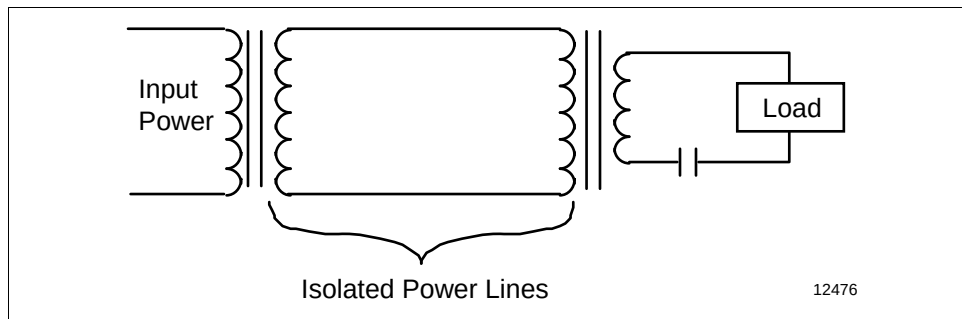
Figure 6-2 Open Twisted Power Lines



Isolated Open Power Lines

For open power lines with switched loads isolated by a step down transformer, use a separation of at least 30 cm (1 ft.)/kV.

Figure 6-3 Isolated Open Power Lines



Continued on next page

6.2 Routing and Separation Techniques, Continued

Enclosed power lines	With power lines enclosed in metal trays or conduit, eddy currents in the enclosure wall eliminate high frequency magnetic radiation. Keep a minimum separation distance of 30 cm (1 ft.)/kV.
Right angle crossings	For right angle (90 + 5 degrees) crossing of the coax and the power line, reduce the separation distances to one-third (3:1), if the distance to the first bend of either cable is equal to or greater than specified for the previous examples.
Transformers and motors	Transformers and motors generate only a small amount of magnetic radiation. Large transformers frequently have short lengths of open wire on their high tension posts; however, the transformer inductance prevents the generation of a high-frequency magnetic field around them. When routing the coax past transformers and motors, keep a separation of at least 30 cm (1 ft.)/kV.
Ground grid	Consider installing the coax in a bonded conduit or on a ground grid to reduce the separation distances of the previous examples to one-third (3:1). 1. A ground grid is a 75 to 100 mm (3 to 4 inch) wire mesh made of 2.6 mm (No. 10 AWG) minimum, bare, galvanized steel wire, welded at each intersection. 2. At power line crossings, the grid must be at least 1.2 m (4 ft.) wide and 23 m (75 ft.) long. 3. The grid must be in direct contact with earth. 4. If the grid is vinyl coated or is imbedded in concrete, it must be connected to earth ground at each end and each pull box.
Conclusion	Digital signals of any voltage in TotalPlant Solution (TPS) System cabinets are best handled by using twisted, shielded wiring pairs. Analog signals should be shielded.

Section 7 – Temperature and Humidity

7.1 Overview

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7.2 Ratings

Introduction

TotalPlant Solution (TPS) System equipment operate over a wide range of temperature and humidity; however, corrosive process materials, contaminants in the air and other realistic considerations may reduce the environmental range that can be placed on the equipment. Table 7-1 shows the operating ranges of TPS devices installed in an ideal, noncorrosive environment.

Table 7-1 Temperature and Humidity Ranges of TPS Devices

Device	Temperature	Humidity
Local Control Network (LCN)	External 0° to 50°C (32° to 122°F) Internal 0° to 70°C (32° to 158°F)	5 to 95%, noncondensing
Process Manager (PM)	External 0° to 50°C (32° to 122°F) Internal 0° to 70°C (32° to 158°F)	5 to 95%, noncondensing
Advanced Process Manager (APM)	External 0° to 50°C (32° to 122°F) Internal 0° to 70°C (32° to 158°F)	5 to 95%, noncondensing
High-Performance Process Manager (HPM)	External 0° to 50°C (32° to 122°F) Internal 0° to 70°C (32° to 158°F)	5 to 95%, noncondensing
Logic Manager (LM)	External 0° to 60°C (32° to 140° F)	5 to 95%, noncondensing
Data Hiway (BASIC)	External 0° to 50°C (32° to 122°F) Internal 0° to 70°C (32° to 158°F)	5 to 95%, noncondensing

Reference

Refer to the *Local Control Network Planning*, *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Planning*, *Logic Manager Planning*, and *Data Hiway Planning* manuals for complete specifications.

Continued on next page

7.2 Ratings, Continued

Guideline

Control of relative humidity is a very important factor to consider in the prevention of corrosion failures. Corrosive effects on electronic equipment are greatly accelerated, even in the presence of a few parts per billion of atmospheric contaminants, by humidity in excess of 50% or by rapidly changing or cyclically fluctuating humidity. On the other hand, excessively low humidity (less than 40%) may cause static discharge problems with electronic equipment. To minimize the environmental impact of corrosives and electrostatic discharge, the device operating ranges must be limited. The following guidelines limiting the operating ranges are recommended:

- Maintain relative humidity levels between 40% and 50%.
 - Control humidity fluctuations to less than 6% per hour rate-of-change.
-

7.3 Dehumidification

Why are special controls necessary?

The life of electronic circuits degrades in corrosive atmospheres at high humidities. Low concentrations of atmospheric sulfides and chlorides condense on the circuit boards (as dissolved electrolytes) each time the room temperature drops a degree or two. The corrosives accumulate with each temperature cycle below the dew point until circuit failure occurs. For example, indicated process variables may shift each time the air conditioner compressor turns on.

When the facility's history for corrosion of circuit boards is not good, limit the maximum relative humidity (RH) to 60% at the boards as follows:

- Dry the electronics room air. Run the air conditioner compressor continuously to keep the room dry. Use heat to keep the room comfortable. Commercial air dryers are also available.
- Limit the temperature rate-of-change to 1°C/minute (1.8°F/minute) on actuation of the air conditioner, or use staged air conditioner compressors with the first having less than 1/4 of the capacity of the second. Alternatively, use heat to keep the room comfortable.

Corrosion and environment is discussed further in the corrosion section of this manual.

Considering the above requirements, set the humidity control system to 50% RH and monitor that the room does not fall below 40%, or rise above 60%.

7.4 Humidification

Humidification controls ESD and extends the life of equipment

In a manned electronics room, maintain a minimum relative humidity of 40%. An excessively dry room air causes personnel to generate high amounts of electrostatic discharge (ESD). The discharge spark causes Electromagnetic Interference (EMI), and excessive EMI can damage control electronics or affect its operation. This problem is particularly severe when personnel are required to wear flame-retardant clothing made from 100% synthetic fibers. Synthetics blended with cotton or stainless steel do not create as much of a problem.

Extra humidifier capacity allows for frequent room air changes from doors opening. Use process steam or purchase a heating system that includes an evaporative humidifier. Avoid using faucet water directly in a spray or mist. This may inject dissolved solids into the air.

7.5 Thermal Sensitivity

Why maintain close control of temperature?

Whenever millivolt and thermocouple circuits are used in an electronics area, limit the room temperature rate-of-change to 1°C/minute (1.8°F/minute) maximum to prevent the generation of thermal gradients at terminal blocks. Also restrict drafts and air currents around FTAs where thermocouple wires connect.

7.6 Air Conditioner Loading

Introduction

The load an air conditioner experiences is caused by the infiltration of heat from outside sources and from the heat generated by equipment in the room. The *Local Control Network Planning*, *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Planning*, *Logic Manager Planning*, and *Data Hiway Planning* manuals have figures for estimating heat infiltration from standard sources. There are two ways for estimating the equipment heat input to the room.

How to estimate HVAC loading

The first and quickest way to estimate heat generation by equipment is to assume that the full ac rms input to the power supplies is converted to heat as follows:

- Each PM, APM, or HPM, for example, has a primary Power Supply Module and often a backup (redundant) Power Supply Module. The primary module, at full load, adds heat that is calculated as follows:
 - Power = voltage x current x Power Factor (primary supply)
= $120 \times 7.6 \times 0.8$
= 730 watts

The redundant or backup module (if used) adds:

- Power = $120 \times 1 \times 0.5$ (backup supply)
= 60 watts

Each fully-loaded PM, APM, or HPM with dual Power Supply Modules adds at the most 790 (730 + 60) watts of heat to the room. Using the calculation

$$\frac{\text{watts}}{3.5168 \times 10^3} = \frac{790}{3.5168 \times 10^3} = 0.225 \text{ tons of air conditioning for}$$

each PM, APM, or HPM with redundant Power Supply Modules.

- PM, APM, and HPM heat generation can be calculated by a second method, using the component list information in the *Process Manager/Advanced Process Manager Planning* or *High-Performance Process Manager Installation* manual on ac power consumption to calculate the exact losses. This method is accurate for any level of system complexity.

An undersized air conditioner is preferred in wet climates because it will tend to run continuously and dry out the air. This minimizes corrosion. An oversized air conditioner is preferred in dry climates because it runs less often. This helps to minimize ESD problems by leaving moisture in the air.

Section 8 – Ventilation and Filtration

8.1 Overview

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8.2 Air Distribution

Specification

LCN Control rooms require air filtration of particles to a minimum of 19 microns. See the *Local Control Network Planning* manual.

Recommendation

An air distribution system is recommended when atmospheric contaminants from process fumes, road dust, or cooking fumes are present. Heating and cooling are discussed in the previous section. To protect the electronics from various fumes, arrange an air flow moving towards the sources, such as towards doorways and the cooking area.

Guidelines

Use ducts to handle the air. Do not use the area under a raised floor as a plenum. The resulting vacuum/pressure under the floor draws air through the PM, APM, or HPM cabinet cable cutouts. The unnecessary flow causes the following problems:

- The electronic circuits are continuously exposed to dirt, corrosives, and cooking fumes.
- The temperature drop caused by the air-conditioning compressor cycling on may cause condensation of water soluble corrosives on the circuit boards.
- Leakage of heated or cooled air may push electronic equipment to its temperature limit.
- Temperature rate-of-change from heating and cooling may cause thermal gradients in millivolt circuits.

An air distribution system should use a minimum flow velocity and a minimum of makeup air. See Section 10 in this manual concerning corrosion.

8.3 Air Filtration

Recommendation

All air flow should be routed through a mechanical (nonelectronic) dust filter. The air can then be treated further for corrosives as discussed in Section 10 of this manual.

The mechanical filters should have an efficiency rating of not less than 20% after performing the Bureau of Standards discoloration test. This test consists mainly of measuring the total cross-sectional area of dust particles removed from the filter. The filter should meet all local fire codes.

Section 9 – Lightning Protection

9.1 Overview

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Philosophy

Lightning ground safely dissipates lightning energy to protect personnel, the control system equipment, and the building. Lightning is intercepted by air terminals and/or the building's frame and conducted through a dedicated 5.2 mm (#4 AWG) wire to the lightning-ground rods or grid. This ground system must conform to applicable codes and design-construction criteria.

- Typically, a lightning-ground system consists of 3 meter (10 ft.) ground rods bonded (connected) to vertical structural members every 30 meters (100 ft.) along the building's perimeter.
 - If possible, locate the rods near wet areas.
 - Allow enough space for future access to existing rods or for expansion.
 - Normally, the ac safety ground (main) rod is connected to the lightning ground-rod system. A lightning-ground resistance of 0.1 ohms to true earth maximum is recommended.
 - To help ensure the safety of all equipment and personnel, the primary transformers should be protected by lightning arresters.
 - It is highly desirable that similar protection be provided at the service entrance to the building.
 - These arresters will reduce the possibilities that excessive voltage and currents caused by lightning strikes will seek some indeterminate, low-impedance path to ground, such as the system power circuits.
 - There is always the possibility of arcing (flash) to an isolated ground "systems" when lightning strikes. To prevent this, some electrical codes require all isolated ground systems to be connected to building steel through spark-gap devices. These spark-gap devices can be installed at any convenient location, usually where the ground wire leaves the building.
-

Continued on next page

9.1 Overview, Continued

Philosophy,
continued

- Soil resistivity can vary a great deal, depending upon the water table, chemical content of the soil, and geographical characteristics of the site. In most cases, deep grounding is preferred because lower soil resistivities are usually found and seasonal variations are minimized.
- If soil resistivity variations are a potential problem, periodically measure and record the resistance of all ground systems to determine whether the resistance meets specifications.

If later tests (refer to Measuring Ground System Resistance in this section) show that the resistance is increasing to an undesirable value (i.e., generally greater than 5 ohms, and greater than 1 ohm for intrinsically safe systems), reduce the resistance by increasing the moisture content of the soil, by chemical treatment, or by adding ground rods.

Using lightning protectors on the process signal twisted pair wires can be expensive. A suitable 10 kA protector costs \$100, and one is required at each end of the twisted pair wires. Protection with a large lightning ground can also be expensive. When the ground is suitably large, no additional lightning protection is required because significant voltage cannot be induced by any kind of strike. The two fundamental lightning protection approaches, protectors and grounding, are proportioned to provide performance and economy as follows:

1. Ground the facility lightning antennas to an electrode system that has a contact resistance to the earth of 0.1 ohms or less.
2. Install 150 kA protectors on the system power and grounds. If a strike occurs, it will be present here first.
3. Install protection on circuit twisted pair wires only as damage occurs. See Figure 9-1.

Most facilities do not have to go beyond step 2; however, more discussion of this philosophy follows.

Continued on next page

9.1 Overview, Continued

Lightning Ground

Lightning ground is little more than an augmented AC Safety Ground as shown in Figure 3-5. A process control facility does not always require special construction for lightning if the two following requirements are met:

- if the process facility high-line to the transformer yard can serve reliably as the facility lightning rod
- if the substation ground has a maximum resistance to earth of 0.1 ohms

As an example, a tall distillation column can serve as a process and lightning rod. The column would require its own 0.1 ohm electrode system. Otherwise, the lightning would travel through the process connecting piping to find a suitable ground. If control wiring uses the same pipe rack, control system failure can occur.

Section 4 discusses the design of the electrode system.

9.2 Power Protection

Practice

The preferred power line protector is based on a Metallic Oxide Varistor (MOV). For power lines, the MOV is preferred because it does not short the power along with the transient. Use a 150 kA unit. Overcapacity does not carry a penalty. A suitable protector can be purchased from Lightning Protection Corporation in Santa Barbara, CA, at telephone number (805) 967-5089. For a 120 Vac system, use model LPC 20208 and a model LPC 20618 for a 240 Vac system.

ATTENTION

ATTENTION—Install a 150 kA power line surge protector as shown in Figures 3-2 and 3-3. In Figures 3-4, 3-5, and 3-6 the MOV is located on the output of the utility power transformer.

9.3 MRG Protection

Practice

Should lightning strike, AC Safety Ground can develop over 10 kV relative to the isolated Master Reference Ground (MRG). Damage is possible inside the cabinet that houses the process control equipment. Use a 90 V, 150 kA spark gap as shown in Figures 3-7, and 4-3.

Spark gaps are preferred for grounds because of their very low voltage drop while conducting a surge. A suitable spark gap assembly can be purchased from Lightning Protection Corporation in Santa Barbara, California, at telephone number (805) 967-5089. Use model LPC 11001. Suitable protectors can be purchased from local or national suppliers. It is recommended to consult lightning specialists for proper model choice and correct installation.

ATTENTION

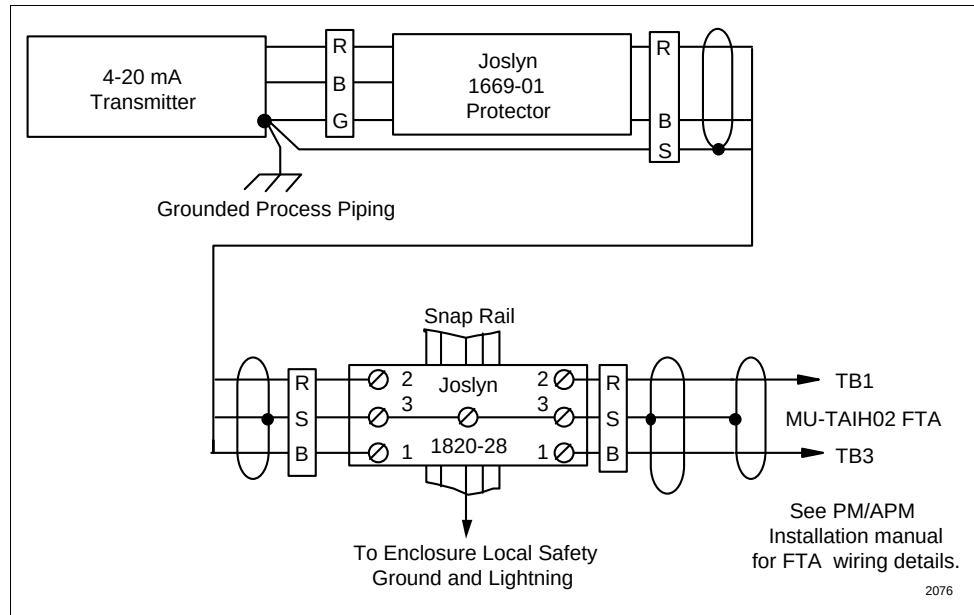
ATTENTION—Incorrect installation of lightning protection devices can reduce their effectivity—Honeywell strongly advises installation by experts.

9.4 Twisted Pair Protection

Practice

When a particular 4-20 mA/1-5 V twisted pair requires lightning protection, use the protection system shown in Figure 9-1. Suitable protectors are available from Joslyn Electronic Systems in Goleta, California, telephone 805-968-3551. Use model 1669-01 at the transmitter and model 1820-28 in the electronics room.

Figure 9-1 Twisted Pair Lightning Protection



9.5 UCN Protection

Practice

Except when routed on poles, the Universal Control Network (UCN) coaxial cable system requires no added protection for lightning. The coaxial cable shield already grounds to AC Safety Ground at each tap, and each process control equipment UCN coaxial cable interface already has input protection.

When the UCN cables are outdoors on poles, place a counterpoise 2 meters (6 feet) above the coaxial cable for the full run. Additionally, bury another counterpoise in the ground below. Connect both counterpoises and the coax messenger together at each pole. Ground the counterpoises and the coax messenger to AC Safety Ground at each building entry and at each end. Use 5.3 mm² (4 AWG) wire throughout.

Section 10 – Corrosion and Dust

10.1 Overview

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10.2 Corrosion

Introduction

Atmospheric pollutants, such as corrosives, can cause premature failure of control electronics. Corrosives destroy the electronic components, while oils and dust cause connector contacts to lose continuity.

Placement of the Deviceequipment must comply with the environmental specifications found in the *Local Control Network Planning*, *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager*, *Logic Manager Planning*, and *Data Hiway Planning* manuals.

Effects

Corrosion of electronic circuit boards can cause operating problems that do not indicate the root cause. That is, a replacement circuit board may temporarily solve a problem, but not prevent the problem from recurring. This can be prevented by measuring the degree of the corrosion, then applying proportional protection.

Continued on next page

10.2 Corrosion, Continued

Concentrations

The most prevalent compounds causing corrosion in the process industry are active sulfurs and inorganic chlorides as shown in Table 10-1.

Table 10-1 Corrosive Compounds Found in Industry

Type of Compound	Effect
Inorganic chlorine compounds (chlorine, chlorine dioxide, hydrochloric acid)	These compounds are seldom absent in major installations, and in the presence of moisture, they readily react with copper, tin, silver, and iron alloys found in electronic components. These reactions can be significant even at chlorine levels in the low Parts-Per-Billion (PPB).
Active sulfurs (hydrogen sulfide, elemental sulfur, organics such as mercaptans)	In an environment with mild reactivity, the concentration of active sulfurs would typically be 2-3 PPB. The presence of moisture and a few tenths PPB of chlorides could increase reactivity into the moderate class or above.
Sulfur oxides (SO ₂ , SO ₃)	These compounds are generated as combustion products of sulfur-bearing fossil fuels, and from coking and nonferrous smelting operations. A few PPB of sulfur oxides can passivate reactive metals and retard attack by other contaminants. When dissolved in water at high levels (>200 PPB), forming sulfurous and sulfuric acid, they can attack certain types of masonry, elastomers, and plastics.
Other compounds	The effects of other compounds such as nitrogen oxides, ammonia, oxidants, and fluorides are either not precisely known or typically not present in sufficient quantities to warrant special precautions. If a problem with any specific constituent is suspected, a corrosion specialist should be consulted.

Continued on next page

10.2 Corrosion, Continued

Reactivity

Copper reactivity has been found to be more accurate and easier than measurement of individual gas concentrations for measuring environmental severity because it integrates the effect of all of the variables. By placing a small copper strip, called a coupon, in the equipment enclosures and measuring the corrosive-film thickness that develops on the surface over a known period of time, data can be obtained with which reliability predictions can be made.

Four classifications for copper reactivity (angstroms/month) have been established by ISA-571.04-1985 and are listed in Table 10-2.

Table 10-2 Established Classifications of Copper Reactivity

Classification Angstroms/Month (ISA Class)	Effect
Mild: 0 - 299 (G1)	Corrosion is not a factor in determining equipment reliability.
Moderate: 300 - 999 (G2)	The effects of corrosion are measurable and may be a factor in determining equipment reliability.
Harsh: 1000 - 1999 (G3)	There is a high probability that corrosive attack will occur. This harsh level should prompt additional evaluation and result in environmental controls or specially designed and packaged equipment.
Severe: ≥ 2000 (GX)	Only specially designed and packaged equipment would be expected to survive. Specifications for equipment in this class are a matter of negotiation between user and supplier.

Reactivity greater than 300 angstroms each month does not result in immediate equipment failure; however, problems may start intermittently from the formation of thin corrosion films on contacts or localized dendritic growth between printed wiring-board tracks. These phenomena often occur during brief periods of high humidity and may be temporarily cleared when humidity returns to a lower level. As reactivity approaches 1000 angstroms per month, intermittent failures become more frequent. Above this level, significant degradation of equipment reliability occurs.

10.3 Gaseous Contaminants

Overview

Listed are the most common gaseous pollutants found in process control environments along with typical concentration levels. Note, however, that **gas levels should not be used exclusively as a means to determine the severity of atmospheric conditions.**

- Inorganic chlorine compounds (chlorine, chlorine dioxide, hydrochloric acid)
- Active sulfurs (hydrogen sulfide, elemental sulfur, organics such as mercaptans)
- Sulfur oxides (SO₂, SO₃)

Inorganic chlorine compounds

In the presence of moisture, these gases generate chloride ions that readily react with the various metals found in electronic components. These reactions are significant even at levels in the low parts per billion (PPB). **While moderate reaction occurs at levels of 1 PPB, a few tenths of a PPB combined with very high humidity increases reaction well above the mild limit (Class G1) recommended for TotalPlant Solution (TPS) System equipment.** It is important to consider contaminants from secondary sources (e.g., cooling towers, cleaning solvents, etc.), which often produce sufficient quantities (a few tenths PPB) to cause significant reactivity when combined with moisture and other gases.

Active sulfurs

In a mild (G1) environment, the concentration of active sulfurs typically is 2-3 PPB. The presence of moisture and a few tents PPB of chlorides increases reactivity into the moderate (G2) class or above. Field studies show that reactivities from mild (G1) to very harsh (GX) could be found with concentrations of H₂S less than 20 PPB and Cl less than 3 PPB, with the significant variable being the magnitude and rate of change of humidity. Active sulfurs and inorganic chlorides are predominant causes of atmospheric corrosion in the process industry.

Sulfur oxides

These compounds are generated as combustion products of sulfur-bearing fossil fuels, and from coking and nonferrous smelting operations. A few PPB of sulfur oxides can retard attack by other contaminants on reactive metals. At higher levels (>200 PPB), when dissolved in water to form acids, they can attack certain types of masonry, metals, elastomers, and plastics. Natural attenuation of sulfur oxides by permanent structures usually causes this family to be of little adverse consequence indoors.

Other compounds

The compounds described here are believed to represent the most common and significant factors in atmospheric corrosion of electronic equipment. The effect of other compounds such as nitrogen oxides, ammonia, oxidants, and fluorides are either not precisely known or typically not present in sufficient quantities to warrant special precautions. If a problem with any special constituent is suspected, a corrosion specialist should be consulted.

10.4 Dirt and Oil

Effects

Dirt and airborne oils cause electronics to fail the same way. The deposited material causes the electrical connectors in the equipment to lose their continuity. Fumes from a cooking area can be just as much a problem as dust from a dirt road.

10.5 Protective Guidelines

The following guidelines minimize equipment failure from atmospheric pollutants:

Humidity

Relative humidity (RH) is a very important factor in the prevention of corrosion failures. A few parts-per-billion of atmospheric contaminants combined with a humidity in excess of 60% becomes very corrosive for electronic equipment; however, low humidity (less than 40%) may cause electrostatic discharge (ESD) problems. Therefore, Honeywell recommends the following:

- Maintain humidity levels between 40% and 60%.
 - Control humidity fluctuations to less than 6% rate-of-change per hour.
-

Purification systems

Chemical purification systems may be required to provide acceptable air quality in industrial environments. The system should be chosen based on the manufacturer's ability to provide the necessary low levels of pollutants and verification by reactivity monitoring.

Air locks should be considered to minimize the ingress of untreated air. Positive pressure in equipment rooms should be maintained at approximately 0.1 inches of water.

Minimize the quantity of particulates reaching equipment to less than 100 ug/m³. The accumulation of particulates on equipment can be a problem in several ways. They can form leakage paths between conductors, interfere with contacts and switches, or result in corrosion.

Diffuse air away from electronic equipment and towards doors and other entrances. Do not duct air directly into cabinets. Temperature cycling and unnecessary air flow cause a number of problems.

Continued on next page

10.5 Protective Guidelines, Continued

General operating practices

Do not defeat temperature and humidity controls by opening doors and windows to enhance comfort.

Provide a means to close air inlets during periods of unusually high external pollutant levels.

During process upset or air conditioner breakdown, priority should be given to maintaining low relative humidity, minimizing air velocity, and preventing high concentrations of gaseous contaminants from reaching the equipment. Raising the temperature lowers the humidity and does not cause significant short-term reliability problems.

Chemical purification systems should be placed on a strict analysis/replacement schedule.

Keep the equipment doors closed at all times, except during times of actual human access.

Traffic through the equipment rooms should be restricted because of the dust being churned up.

Section 11 – Electrostatic Discharge

11.1 Overview

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Introduction

Most people are aware that Electrostatic Discharge (ESD) exists, especially with low humidity conditions. This discharge generates Electromagnetic Interference (EMI) that can cause control system problems. Early planning will bring ESD under control at little cost.

11.2 ESD Guidelines

Humidity

The most effective method of eliminating ESD in the operating environment is to control the temperature and relative humidity (RH) of the room.

Relative humidity (RH) control costs are reasonable when installed with the heating and air conditioning system. If an existing heating/air conditioning system does not contain RH control, conversion kits are usually available. Also, individual room humidifiers can be installed. Maintaining the RH of a room at 40 to 60 percent RH, for all practical purposes, eliminates ESD and provides a very comfortable atmosphere for people to work in.

Furniture and floors

Attention to furniture coverings and floor coverings reduces ESD with little or no extra costs. Avoid cut-pile or other (fuzzy) fabric with exposed thread ends; also 100% synthetic covers such as vinyl. Periodic treatment with a commercially available antistatic spray helps.

Chair casters are often made of a material that generates a charge as the chair is rolled. An antistatic mat does not help because the caster insulates the chair from the mat. Antistatic casters are available.

The floor tile wax is an active source of charge. It also prevents dissipation of static to the subfloor. Never allow the use of a floor tile wax.

Any carpeting should include conductive strands to bleed the charge to the subfloor below.

Clothing

Clothing is one of the most common contributors of ESD. Fire retardant polyamide uniforms may be required dress in certain facilities, and this material is also a rich source of ESD. It is not necessary for personnel wearing such clothing to touch grounded metal in order to draw ESD. The clothing itself causes an electrostatic discharge or sparks during walking or other movement. When fire retardant clothing is required, use fabric with woven-in stainless steel thread.

11.3 ESD Standard

ESD design methodology

All **TotalPlant** Solution (TPS) System products are designed under strict standards of control using test equipment that simulates electrostatic discharge. The discharge testing is in accordance with International Electrotechnical Commission (IEC) Standard 801-2 (1984). Testing is done at 15 kV with a 150-picofarad storage capacitor and a 150-ohm discharge resistor.

11.4 Equipment Response

Electrostatic discharges

Electrostatic discharge to process control equipment can be classified by how the discharge is applied; therefore, various failures can be seen, up to and including permanent damage.

Table 11-1 Categories of ESD and the Errors That Can Occur

Category	Description
Operational Discharge	An operational discharge is defined as a discharge applied to any surface of the enclosed equipment that can be touched by an operator. • Equipment errors that are not acceptable are – A reset or halting of the normal operation of the device that requires manual restarting, software reloading, or manual reprogramming to regain normal operation. – A deviation in an output of the device that exceeds the specified tolerances of the device's accuracy, or a deviation of a sufficient amount to cause failure, or create an abnormal or hazardous condition. – A detected change at the inputs of the device that causes raised indications (for example, alarms), or causes outputs to permanently change to an abnormal value or to change long enough to cause improper control. – Permanent damage to hardware or software that requires replacement to correct the failure. • Acceptable errors are – Display or indication flashes that are very short in duration (for example, 1 second or less) and return to normal without manual intervention. – A display or indication change that is recoverable by actuating available apparatus (for example, keys, switches, or buttons) used during normal operation and readily accessible to the operator.

Continued on next page

11.4 Equipment Response, Continued

Electrostatic discharges, continued

Table 11-1 Categories of ESD and the Errors That Can Occur, Continued

Maintenance Discharges	A maintenance discharge is defined as follows: • A discharge applied to any surface of a subassembly housing (for example, chassis, card file, or mounting/support rails/members) that is exposed when the device is withdrawn from its final housing, or when panels or doors are opened or removed. • A discharge to a field/customer accessible termination (for example, screw terminals for field/customer wiring, or cable connections for field/customer termination). • A discharge to a device that is used to interrupt power or preset/set/restart the unit. This includes mechanical adjusting devices that are used in field calibration, or test points that are made available for field calibration of the unit. Direct discharges to printed circuit tracks or components other than field/customer calibration devices are not maintenance discharges. See the section on troubleshooting discharges in the <i>Process Manager/Advanced Process Manager Service or High-Performance Process Manager Service</i> manual. • Unacceptable equipment errors are – The permanent damage to hardware or software media that requires replacement to correct the failure. – The opening of fuses/breakers that are power protection devices. • Acceptable equipment errors are: – Resets or halting of the equipment that requires manual restarting, software reloading, or manual reprogramming to regain normal operation.
Troubleshooting/Replacement Discharges	Troubleshooting/Replacement Discharges—A troubleshooting/replacement discharge is defined as a discharge that is applied to any component/printed circuit disconnected or removed from the final assembly of a device and are not designated as field/customer accessible. Handling/unpacking of replacement parts/subassemblies is included in this category. Any damage to the devices that occurs in the maintenance mode remains the responsibility of the maintenance personnel. Proper techniques require the use of grounding straps, correct handling, and a general knowledge of electrostatic discharge and its effects on devices if damage is to be prevented.

11.5 Circuit Board Handling

Circuit Board Handling

Avoid static damage when an electronic circuit board is handled outside its enclosure. The following practices and procedures are recommended:

- Handle a board only by its edges. Do not touch the lands, connectors, or components unless you are wearing a grounding strap, and the board is on a conductive work surface.
 - Stand on a grounded floor mat or touch a grounded surface before handling boards.
 - Do not carry unprotected boards across carpeting.
 - Ground all test equipment and handling equipment/tools (for example, screwdrivers or pliers) by touching the metal chassis with the tool before touching the boards or internal wiring on the enclosure or back panel.
 - Handle/transport boards to and from the job site in their protective bags.
 - Before removing the boards from their protective bag or from a card slot, touch the hand to the chassis to discharge any static charge buildup.
 - When shipping a suspected defective board, pack it in its protective bag before placing it in the shipping carton.
 - Do not use standard bubble pack mailers
 - Do not allow unprotected boards to come in contact with Styrofoam packing material.
-

Section 12 – Electrical Interference

12.1 Overview

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12.2 Electromagnetic Interference (EMI)

Effects

Electrical radiation from radio sources such as transmitter antennas may be so powerful as to alter the electronic signals in a Process Manager (PM), Advanced Process Manager (APM), or High-Performance Process Manager (HPM). This can affect control of the process. The following discussion goes into sources, susceptibility, and control.

12.3 Sources of EMI

Causes

Electromagnetic Interference (EMI) signals can be either steady-state or transient and are most commonly caused by the following:

- Electrical storms
 - Broadcast stations: AM, FM, TV, also ship and air radio beacons
 - Mobile 2-way radios: aircraft, ship, taxi, truck, police, and emergency
 - Hand-held 2-way radios: Citizens Band, 440 MHz industrial and security
 - Power tools and appliances: residential, commercial, and industrial
 - Electrical substations and transmission lines
 - Large rotating electrical machinery
 - Relays, contractors, or SCR circuits
 - Business machines: computers
-

Causes, continued

- Static electricity
 - Industrial: arc and induction furnace, heat sealer
 - Arc lighting: fluorescent, mercury, sodium
 - Microwave oven
-

12.4 Susceptibility to EMI

Specification

Excessive Electromagnetic Interference (EMI) from sparking contacts, radio frequency sources, or Electrostatic Discharge (ESD) can cause control system failures, such as a momentary shift in an indicated process variable or an unaccountable stop of a microprocessor.

For a frequency range of 10 kHz to 500 MHz, the Process Manager, Advanced Process Manager, and High-Performance Manager are rated to operate in a field strength of 15 V/m at the surface of the enclosure. This criteria is explored in the following subsections.

12.5 Control of EMI

Filtering

EMI is best stopped at its source. This usually requires little more than redundantly grounding the offending device and putting filters on all lines entering and leaving the enclosure; otherwise, the radiation source or electronics room may have to be shielded or relocated.

Shielding

The design of shielding for interference is based on knowing the skin-depth "d" for radio waves impinging on a metal shield-plate. Once "d" is known for a certain material and a certain frequency, the induced current at depth "z" in the shield is calculated by the formula

$$\frac{\text{Current at depth } z}{\text{Current at surface}} = e^{-\frac{z}{d}}$$

Both z and d are to be in the same units.

Skin depth "d" for various materials is calculated by the formula

$$d = 5033 \sqrt{\frac{\rho}{\mu_r f}}$$

where: d is the skin depth in cm

ρ is the resistivity of metal in ohm/cm

f is the frequency in Hertz

μ_r is the metal relative magnetic permeability (air = 1)

As an example, a Process Manager, Advanced Process Manager, or High-Performance Process Manager is to be placed on the second floor of an electrical substation. One meter (3 feet) below are 10 kV directly switched bus bars that have been attached to the ceiling. How thick should the cold-rolled steel enclosure be that is placed around the PM, APM, or HPM and its coaxial cables to satisfy the UCN coaxial cable separation needs?

Using the power line separation rules, the 10 kV bus bar system falls in the category of open parallel directly switched lines. This calls for a separation of 15 m/kV (50 ft/kV), corresponding to 150 meters (500 feet). The available separation is only 1 meter, and the resulting field strength at the PM, APM, or HPM is inversely proportional to the distance. As such, the field calculates to be (150 m ÷ 1 m) equal to 150 times greater than desired. The difference is made up by placing the PM, APM, or HPM in a walk-in enclosure of adequate thickness cold-rolled steel.

Continued on next page

12.5 Control of EMI, Continued

Shielding, continued

The effectiveness of a metal shield is at its worst at the lowest frequency of interest. For the Process Manager, Advanced Process Manager, or High-Performance Process Manager and the UCN, the lowest frequency of interest is 2 MHz. The skin depth for cold-rolled steel at this frequency is calculated as follows:

$$\begin{aligned}d &= 5033 \sqrt{\frac{\rho}{\mu_r f}} \\&= 5033 \sqrt{\frac{2.2 \times 10^{-6}}{100 \times 2 \times 10^6}} \\&= 0.0005 \text{ cm}\end{aligned}$$

We know "d" and the need for an attenuation of 150:1. The steel thickness "z" needed is calculated as follows:

$$\frac{\text{Current at depth } z}{\text{Current at surface}} = e^{-\frac{z}{d}}$$

$$\frac{1}{150} = e^{-\frac{z}{0.0005}}$$

$$\begin{aligned}\ln \frac{1}{150} &= -\frac{z}{0.0005} \ln e \\-5.0106 &= -\frac{z}{0.0005} (1)\end{aligned}$$

$$z = 0.0025 \text{ cm (0.001 inches)}$$

It takes very little ferrous material to do an excellent job of shielding at 2 MHz, 0.0025 cm (0.001 inches) in this case.

The shield enclosure has a doorway, it may have seams, and it may have openings for air. Keep the PM, APM, or HPM and its coaxial cables inside and back by a distance equal to the longest diagonal break (seam and door) in the shielding. Wire mesh can be substituted for sheet metal. Calculate the equivalent solid-sheet thickness based on the kg/m² (lb/ft²) weight.

The coaxial cable conduits exiting the enclosure are to run for a distance of (in this case) 15 m/kV (50 ft/kV) past the source of interference, and are to be bonded and grounded to safety grounds at both ends.

Continued on next page

12.5 Control of EMI, Continued

Separation

Sufficient separation between a source of Electromagnetic Interference (EMI) and the PM, APM, or HPM reduces electrical interference at the PM, APM, or HPM to an acceptable level. The EMI field strength of a radio transmitter at a certain distance is calculated by the equation:

$$E = \frac{7.02 \sqrt{P}}{d}$$

where: E is the field strength

d is the distance to the antenna in meters, $d \geq \text{wavelength}/(2\pi)$

P is the antenna radiated power in watts

As an example, a 5-watt trade rating, 440 MHz transceiver at 1 meter (3.3 feet) generates what strength at 100% efficiency?

$$\begin{aligned} E &= \frac{7.02 \sqrt{5 \text{ w}}}{1 \text{ m}} \\ &= 15.7 \text{ v/m} \end{aligned}$$

Because it is difficult for a hand-held unit to have a radiation efficiency of even 50%, keeping a 5-watt unit 1 m (3.3 ft.) from a PM, APM, or HPM works well. All other **TotalPlant** Solution (TPS) System products require a separation of 3 m (10 ft.).

Section 13 – Fire, Safety, and Emergency Lighting

13.1 Overview

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13.2 Safety

Guidelines

During construction, startup, and operation of the equipment, safety should be a prime consideration. Review your plans and operating procedures to make sure that

- All vital control areas, service areas, and data storage areas are continually monitored.
 - All service piping located above raised ceilings and below raised flooring is periodically inspected for damage.
 - Waterproof electrical connections are used when any power connections are below grade, underneath a raised floor. It would be embarrassing to be shut down by a minor flood.
 - Safety grounds to equipment cabinets and frames are required to protect personnel.
 - Emergency exits are provided at all control areas. The number and size are determined by the size of the area.
 - Personnel are properly trained in all emergency measures and safe shutdown procedures. These include shutting off electrical power (in the proper method and sequence), turning the battery switch off on each Uninterruptible Power Supply (UPS) and the Process Manager (PM), Advanced Process Manager (APM), or High-Performance Process Manager (HPM), shutting down the air conditioning system, notifying the fire department clearly and promptly, operating fire extinguishing equipment, and promptly evacuating personnel.
-

13.3 Fire Control

Scope	Providing a suitable gas flooding fire control system and hand-held extinguishers can prevent a fire from shutting down an operating control system.
Area Extinguisher	Honeywell recommends that the fire control equipment installed at a new or existing site be compliant with prevailing and existing fire safety and regulatory codes for your location. The recommendations for the type of area fire control equipment, location of the equipment, and general usage should come from agencies such as Factory Mutual Research (FM), Underwriters Laboratory (UL), etc. These Nationally Recognized Testing Laboratories (NRTLs) are very qualified in recommending the choice and use of fire control equipment. Local and national electrical codes, such as the National Electrical Code (NEC), the Canadian Electrical Code (CEC), and the International Electrotechnical Commission (IEC), should also be referenced. The National Fire Prevention Agency (NFPA) is also another source of assistance.
Portable Fire Extinguisher - Carbon Dioxide	Fire extinguishers filled with liquid CO ₂ are suitable for electrical fires. They must be carefully used because CO ₂ replaces oxygen in the air and can be hazardous to personnel. If the nozzle is held close to printed electronic circuitry, the cold CO ₂ can cause damage by thermal shock.

13.4 Emergency Lighting

Guideline	Most electronics buildings are windowless for security. For safety during a power loss, provide some kind of emergency lighting, such as automatic-recharge, wall-mounted lights.
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Section 14 – Planning for Hazardous Environment

14.1 Overview

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14.2 Introduction

Types of process control equipment

This section details safety requirements for Process Control Equipment (PCE), such as a Process Manager (PM), Advanced Process Manager (APM), or High-Performance Process Manager (HPM) that is located in areas where ignitable or explosive materials may be present. The Logic Manager (LM) and Data Hiway equipment are not approved for location in hazardous environments. Wiring to hazardous areas can be accomplished through zener barriers.

Certain processes handle ignitable or explosive materials. Local electrical codes require that electrical devices that are located in, or connected to, such process areas have some type of control to prevent accidental ignition of the process material. Terminology for discussing such installations is defined next, followed by specific requirements for each type process control equipment. For information about intrinsic safety applications, see your Honeywell Sales Engineer.

14.3 Hazardous Area Classifications

Terms

Hazardous materials are classified by a variety of terms. The terminology for two commonly used electrical codes is summarized in Table 14-1. These are the National Electrical Code (NEC) in the USA, and the standards by the International Electrotechnical Commission (IEC) in Switzerland. This table is not complete. Check your own electrical codes for more information.

In Table 14-1, the divisions (zones) are shown to be subject to interpretation. For example, an area that has a hazardous atmosphere for 1 hour in 10,000 hours (0.01% of the time) is far off the chart for hazardous areas. As such, it seems to be a safe and nonhazardous place to put electronic equipment; however, when the equipment has a hot component that is capable of causing ignition 100% of the time, explosion is imminent upon any process emission. This is a bad environment to install electronic equipment that is not approved for hazardous areas and it should be classified as in Division 2 (Zone 2), regardless of Table 14-1.

From Table 14-1, the approach to use when planning the installation of process control equipment depends on whether the process is classified as hazardous or nonhazardous, and the designated class, group, and division of the electrical installation and material handling areas. The PM, APM, or HPM housing area does not necessarily have the same categories as the material handling area of the process. For example, in using this table, oil refiners use materials that are defined by the NEC as being hazardous and belong to Class 1, Groups B, C, and D. The installation has to be appropriately planned.

14.4 PCE Hazardous Area Requirements

FM approval

Many Process Control Equipment (PCE) system configurations are approved by Factory Mutual for installing in, or interfacing to, Class 1, Division 2 locations. See Section 5 in the *Process Manager/Advanced Process Manager Planning* or *High-Performance Process Manager Planning* manual for a list of the approved components. When a cabinet configuration is approved, a label is attached to the cabinet containing the approved components specified in the initial purchase order. Certain components, such as FTA containing relays, are not approved for these applications because the relay contacts may produce sparks.

14.5 PM/APM/HPM Hazardous Area Placement

Guidelines

Many PM, APM, and HPM subsystem configurations are approved by Factory Mutual for installing in, or interfacing to, Class 1, Division 2 locations. See Section 5 in the *Process Manager/Advanced Process Manager Planning* or *High-Performance Process Manager Planning* manual for a list of the approved components. When a cabinet configuration is approved, a label is attached to the cabinet containing the approved components specified in the initial purchase order.

Certain components, such as FTA containing relays, are not approved for these applications because the relay contacts may produce sparks.

- Use only those PM, APM, and HPM devices that have been certified for safe use in Division 2 locations.
 - Locate the PM, APM, and HPM in a nonhazardous area, or
 - Purge the PM, APM, or HPM enclosure or the PM, APM, or HPM room of explosive gases or vapors.
-

Continued on next page

14.5 PM/APM/HPM Hazardous Area Placement, Continued

Hazardous location classifications

Table 14-1 is a list of hazardous location classifications.

Table 14-1 Hazardous Location Classifications

NEC	IEC	Environment
Class I		Explosive gases or vapors are present.
Class II		Combustible dusts are present.
Class III		Ignitable fibers or flyings are present.
Division 1	Zone 0	A location where a hazardous concentration of gases or vapors exist approximately 10-100% of the time (subject to interpretation as above)
Division 1	Zone 1	A location where a hazardous concentration of gases or vapors exist approximately 1-10% of the time (subject to interpretation as above)
Division 2	Zone 2	A location where a hazardous concentration of gases or vapors exist approximately 0.1-1% of the time (subject to interpretation as above)
Non-hazardous	Non-hazardous	Hazardous vapors exist less than .1% of the time.
Group A	Group IIC	A hazardous atmosphere containing acetylene or other similar gases or vapors.
Group B	Group IIC	A hazardous atmosphere containing hydrogen or other similar gases or vapors.
Group C	Group IIB	A hazardous atmosphere containing ethylene or other similar gases or vapors.
Group D	Group IIA	A hazardous atmosphere containing pentane or other similar gases or vapors.
Group E		A hazardous atmosphere containing metal dust, such as aluminum.
Group F		A hazardous atmosphere containing carbon black, coal, or coke dust.

USA classifications

For the USA, the nonhazardous area for placing PM, APM, and HPM enclosures can be selected by using publications such as the National Fire Protection Agency (NFPA) publication #497, *Classification of Class I Hazardous Locations for Electrical Installations in Chemical Plants*. Should no nonhazardous area exist, purged/pressurized enclosures or buildings are needed. Suitable purge techniques are described in local electrical code documents such as NFPA #496, *Purged and Pressurized Enclosures for Electrical Equipment*.

14.6 Division 2 (Zone 2) Wiring

Method

For certain electrical codes, the PM, APM, and HPM can connect directly to devices placed in Division 2 (Zone 2) areas, and the wiring techniques and materials can be those approved for use in ordinary (nonhazardous) locations.

Refer to the *Process Manager I/O Installation* manual for connection diagrams.

Use of the IOPs and Field wiring to the FTAs in Section 5 of the *Process Manager/Advanced Process Manager Planning* or *High-Performance Process Manager Planning* manual has been approved as nonincendive wiring by Factory Mutual Research; that is, when opened, shorted or grounded and with the PM, APM, or HPM operating normally, the wiring cannot release enough energy to cause ignition in a flammable atmosphere.

In some jurisdictions, this wiring need not conform to normal Division 2 wiring rules, but can be any wiring suitable for ordinary locations.

Wiring to other FTAs must conform to the Division 2 wiring rules, except that when digital output circuits have been current and voltage limited to nonincendiary levels by the customer, these circuits may also be nonincendiary. See ANSI/ISA S12.12 “Electrical Equipment for Use in Class I, Division 2 Hazardous [Classified] Location.”

To ensure that the circuits are incapable of igniting a flammable atmosphere, the size of cable and load parameters must be controlled. The maximum values are given in Section 5 of the *Process Manager/Advanced Process Manager Planning* or *High-Performance Process Manager Planning* manual.

FTA current limiting

To avoid ignition of hazardous process materials, the unpowered FTAs in the PM, APM, or HPM include a certain level of current limiting on the conductors going to the process. The limiting prevents ignition by the opening, shorting, or grounding of the process circuits. The current limiting resistors are shown on the FTA schematics in the *Process Manager I/O Installation* manual.

Resistor size

The resistor value is selected to assure a worst case short circuit current in the hazardous area of less than 150 mA for normally operating equipment. In NFPA publication #493, *Intrinsically Safe Apparatus for Use in Division 1 Hazardous Locations*, 150 mA is the ignition threshold for gases in Groups A through D. The NEC permits the use of ordinary location switches, buttons, wiring, and enclosures in Division 2 areas. Division 1 processes usually require zener barriers or explosion proof conduit.

Continued on next page

14.6 Division 2 (Zone 2) Wiring, Continued

Wiring Capacitance

There are restrictions for the wiring as documented in NFPA #493 above. Otherwise, energy stored in the wire capacitance and inductance can cause ignition of the process material. The signal circuit wiring from the PM, APM, or HPM to the process element must have an inductance of less than 3.6 millihenries and a capacitance of less than 0.3 microfarads. A typical twisted copper-pair wire rarely has an inductance of over 0.65 $\mu\text{h}/\text{m}$ (0.2 $\mu\text{h}/\text{ft}$) and capacitance of over 175 pf/m (60 pf/ft). This allows enough conductor to connect most processes.

Explosion proof conduits may be required for other wiring situations and electrical codes. See NFPA #493 or similar local electrical code documents for more guidance.

Power limited tray cable

The current limiting in the FTA 24 volt field wiring is superior to the Class 2 Power Limited signaling circuit definitions in the NEC. This allows the use of a Power Limited Tray Cable (PLTC). Note that PLTC is recognized by the NEC, but not the Canadian Electrical Code (CEC). When allowed, PLTC can be installed with other similar circuits on open ladder racks in Division 2 hazardous (classified) locations.

Coaxial cable

UCN coaxial cable has a maximum normal operating signal level of 66 dBmV (2 Vac rms); therefore, the normal signal level is not enough to cause ignition from the opening, shorting, or grounding of the coaxial cable circuit. The standard coaxial cables usually can be routed through a Division 2 area on open ladder racks.

Surface temperature tests Class 1

Honeywell adheres to Factory Mutual Research Corporation's *Standard 3611* when marking equipment as a result of surface temperature tests. The specification is as follows:

The maximum temperature of an external or internal surface to which a surrounding specified flammable gas or vapor-in-air mixture has access shall be determined under normal operational conditions. Such measurements need not be made on the internal parts of sealed devices. Measurements shall be made at any convenient ambient temperature normally encountered in a laboratory, corrected linearly to 40°C (104°F).

Continued on next page

14.6 Division 2 (Zone 2) Wiring, Continued

Temperatures greater than 100°

Equipment that attains temperatures higher than 100°C (212°F), based on a 40°C (104°F) ambient, shall be marked by the temperature code as given in Table 14-2 and corrected linearly to a 40°C (104°F) ambient. Component surface temperature may exceed the marked rating if it can be demonstrated by testing that no hazard exists.

Table 14-2 Equipment Maximum Surface Temperature Codes

Degrees C	Degrees F	Temperature Code
450	842	T1 *
300	572	T2
280	536	T2A
260	500	T2B
230	446	T2C
215	419	T2D
200	392	T3 *
180	356	T3A
165	329	T3B
160	320	T3C
135	275	T4 *
120	248	T4A
100	212	T5 *
85	185	T6 *

*Temperature codes marked with an asterisk are preferred and agree with the temperature identification system specified by the International Electrotechnical Commission (IEC 79 Series Publications).

14.7 Division 1 (Zone 0, 1) Wiring

Guideline

When permitted by local electrical codes, the PM, APM, or HPM connects to Division 1 (Zone 0, 1) areas most economically using the following techniques:

- Zener barriers for the process signals
- Explosion proof conduit for the UCN coax
- Fiber optic isolation of CRT equipment

Electrical codes that are not satisfied by any or all of the above techniques usually permit explosion proof conduit and housings.

Explosion proofing

Enclose the electrical device and conductors throughout the Division 1 (Zone 0, 1) hazardous area in housings designed to contain the explosion. This is the most common method of connecting electrical devices to a hazardous process.

The UCN coaxial cables ordinarily cannot cause ignition of hazardous material if broken; however, the coaxial cable drivers do not have any special features to limit energy during a circuit breakdown. If it is not possible to go around or go underground past a Division 1 (Zone 0, 1) area, place the UCN coax in explosion proof conduit.

Zener barriers

Zener barriers install in series with process wiring going into Division 1 (Zone 0, 1) areas. They are installed near or in the PM, APM, or HPM enclosure. The secured hazardous side wiring must then be segregated from all other facility wiring as specified by local electrical codes.

A zener barrier's special feature is that it limits energy into the process when an abnormal failure occurs in the electronics. Use only zener barriers approved for the PM, APM, or HPM and the process. A barrier commonly used with the PM, APM, or HPM is the 700 series from MTL Incorporated, Manassas, Virginia, telephone number (703) 361-0111, or consult your local MTL Incorporated office.

Continued on next page

14.7 Division 1 (Zone 0, 1) Wiring, Continued

Considerations for zener barriers

The following considerations are important when installing zener barriers:

- In a system using zener barriers, no one piece of equipment by itself is intrinsically safe. After evaluating each device with the barriers, the test agency lists or certifies each device or loop as intrinsically safe for a specific type of flammable group when used with the listed or certified zener barrier.
- Any type of equipment can be used in a nonhazardous location (safe side of the barrier) provided it does not contain a potential source greater than 250 Vdc or 250 Vac rms.
- Equipment containing voltage greater than 250 Vdc or 250 Vac rms can be used if specifically approved by a third party testing agency for use with the listed or certified zener barrier.
- The substation power transformer must be an isolating type. Three-wire auto transformers are not permitted.
- Any combination of equipment located in nonhazardous areas, interfaced with specified zener barriers and equipment located in hazardous areas, must be analyzed based on the intrinsic safety of each specified combination.

Zener barrier grounding

Zener barriers always mount on an approved/listed metal rail. The hanger rail for each is grounded with a 1.67 mm (12 AWG) wire to the local Zener Barrier Ground bus bar at the bottom of the cabinet. The bus, in turn, connects to Master Reference Ground (MRG) as shown in Figure 3-2.

Ground buses

The local Zener Barrier Ground bus bar is one of up to three bus bars that provide local grounding for the PM, APM, or HPM cabinet. All three buses interact in their requirements for zener barrier PM, APM, or HPM subsystems as explained in the three following sections.

Continued on next page

14.7 Division 1 (Zone 0, 1) Wiring, Continued

Local AC safety ground

Local AC Safety Ground — There is a local AC Safety Ground bus bar at the floor of each PM, APM, or HPM enclosure. All enclosure metal and all AC Safety Ground circuits for the cabinet connect to this bus. The bus always connects to the facility's AC Safety Ground. Connection is either through

- the power cable to cabinet
- by deliberately connecting the bus to the AC Safety Ground in the nearest power distribution panel
- by connection of the bus to a AC Safety Ground star plate that connects to the AC Safety Ground in a power distribution panel

Redundant or multiple connections are acceptable.

When a facility has zener barriers, the contact resistance of the AC Safety Ground electrode array to earth normally must be less than 0.1 ohms.

Local zener barrier ground

Local Zener Barrier Ground—There is a local Zener Barrier Ground bus bar near the floor of each PM, APM, or HPM cabinet. All zener barrier mounting rails in the enclosure separately connect with a 1.67 mm (12 AWG) green insulated conductor to this bus. The bus, in turn, connects with redundant 5.3 mm (4 AWG) green insulated wires to the Master Reference Ground (MRG) entry star plate and electrode as shown in Figure 3-2.

Local electrical codes normally do not allow the convenience of daisy chaining this bus to a similar one in the next cabinet of a complex of cabinets, nor do they allow connection to the nearest MRG star plate. Only the entry plate is acceptable. Most codes do not allow substituting the local Zener Barrier Ground for the local Master Reference Ground bus in the same cabinet, which usually connects to the same MRG star plate. Codes generally require clarity and identification of zener barrier grounding to the final grounding point.

Continued on next page

14.7 Division 1 (Zone 0, 1) Wiring, Continued

Local zener barrier ground, continued

Certain local electrical codes require that the ground wire for each barrier hanger strap be wired directly to the MRG electrode nailhead. They do not accept the consolidation of circuits at buses and star plates because the extra connections represent reliability problems; however, the running of a multiplicity of conductors through the earth to a distant electrode array also has reliability problems. Check with your local electrical safety authority and see if something more practical can be worked out, such as bridging or paralleling the circuits for reliability.

The local Zener Barrier Ground bus may have to accept amperes of direct current with zener barriers that have built-in 4-20 mA load resistors. This return current for each circuit goes into the local Zener Barrier Ground rather than to the usual local Master Reference Ground. Connect a short 1.67 mm (12 AWG) jumper between the local Zener Barrier Ground and the local Master Reference Ground. This creates a ground loop, but this is less of a problem than the voltage drops that would otherwise be created in the MRG system. The jumper is not shown in Figure 3-2.

Local electrical codes can accept using only the facility's AC Safety Ground and the service neutral for grounding the zener barriers. The procedure and installation changes are discussed in more detail in the grounding section of this manual. Also, you can contact your Honeywell representative for help.

MRG electrode

An MRG electrode that grounds zener barriers has special demands as follows:

- Local Master Reference Ground — All signal grounding and logic circuits in an PM, APM, or HPM cabinet reference to the local Master Reference Ground (MRG) bus bar. The MRG is a Honeywell-only isolated electrode, discussed in the grounding section in this manual. There is at least one, and usually two, specific performance requirements placed on a MRG electrode that grounds zener barriers.

First, any MRG that grounds zener barriers is to have a contact resistance to earth of less than 1 ohm. Without zener barriers, the MRG electrode has an earth contact resistance of up to 5 ohms. The 1-ohm zener barrier requirement comes from the code requirement that installed barriers have to be checked for grounding with an ohmmeter. One conductor for the ohmmeter connects to the barrier hanger bar. The other conductor connects to the AC Safety Ground bus in the nearest power distribution panel. The maximum resistance reading for this test is to be no more than 1 ohm. The **TotalPlant** Solution (TPS) System best meets this requirement by using an AC Safety Ground with a resistance to earth of 0.1 ohms or less, and with an MRG resistance to earth of 1.0 ohms or less. For more information, see the grounding section in this manual.

Appendix A – Conversion Factors and Tables

A.1 Overview

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Source

In 1973, ISO (the International Organization for Standardization) published measurement standard *International System of Units (SI)*. The following tables are from this system.

Conversion factors

Table A-1 Conversion Factors

Parameter	Given	Multiply By		To Obtain
Length				
in	Inches	25.4*	millimeters	mm
ft	feet	30.48	centimeters	cm
yd	yards	0.9144	meters	m
mi	miles (statute)	1.609	kilometers	km
nmi	miles (Nautical)	1.852	kilometers	km
	micron	1.0*	micrometers	μm
Å	angstrom	0.1*	nanometers	nm
Area				
cmil	circular miles	0.0005067	sq. millimeters	mm ²
in ²	square inches	6.452	sq. centimeters	cm ²
ft ²	square feet	0.09290	sq. meters	m ²
yd ²	square yards	0.8361	sq. meters	m ²
	sq. miles (statute)	2.590	sq. kilometers	m ²
mi ²	acres	0.4047	hectares (104 m ²)	km ² ha
Volume				
fl oz	fluid ounces (US)	29.57	cubic cm (ml)	cm ³ or ml
gal	gallon (US liq)	3.785	liters	l
gal	gallon (Canada)	4.546	liters	l
in ³	cubic inch	16.39	cubic centimeters	cm ³
ft ³	cubic foot	0.02832	cubic meters	m ³
	cubic yard	0.7646	cubic meters	m ³
yd ³	barrel (US petro)	0.1590	cubic meters	m ³
bbl	acre foot	1233.5	cubic meters	m ³
* Indicates exact value. Extracted from <i>IEEE Std 286-1973, 280-1968, and 322-1971</i>				

Continued on next page

A.1 Overview, Continued**Conversion factors,**
continued

Table A-1 Conversion Factors, Continued

Parameter	Given	Multiply By		To Obtain
Speed				
ft/min	feet per minute	5.080	millimeters per sec	mm/s
mi/h	miles per hour	0.4470	meters per second	m/s
km/h	kilometers per hour	0.2778	meters per second	m/s
kn	knots	0.5144	meters per second	m/s

Mass oz lb ton	ounces (avdp) pounds (avdp) short tons (2000 lb)	28.3 0.4536 0.9072	grams kilograms metric tons (1000 kg)	g kg t
Density lb/ft ³	pounds per cubic foot	16.02	kilograms per cubic meter	kg/m ³
Force oz f lb f kg f dyne	ounces-force pounds-force kilograms-force dynes	1.356 4.448 9.807 10-50	newtons newtons newtons newtons	N N N N
Work Energy - Power ft-lb f cal Btu hp ft-lb f /s Btu/h	foot pounds-force calorie(thermochem) British thermal Units (Intl) horsepower (elec) foot pound-force per second British thermal units per hour (Intl)	1.356 4.184 1055.0 746.0 1.356 0.2931	joules joules joules watts watts watts	J J J W W W
Pressure lb f /in ² lb f /in ² kg f /m ² mb mm Hg in H ₂ O ft H ₂ O	pounds-force/inch ² pounds-force/foot ² kilograms-force/m ² millibars millimeters of Hg inches of water (39°) feet of water	6.89 47.88 9.807 100* 133.3 0.2491 2.989	kilopascals pascals pascals pascals pascals kilopascals kilopascals	kPa Pa Pa Pa Pa kPa kPa
Light fC fL	footcandles footlamberts	10.76 3.426	lux candles per	lx cd/m ²
* Indicates exact value. Extracted from <i>IEEE Std 286-1973, 280-1968, and 322-1971</i>				

Continued on next page

A.1 Overview, Continued

Conversion factors, continued

Table A-1 Conversion Factors, Continued

Temperature Symbol	Given	Compute By	To Obtain	Symbol
°F	°Fahrenheit	$(^{\circ}\text{F} - 32) \times 5/9$	°Celsius	°C
°C	°Celsius	$(^{\circ}\text{C} \times 9/5) + 32$	°Fahrenheit	°F

AWG/metric

Table A-2 American Wire Gauge/Metric Equivalents

AWG	Metric (mm) Including Jacket	Copper Cross-Section, (mm ²)
0000	11.8	107.2
000	10.6	85.0
00	9.5	67.4
0	8.5	53.5
1	7.5	42.4
2	6.7	33.6
3	6.0	26.7
4	5.30	21.2
6	4.25	13.3
8	3.35	8.37
10	2.65	5.26
12	2.12	3.31
14	1.67	2.08
16	1.32	1.31
18	1.06	0.823
20	0.8	0.519
24	0.5	0.205

Continued on next page

A.1 Overview, Continued

Metric wire size table

Table A-3 Metric Wire Sizes

Cross Section in mm ²	Diameter mm (Approx)	Fuse (Amps)	Circuit Breaker (Amps)	Resistance (Ohms/100 m)
1.0	1.13	6	10	1.9
1.5	1.38	10	16	1.27
2.5	1.78	16	20	0.76
4.0	2.16	20	25	0.48
6.0	2.76	25/32	32/40	0.32
10.0	3.57			0.19
16.0	4.51			0.12
25.0	5.64			0.076
35.0	6.67			0.054
50.0	7.98			0.038
70.0	9.77			0.027

ATTENTION

ATTENTION—Wire with a cross-sectional area up to 16 mm² are mostly solid wire. Wire 16 mm² and greater are normally stranded wire.

Fuse and circuit breaker ratings are for insulated copper wire and cable. No values are given for wire sizes 10mm² and greater because those values are installation dependant.

Miscellaneous

Table A-4 Miscellaneous Conversion Factors

Multiply	By	To Obtain
bars	14.50	pounds/sq in
Btu/hr	0.2520	kilocalories
watts	3.413	Btu/hr
kilograms	2.205	pounds
kilograms/cm ²	14.22	pounds/sq in
kilowatts	860.5	kilocalories
kilowatts	3415.0	Btu/hr
millimeters	0.03937	square inches
meters	10.76	square feet

Appendix B – Temperature and Humidity for Data Hiway

B.1 Overview

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These are the topics covered in this section:

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-------	----------

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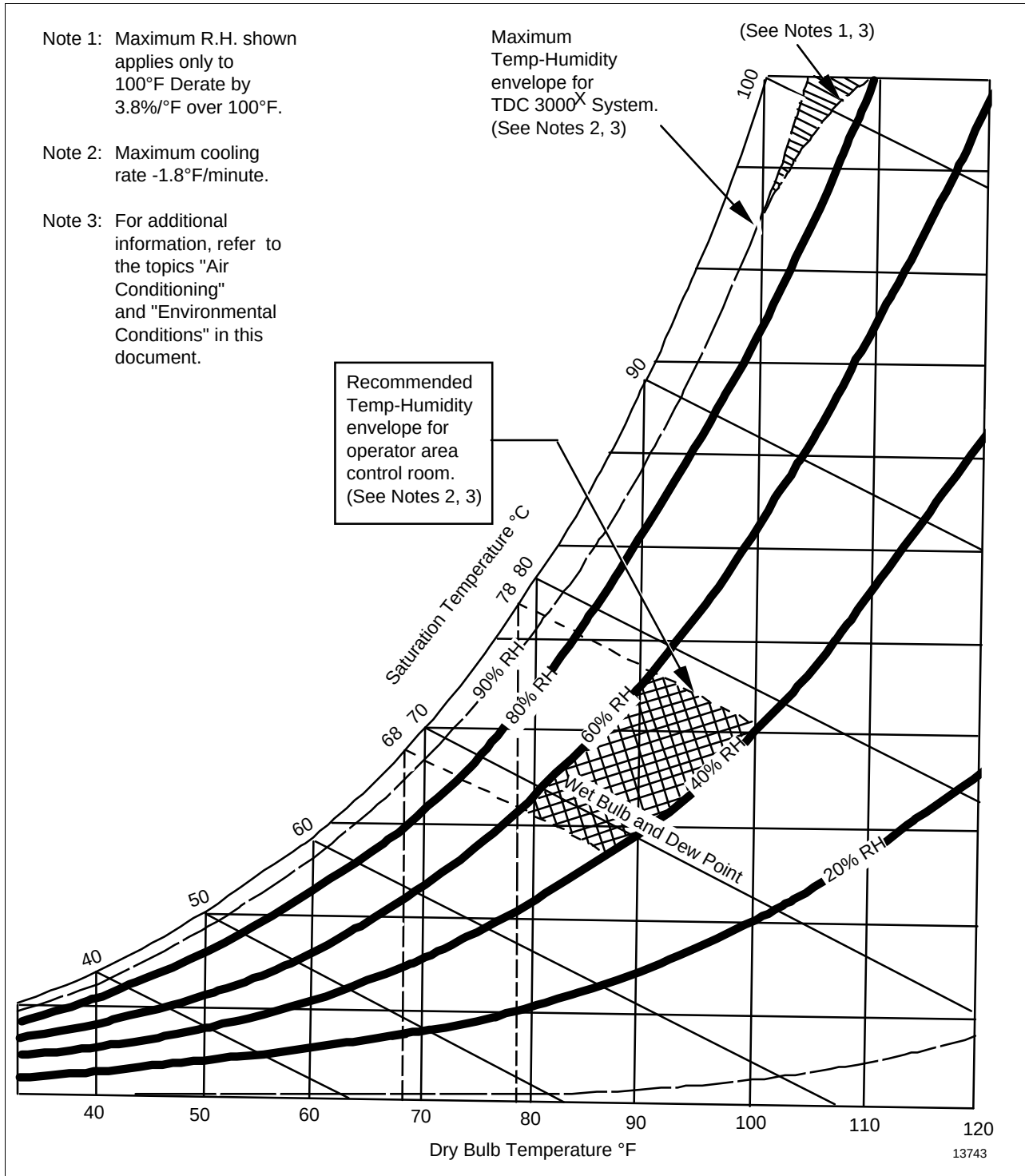
Scope

The charts in Figure B-1 and B-2 are used to determine temperature and humidity ranges for Data Hiway installations.

B.2 Temperature and Humidity Charts

Illustration

Figure B-1 Psychrometric Chart (English Units)

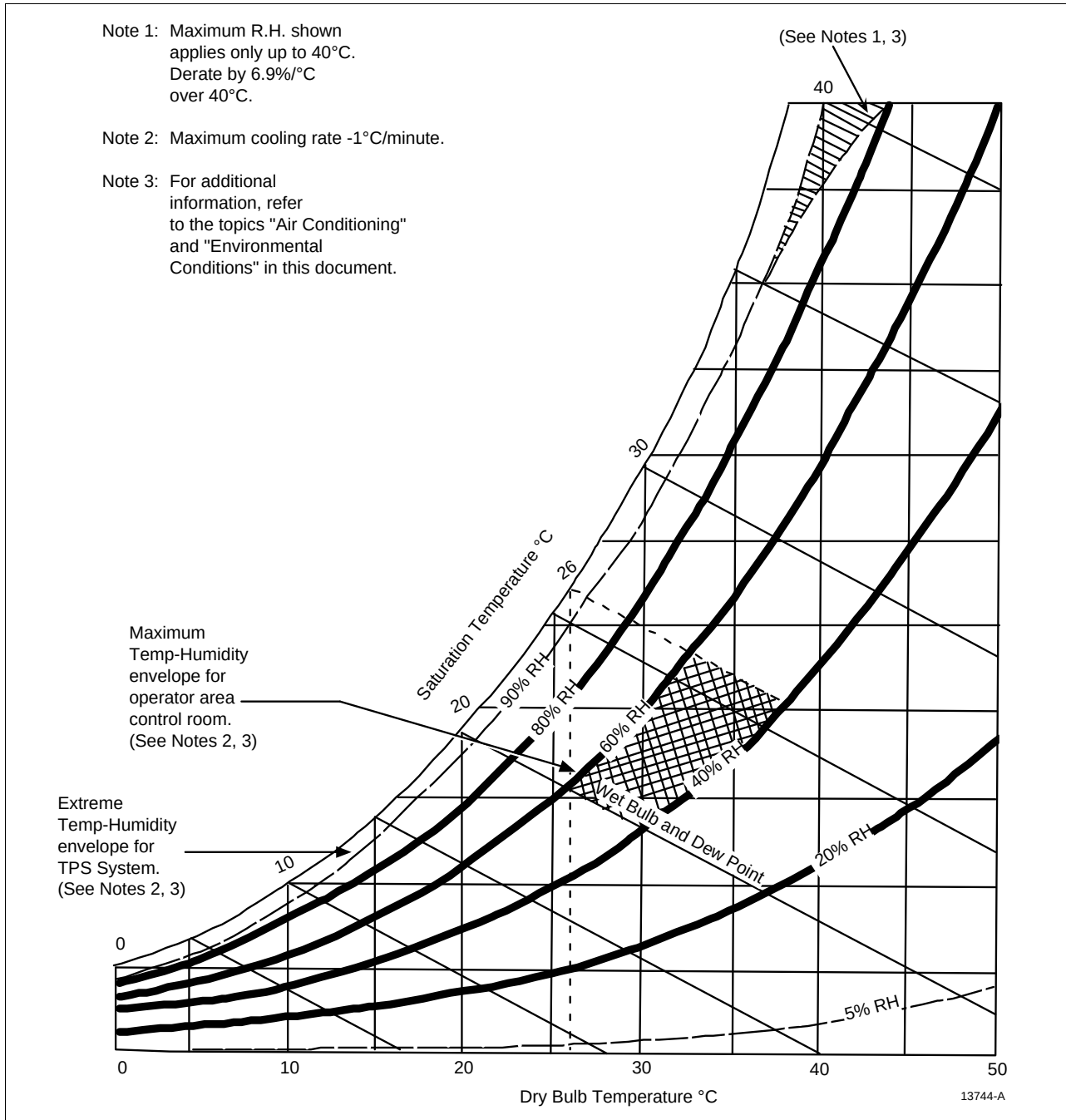


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B.2 Temperature and Humidity Charts, Continued

Illustration

Figure B-2 Psychrometric Chart (Metric Units)



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